KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION **v. 1.3**

RELATED SUPPORT

This document contains the following Sections

Key Project Information

SECTION A – Description of project

SECTION B – Application of approved Gold Standard Methodology(ies) and/or demonstration of SDG Contributions

SECTION C – Duration and crediting period

SECTION D – Summary of Safeguarding Principles and Gender Sensitive Assessment

SECTION E – Outcome of Stakeholder Consultations

- Appendix 1 – Safeguarding Principles Assessment (mandatory)
- Appendix 2 – Contact information of Project participants (mandatory)
- Appendix 3 – LUF Additional Information (project specific)
- Appendix 4 – Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of project	GS11131
Title of project	Conversion of intensive agricultural systems to dynamic agroforestry (DAF) systems for sustainable cocoa production in Ecuador
Time of first submission date	31/05/2021
Date of design certification	
Version number of the PDD	1
Completion date of version	30/09/2021
Project developer	UNOCACE – Unión de Organizaciones Campesinas Cacaoteras
Project representative	South Pole Carbon Asset Management Ltd.
Project participants and any communities involved	UNOCACE – Unión de Organizaciones Campesinas Cacaoteras – Developer entity South Pole – Entity providing support in the fulfilment of the GS4GG requirements
Host country(ies)	Ecuador
Activity requirements applied	☐ Community Services Activities ☐ Renewable Energy Activities ☒ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small scale ☐ Large scale
Other requirements applied	N/A
Methodology(ies) applied and version number	Gold Standard Afforestation/Reforestation (A/R) GHG Emissions Reduction & Sequestration Methodology. V1 – 2017
Product requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A

Project Cycle:	☐ Regular ☒ Retroactive

Land use & forest key project information[1]

<table>
<tr><td>Scope:</td><td>☒ Forestry
☐ Agriculture</td></tr>
<tr><td>Silvicultural system:</td><td>☐ Conservation (no use of timber)
☒ Selective Harvesting
☐ Rotation Forestry</td></tr>
<tr><td>Project area (ha):</td><td>510 ha</td></tr>
<tr><td>Eligible area (ha):</td><td>431.54 ha for plantation established before 2023.
For future plantations, once they are established, the eligibility will be analyzed.</td></tr>
<tr><td>10% set aside conservation area (ha):</td><td>Not required for micro-scale projects.</td></tr>
<tr><td>Evidence that Project area Boundary is clearly distinguishable in the field:</td><td>The project plots are within the farms of the participants, delimited using wooden stakes at strategic points; in this way they are differentiated from the participants' other production areas.</td></tr>
<tr><td>Planting area</td><td>
<table>
<tr><th>Year</th><th>Area (ha)</th><th>Eligible area (ha)</th></tr>
<tr><td>2018</td><td>46.47</td><td>46.47</td></tr>
<tr><td>2019</td><td>52.33</td><td>52.33</td></tr>
<tr><td>2020</td><td>56.62</td><td>56.62</td></tr>
<tr><td>2021</td><td>178.96</td><td>178.96</td></tr>
<tr><td>2022</td><td>23.29</td><td>23.29</td></tr>
<tr><td>2023</td><td>152.34</td><td>73.88</td></tr>
<tr><td>TOTAL</td><td>510.01</td><td>431.54</td></tr>
</table>
*N.B.: At time of writing, not all 2023 areas have been measured. This number may change at the Design Certification.
</td></tr>
</table>

[1] Please refer to Appendix 3 for detailed information on LUF projects.

How many Modelling Units (MUs) are included in the eligible area:	A total of 36 MUs are identified based on the combination of the year planted, the planting design (pre-2020 and post-2020) and the baseline.

	Cocoa with trees	Cocoa monoculture	Fruit tree and other perennial crops	Grass	Maize	Shrub
2018	X	X	x	X	X	X
2019	X	X	X	X	X	X
2020	X	X	X	X	X	X
2021	X	X	X	X	X	X
2022	X	X	X	X	X	X
2023	X	X	X	X	X	X

Summary of new areas added:	
Size (ha):	N/A
Date Added	N/A

Table 1. Estimated Sustainable Development contributions (Source: South Pole, 2022)

Sustainable Development Goals (SDGs) targeted	SDG impact (defined in B.6)	Estimated annual average	Units or products
13 Climate action (mandatory)	Tonnes of CO_2 removed by the project activities	13,677 tCO_2/year	
2 Zero hunger	Income due to the selling of food production per plot	2001 USD/ha	
15 Life on land	Area reforested annually with DAF systems (ha/year)	17.0 ha/year	

SECTION A. DESCRIPTION OF PROJECT

A.1. **Purpose and general description of project**

The project, "Conversion of Intensive Cocoa Systems to Dynamic Agroforestry Systems DAF in Ecuador", aims to implement AR activities by transforming conventional full-sun cocoa production systems to shade-grown cocoa in combination with other crops. This will be carried out using a Dynamic Agroforestry (DAF) Systems in small cocoa plots in different regions of Ecuador, benefiting approximately 600

associated producers in the ***Unión de Organizaciones Campesinas Cacaotereas***[2] **(UNOCACE)**. This system makes it possible to increase the productivity of cocoa cultivation and diversify production through the introduction of timber trees and food crops. In addition, it generates increased income for producers through the sale of organic cocoa, and production is more stable.

The main entity developing the project is the UNOCACE cooperative, "an organization of integration representative of the Popular and Solidarity Economy (EPS) legally created on February 2, 1999, consisting of 17 EPS associations that bring together 1,172 small cocoa producers from the provinces of Guayas, Los Ríos, El Oro, Cañar, Manabí, Esmeraldas and Pichincha, who own 5,616 hectares of national fine or 'Arriba' aroma cocoa with organic and fair trade certification; marketing and exporting 2,100 metric tonnes annually with the highest quality parameters".

All UNOCACE member associations are placed on an official registry,[3] with each association affiliated with small producers. All producers must comply with organic certification to belong to an organization.

In addition to UNOCACE, the following organizations participate in the development of the project:

- Chocolats Halba[4] purchases the cocoa produced by the project for the COOP supermarket. Halba also provides technical support through the consultant company Ecotop,[5] who developed trainings on the DAF system for the local staff.
- WWF Switzerland[6] manages the project funds and a programme of many other similar projects on behalf of Swiss retailer, Coop.
- South Pole[7] provides technical support in meeting the requirements for carbon project certification.
- GIZ[8] provides support in general administration.

Registering with the Gold Standard for the Global Goals (GS4GG) provides evidence and quantifies the positive impacts generated by the project. However, due to existing regulations in Ecuador, the reduction of emissions reduction of emissions cannot be

[2] http://www.unocace.com/es/
[3] PDD Supporting documents/Associations registries examples/Muisne es Vida and 2 de mayo
[4] http://chocolatshalba.ch/en/home.html
[5] https://www.ecotop-consult.de/
[6] https://www.wwf.ch/fr
[7] https://www.southpole.com/
[8] https://www.giz.de/en/html/index.html

monetized, for which reason this project is will only claim ‘impact statements’. In particular, the National Constitution (2008) states that "Individuals, communities, peoples and nationalities will have the right to benefit from the environment and the natural wealth that allows them to live well. Environmental services shall not be subject to appropriation; their production, provision, use and exploitation shall be regulated by the State". Likewise, the Environmental Organic Code (COA) establishes that "Environmental services shall not be subject to appropriation".

The project is being implemented in an area of 510 ha, where a baseline landcover of cocoa crops (under full sun exposure), alongside various crops such as bananas, corn, grasses and shrubs. Although these areas provide food and economic income to their owners, they are subject to changing market conditions, causing instability and a variable income. In addition, agrochemicals are used, which, if improperly managed, harm peoples' health and the environment.

The project boundaries are presented in Figure 1. The project is comprised of a group of many plots dispersed along the country from Southwest to Northwest, along the Pacific Ocean side of the Andes Mountain range, and includes the following provinces in Ecuador: Bolívar, El Oro, Esmeraldas, Guayas, Los Ríos and Manabí. The project includes only the CO_2 for carbon stock accounting in above-ground and below-ground biomass for the baseline. Also, the project scenario includes only the capture of CO_2 based on the above-ground and below-ground biomass. In accordance with a conservative approach, other emissions and organic soil emissions for the baseline and the project scenario have not been included or assumed.

Technology used to measure progress to meet SDG 13 include KoboToolBox, which will be used to record the measurements of the timber trees and fruit trees incorporated in the plot. For SDG 2, two methods will be used to measure the incomes: the UNOCACE’s accountability system (for Cocoa and Musaceae products) and KoboToolBox to measure complementary foods. For SDG 15, remote sensors or GPS will help to collect data.

A.1.1. Eligibility of the project under Gold Standard

This project is eligible for Gold Standard Certification in accordance with the approved methodology: the Gold Standard Land Use & Forests Activity Requirements, Version 1.2.1, published April 2020.

The project involves afforestation and reforestation (A/R) activities to implement agroforestry systems with cacao. The project presents a historical assessment of non-deforestation intended to support the case of including non-forest areas as eligible project area. This is presented in more detail in section B.2. This historical assessment covers the period of 2008 to 2018 using data from the Forest Reference Emission Level (FREL)[9] of Ecuador and field evidence of the non-presence of forested areas[10].

Aside from this eligibility assessment, it is essential to note that the project area is not currently registered with or part of any other carbon project. This can be confirmed with the list of projects affiliated with Verra and Gold Standard. Each relevant registry indicates only one project in the provinces of Imbabura and Pichincha, and these provinces are not included in the current project boundaries.

At the same time, there is no risk of double counting, as the project does not intend to issue any Verified Emission Reductions (VERs) but only report impacts. Ecuadorean law[11] indicates that all environmental services are governed by the state, which means that non-environmental services may not be appropriated or sold, such as for carbon offsets, including the carbon represented in VERs.[12]

The project is aligned with the national policy[13] of Ecuador, the National Development Plan for a Good Living (2016-2025), which aims for the conservation, presentation and restoration of the natural ecosystems and promotion of the use of clean energies and alternative land uses to reduce the impact of climate change, to achieve the Ecuador Zero Carbon Program, formulated in 2021.[14]

In addition, to integrate to support the transition towards sustainable production free of deforestation, the improvement of sustainable forest management and the use of

[9] https://redd.unfccc.int/files/submission_frel_ecuador.pdf
[10] Please see the supporting documents: Eligibility "second phase".
[11] The Constitution of Ecuador (Article 74) and the Environmental Organic Code (Article 85)
[12] https://www4.unfccc.int/sites/submissions/INDC/Published%20Documents/Ecuador/1/Ecuador%20INDC%2001-10-2015%20-%20english%20unofficial%20translation.pdf
[13] Plan Nacional para el Buen Vivir - PNBV in: "The carbon market in Ecuador" http://repositorio.puce.edu.ec/bitstream/handle/22000/5706/T-PUCE-5861.pdf
[14] https://www.ambiente.gob.ec/ministerio-pone-en-marcha-el-programa-ecuador-carbono-cero/

non-timber forest products, as is the case of the cocoa production presented in this project.[15]

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

i. Full and uncontested legal ownership of all products that are generated under Gold Standard Certification (Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and be discussed during local stakeholder consultations)
This project will not generate GHG products, and there will be no carbon credit commercial transactions on the voluntary/compulsory market. Instead, the project intends to certify impacts.
ii. Legal rights concerning changes in use of resources required to service the project (e.g., water rights)
No permits or authorizations are required by the project participants or UNOCACE to establish small-scale cocoa crops and plant trees on privately owned lands. UNOCACE has been promoting organic cocoa crops since 1999 and also has been exporting organic cocoa in compliance with all relevant regulations.
iii. Full and uncontested legal land title/tenure required to implement the project (e.g., for A/R projects, see Land Use & Forests (LUF) Activity Requirements)
Forms of property ownership in Ecuador are indicated in Article 321 of the Constitution: "The State recognizes and guarantees the right to property in its public, private, community, state, associative, cooperative, and mixed forms, and that it must fulfil its social and environmental function". For this project, all property is private. Most of the participants are owners of their land, and those who are not are in the process of inheriting or have a lease. Full and uncontested land tenure can be demonstrated by each of the participants. In the table below there is the description of land tenure categories, the documents to prove the land tenure and the participation of producers. Landowners may supply land titles to the auditor upon request. It is not common in Ecuador for small landowners to provide copies of their land title to external entities, unless for a bank loan or similar. For those reasons, UNOCACE does not own copies of land titles. For those participants who are holders, due to an inheritance process that is not fully formalized, land tenure can be demonstrated by the “Impuesto Predial”,[16] a legal document that proves the participant inhabits the land and pays the land taxes. Furthermore, those participants who are in the process of inheritance are asked to sign a “Declaración Juramentada”, which is an official declaration registered with a Notary that demonstrates the right to use the land while the inheritance processes are completed. In addition, farmers can prove legal use over the land with a “Certificado de posesión” and “Poder especial”, which are provided by a legal authority of the country. Finally, there are some producers that have the right of use and usufruct granted by any authority of the country and a few producers that are in the process of disengaging the property. For all the participants, UNOCACE has developed a document called "Own testimony of previous land use – cultivation"[17] in which each participant testifies as to which type of tenure applies to their property; thereby tenure records are kept. The steps taken by UNOCACE to assess the land ownership of each participant includes the completion of this “Own testimony of previous land use – cultivation” document, along with the “Declaración Juramentada" and the signing of the Farmers Agreement.

The legal ownership of products generated by the project and legal rights to alter use of resources was evaluated:

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15 https://www.unredd.net/announcements-and-news/2557-ecuador-presenta-su-plan-de-accion-redd-bosques-para-el-buen-vivir.html

16 Supporting documents/Land_ownership/Impuesto predial

17 Supporting_documents/Land_ownership/Testificaciones de uso de la tierra

Table 2. Type of ownership and participation (Source: UNOCACE, 2023; South Pole, 2023)

Type of ownership	Supporting documentations	Participation %
Owner (Propietarios)	Public deed (*escritura*)	46.74%
Holder (Poseedores)	Copy of property tax (*copia impuesto predial*), sworn statement (*declaración juramentada*), effective possession (*posesión efectiva*), certificate of possession issued by the parish council (*certificado de posesión otorgado por la Junta Parroquial*) and special power of attorney (*poder especial*).	30.07%
Tenant (Tenedores)	Rental contract (*contrato de arrendamiento*)	10.14%
Usufruct (Usufructo)	Usufruct (*derecho de uso y usufructo*)	5.43%
Other	Disengagement of inheritance	7.61%

A.2. **Location of project**

The project is located in the Republic of Ecuador. The country is bordered to the north by Colombia, to the east and south by Peru, to the west by the Pacific Ocean and has island territory in the Galapagos Islands. Its capital city is Quito and the official language is Spanish, which is spoken alongside 13 Indigenous languages. Ecuador has an area of 257,217 km^2 and a population of 17,374,000 inhabitants,[18] of which 26% in 2018 live in rural areas.[19] 23.2% of the population live under the poverty line.

Ecuador has four natural regions, according to the physical characteristics of the territory. From east to west they are: The Coast or Litoral region, the Sierra or Inter-Andean region, the Eastern region or Amazon and the Island region or Galapagos.

Administratively, the country is divided into 24 provinces, which are made up of two or more districts (cantones). The project takes place in nine provinces, which are located within the Litoral and Amazon region.

The project developer's (UNOCACE) office is located at Recinto El Deseo Km 30 Autopista Duran – Boliche.

[18] http://data.un.org/en/iso/ec.html
[19] https://datos.bancomundial.org/indicador/SP.RUR.TOTL.ZS?name_desc=false

Host Country	Ecuador
Region/State/Province etc.	Bolívar, El Oro, Esmeraldas, Guayas, Los Ríos, Manabí, Santa Elena, Sucumbios y Orellana.
City/Town/Community etc.	Las Naves, El Guabo, Pasaje, Chilla, Atacames, Muisne, Balzar, Miracle, Quevedo, Palenque, Ventanas, Babahoyo, Montalvo, Pedernales. UNOCACE Ecuador. Address Km 30 vía Duran – Boliche – Milagro, Recinto El Deseo – Ecuador

Region/State/Province

The provinces in the project area are Bolívar, El Oro, Esmeraldas, Guayas, Los Ríos, Manabí, Santa Elena, Sucumbios and Orellana.

Bolívar

This is the smallest province in the country. It is located alongside the Andes Mountain range, offering a diversity of climate that allows diverse agricultural products. The main agriculture product is corn, followed by other short-cycle cultivations such as wheat, barley and potato in the high areas, while in the low areas coffee, banana, cane and citrus fruits are cultivated. The province also has dual-purpose cattle and pig farms.

El Oro

This is one of the richest provinces in the country. It has diverse agricultural production, including bananas, rice, coffee, and cocoa. Cattle are also farmed. Bananas are the top commodity in the province, with the main banana exporting port of the country located in the city of Puerto Bolívar, which is next to El Oro's provincial capital, Machala. Banana production in this province has made Ecuador a benchmark in the world production of this commodity. The province also has mangrove ecosystems, beaches and tourist activity.

Esmeraldas

The economy of this coastal province is based on the export of shrimp and bananas. Its agricultural production also includes cocoa, tobacco, oil palm and coffee. Fishing also forms part of the economy. Tourism has also been developed here, due to the attractive beaches. Further, the rivers are important communication routes. At the same time, the province has abundant forest resources, such as valuable woods and fibers.

Guayas

This is the largest and most populated province in the country, home to approximately 30% of the population. Its capital, Guayaquil, is the largest city in the country. It is the largest financial and commercial center in the country, and is home to Ecuador's main industrial port. The main economic activities include industry, the port, agricultural activity (including rice, sugar, cocoa and coffee) as well as cattle, poultry and pork farming. Tourist activity is also significant, occurring mainly at the beaches and spas of the region.

Los Ríos

The name derives from the abundance of hydrographic resources the province holds, which favors fishing as an economic activity. The province is also recognized as having the most fertile soil in the country. Coffee, cocoa, bananas, rice and tobacco are cultivated, among others. In this province, most of the population lives in rural areas.

Manabí

This province is one of the most important administrative, economic and financial centers of the country. The main economic activities include commerce, cattle raising, industry, fishing and tourism. The agricultural sector focuses on diverse crops such as bananas, corn, rice, cocoa, coffee and fruit trees.

Santa Elena

The economy of the province is based on fishing and tourism. The peninsula has some fishing ports: Santa Rosa, San Pedro and Chanduy are the most important centers of the area, economic potential of this province. The province is gaining prestige as a gastronomic destination, especially for its seafood, but also for its agricultural products.

Sucumbios

Located in the north of the country, in the geographical area known as the Amazon region, mainly on the outer flanks of the western cordillera in the west and the extensive Amazon plain to the east. The province's main activities are oil extraction, commerce and agriculture.

Orellana

It is located in the northeastern part of the country, in the geographical area known as the Amazon region. The main activities of the province are oil extraction, commerce and agriculture.

City/town/community

The administrative divisions of Ecuador consist of firstly the province; one level down is the *cantón*; and the smallest division is the parish. This section will briefly describe the parish context in which project areas are located.

Table 3. Parishes, cantóns and provinces where the project plots are located (Source: South Pole, 2022)

Province	Cantón	Parish	Main activities
Bolívar	Las Naves	Las Naves	Agricultural production, including citrus, cocoa, banana, oil palm, soybean, grass and cassava, among others.
El Oro	El Guabo	Tendales	Agriculture, livestock, fishing and tourism associated with river activities.
		El Guabo	This parish produces the most bananas in the province. Tourist activities are associated with the numerous rivers, waterfalls and beaches. Service activities.
	Pasaje	Progreso	Almost 80% of the population of this small parish is dedicated to agricultural production. The main products are cocoa, bananas and short-cycle products.
		Casacay	The economy of the parish is based mainly on agriculture (cocoa, citrus and fruit trees) and cattle, pig and poultry farming. There are also tourist activities.
	Chilla	Chilla	Extensive altitudinal distribution (200–3,000 meters above sea level) in this parish allows for a wide variety of agricultural and livestock products. It also has tourist activities.
Esmeraldas	Atacames	Tonchigue	Artisanal fishing, livestock, subsistence agriculture and tourism are this parish's main economic activities.
	Muisne	San Francisco	The main economic activities in this parish are fishing, aquaculture, livestock farming and tourism.
		San Gregorio	The main economic activities in this parish are fishing, aquaculture, livestock farming and tourism.
Guayas	Balzar	Balzar	The main economic activity in this parish is agriculture, with corn as the main product, followed by rice and other secondary products. Livestock is also farmed.
	Miracle	Miracle	The main economic activity in this parish is agricultural production, mainly pineapple and sugar cane. An important sugar industry is present.

Province	Cantón	Parish	Main activities
Los Rios	Quevedo	Mocache	The main economic activities in this parish are agricultural, with livestock farming and cocoa cultivation, among others, being significant.
	Palenque	Palenque	The main economic activities in this parish are Agriculture (rice, banana, cocoa, yucca, plantain), livestock farming and fishing.
	Ventanas	Zapotal	The main economic activities in this parish include a variety of agricultural pursuits, significantly bananas, rice, corn, cocoa, coffee, cassava, avocado, fruit trees, among others.
	Babahoyo	La Unión	The main economic activities in this parish are agricultural activities other than cocoa and other fruit trees.
	Montalvo	Montalvo	The main economic activity in this parish is agriculture, significantly fruit trees, citrus fruits and rice.
Manabí	Pedernales	Cojimíes	The main economic activities in this parish are tourism, fishing, agriculture, livestock and wood extraction activities.
Santa Elena	Manglaralto	Manglaralto	The parish contains two geographical sectors: the coastal ridge and the interior, predominantly rural and conservationist, due to the presence of the Chongón-Colonche Protected Forest, which occupies about 40% of the parish's territory. This condition has led the parish to turn to activities related to ecotourism and environmental conservation.
	Colonche	Colonche	Colonche stands out as a strategic sector for crops such as tagua, guadua cane, toquilla straw, grapefruit, among others. However in the area there is a lack of technical training and a shortage of economic resources.
Sucumbios	Lago Agrio	Nueva Loja	The main activities of the parish are the oil industry, commerce, livestock and agriculture.
Orellana	La joya de los Sachas	La joya de los Sachas	It is characterized by the existence of natural spaces in its parishes and communities, with waterfalls, lakes, lagoons, rivers, estuaries and spas, as well as ethnic and cultural representations that constitute a very important tourist attraction.

Physical/geographical location

The working unit is the plot. Each plot is managed by a participant and can be between 0.25 and 2 ha. The coordinates of each plot are provided in the supporting information.[20,21]

The project is located across nine provinces in the country. The following Figure 1 shows the project region:

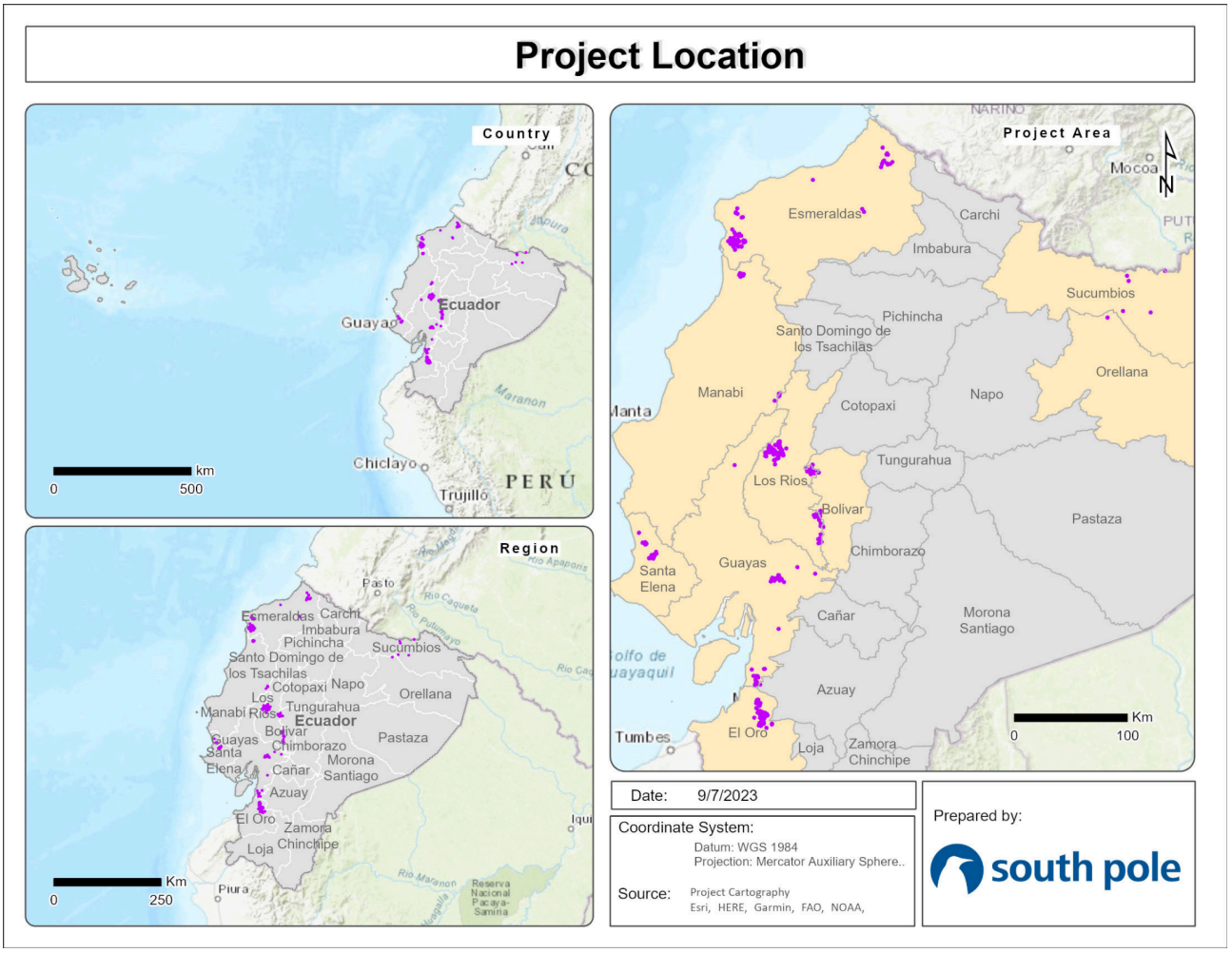


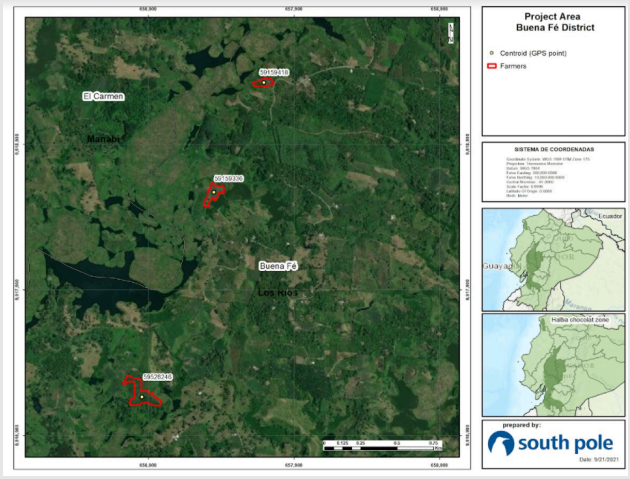
Figure 1. Project region (Source: South Pole, 2023)

Each plot is georeferenced with a GPS point and the perimeter of the area of activities. Since all the plots are dispersed in an important part of the country, fourteen maps were produced to illustrate the location of the project areas. All maps are available in the supporting documents.[22] As an example, Figure 2 below illustrates the location of the plots in six districts ("Cantones"):

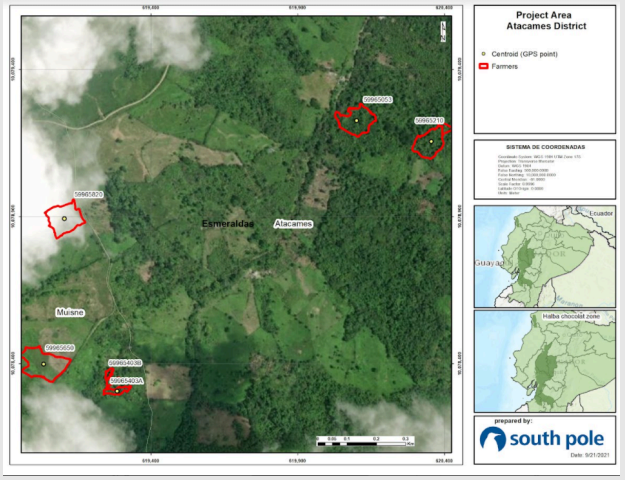
[20] Supporting documents/Technical information/Project coordinates

[21] The number of plots is planned to be increased for the years 2023

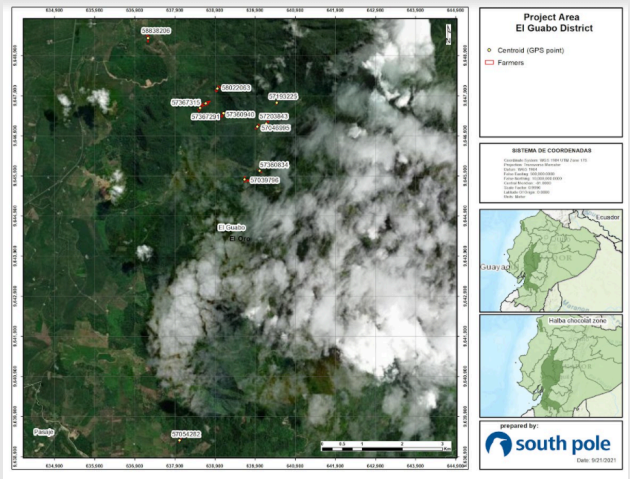
[22] Supporting documents/maps/plots location



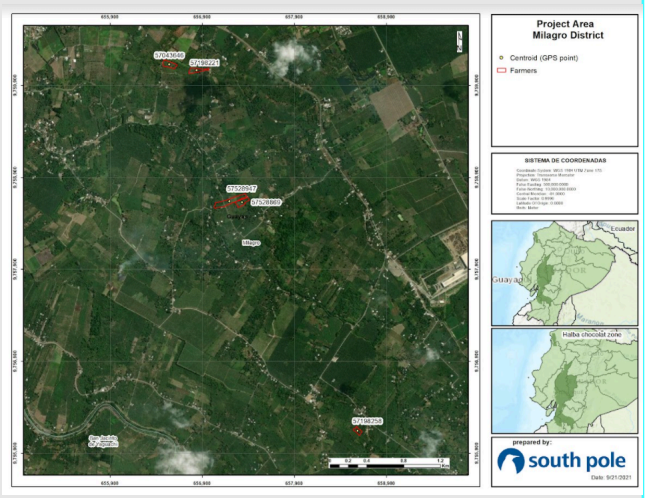
A. Plots in Cantón Buena Fe



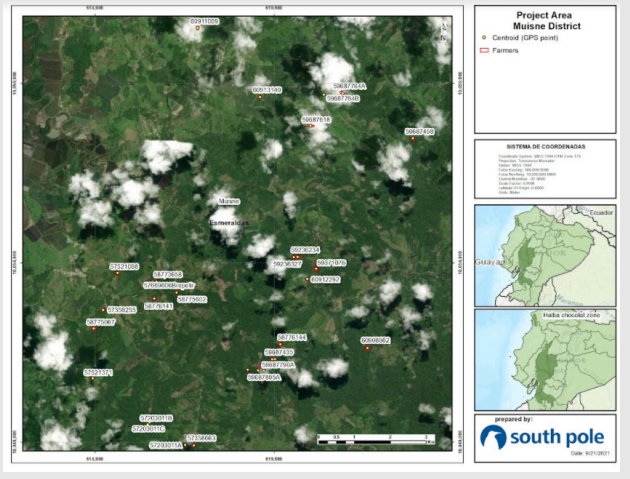
B. Plots in Cantón Atacames



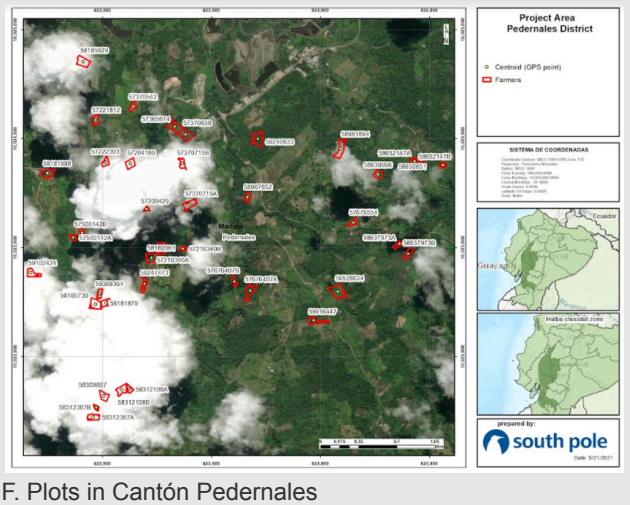
C. Plots in Cantón El Guano



D. Plots in Cantón Milagro



E. Plots in Cantón Muisne



F. Plots in Cantón Pedernales

Figure 2. Plots in Buena Fe, Atacames, El Guano, Milagro, Muisne, and Pedernales (Source: South Pole, 2022)

Maps indicating protected areas, biodiversity areas, infrastructure and permanent water bodies, locations of communities, and sites with special cultural, ecological,

economic religious or spiritual significance, as required in 3.1.2 Project guidelines, are presented in the following Figures 3 to 7:

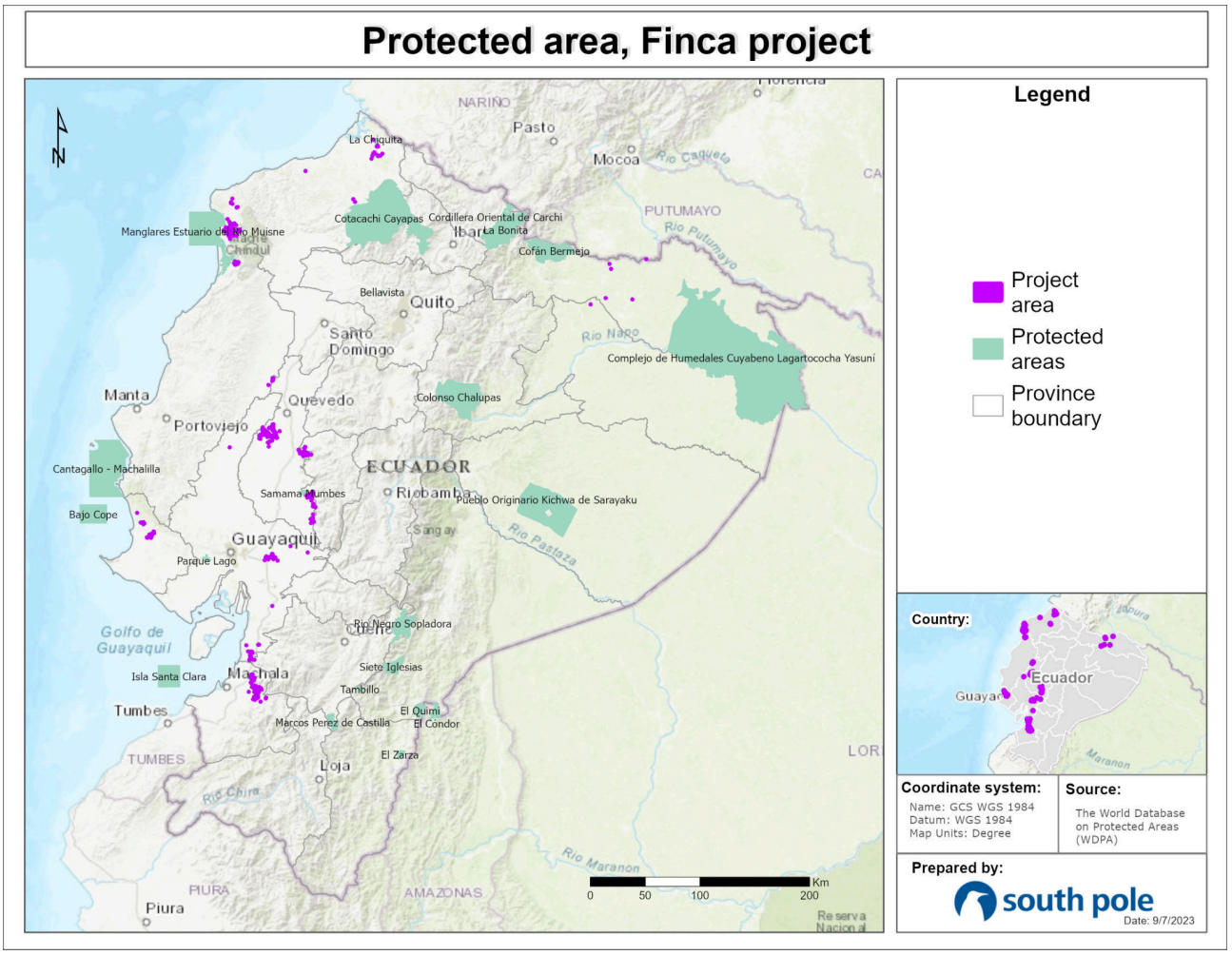


Figure 3. Protected areas and project area (Source: Ministerio del Ambiente, Agua y Transición Ecológica, 2021.)

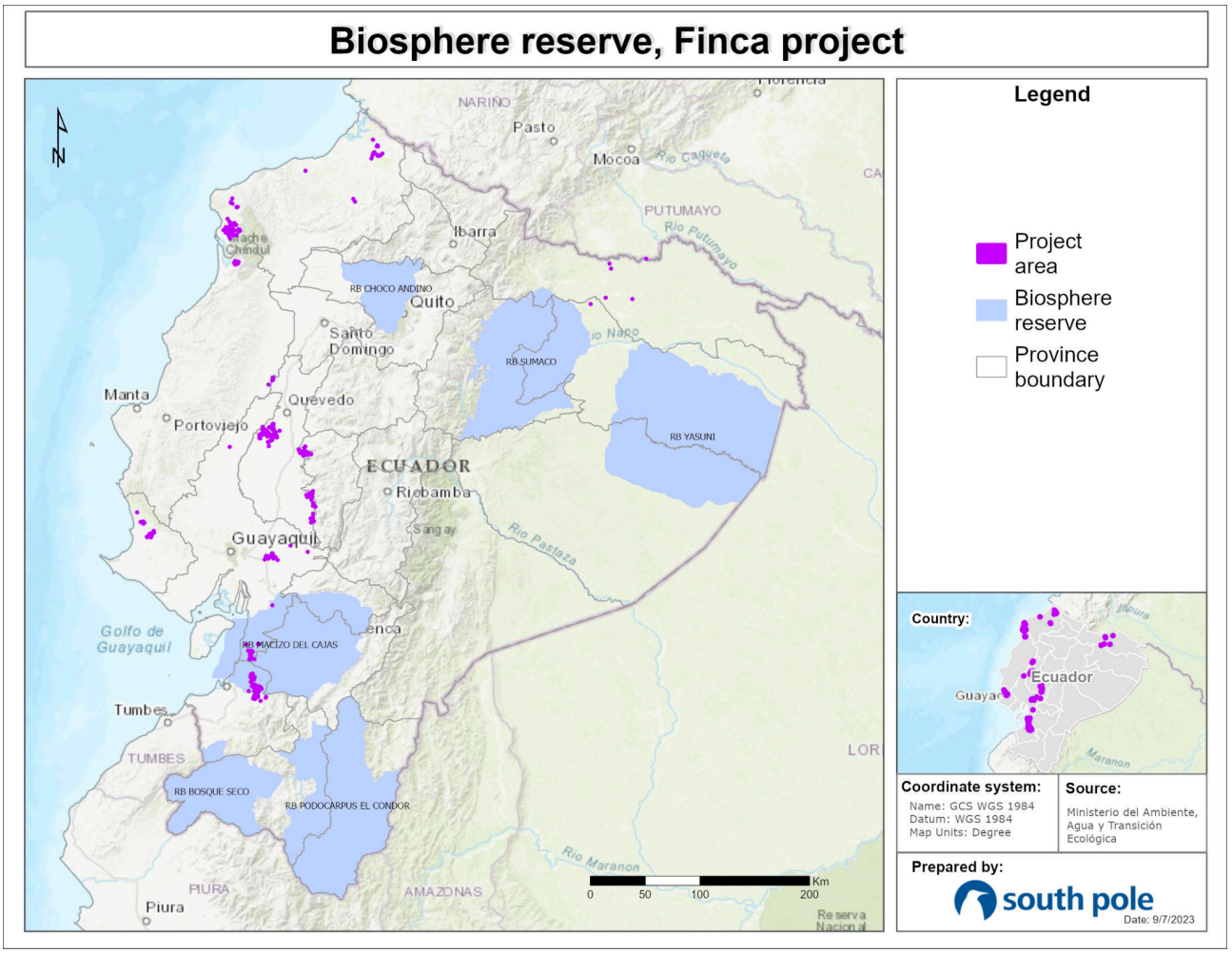


Figure 4. Biodiversity and project area (Source: Ministerio del Ambiente, Agua y Transición Ecológica, 2021.)

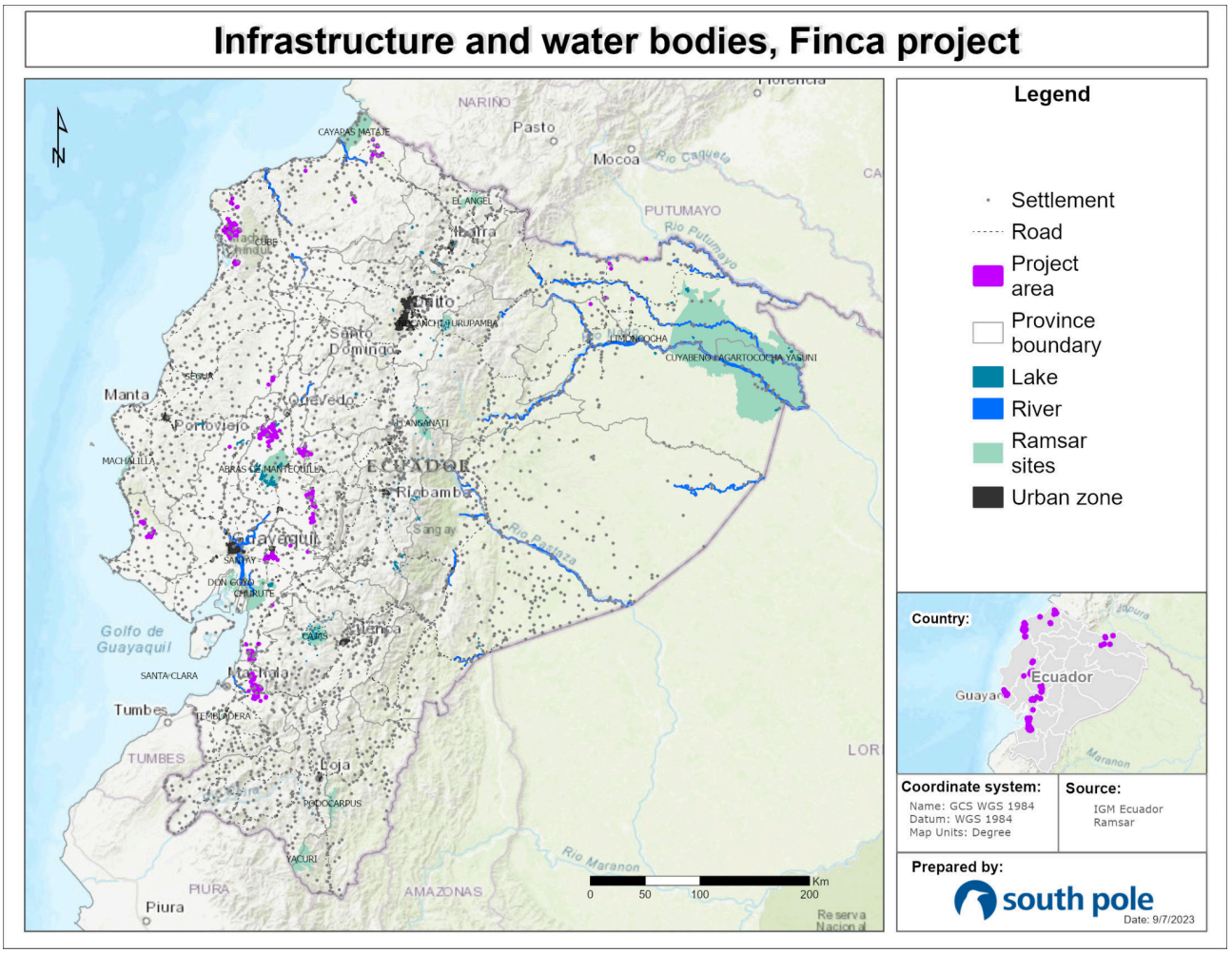


Figure 5. Infrastructure and permanent water bodies (Source: Ministerio del Ambiente, Agua y Transición Ecológica, 2021.)

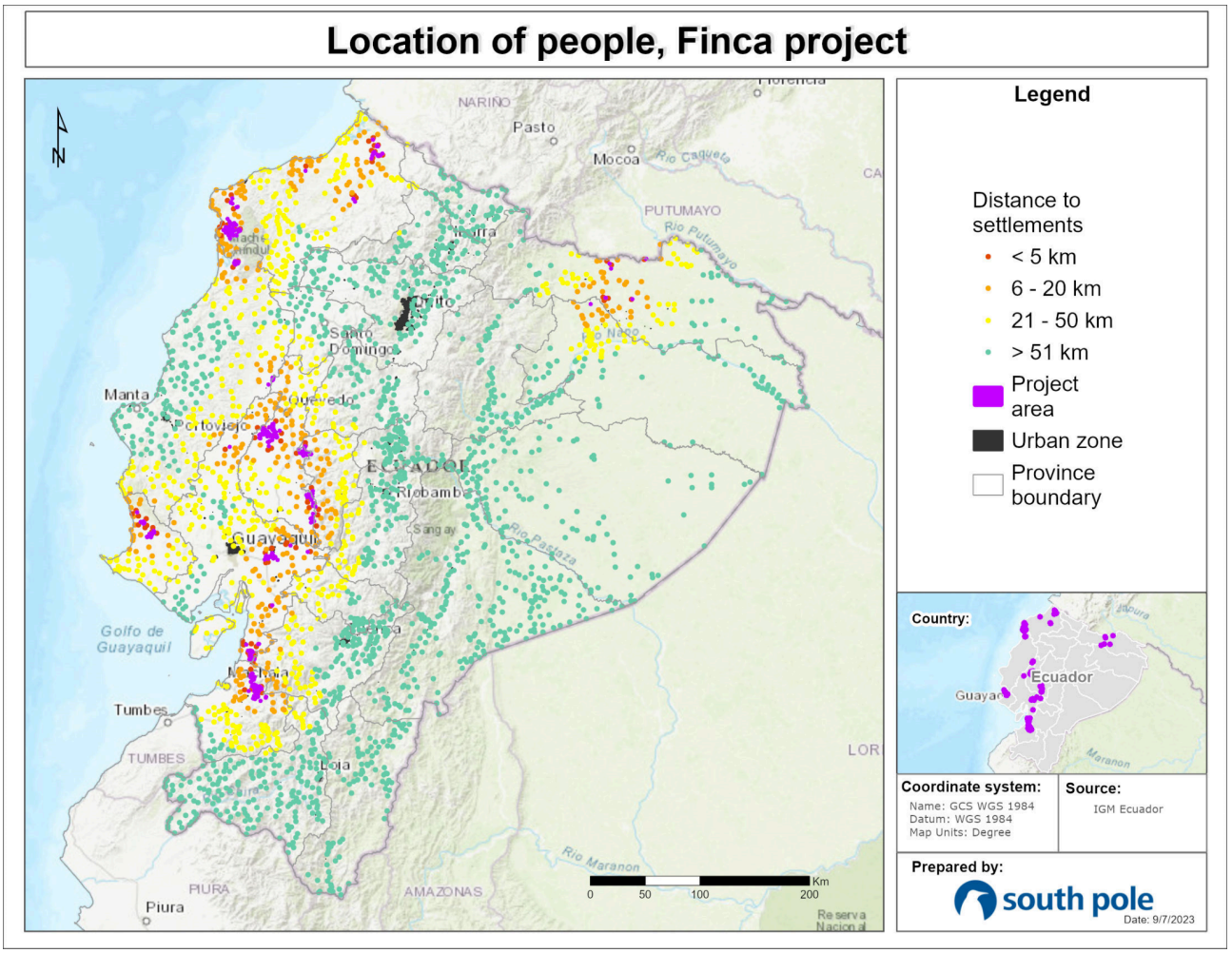


Figure 6. Location of settlements close to the project area. (Source: Ministerio del Ambiente, Agua y Transición Ecológica, 2021.)

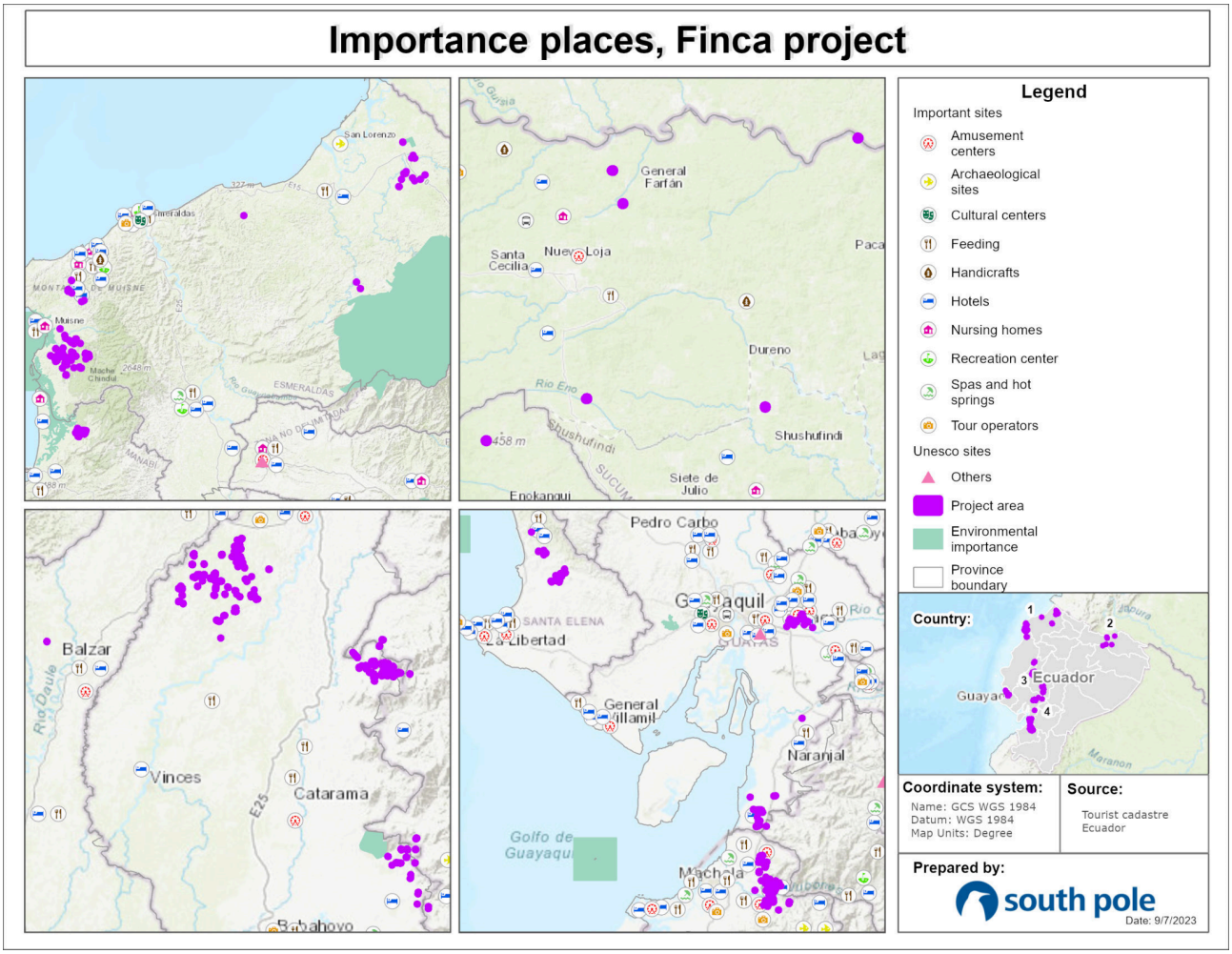


Figure 7. Sites with cultural, economic, spiritual or religious importance. (Source: Ministerio del Ambiente, Agua y Transición Ecológica, 2021.)

The plots, as illustrated in the above maps, are scattered throughout different provinces of the country. Table 4 lists the main climatic and relief characteristics of each province. All provinces cover a wide range of climatic conditions. The cocoa production plots are located in the lower lying areas with tropical climate.

Table 4. Summary of physical information for provinces where project areas are located (Source: South Pole, 2022)

Province	Altitude	Relief	Weather
Bolívar	85–5830 meters above sea level (masl)	Mountain range, plains, valleys	Cold, páramo, dry mesothermic, humid mesothermic and sub humid tropical monsoon.
El Oro	0–3923 masl	Mountain range, plains, coastline	Wet mesothermal, semi-wet mesothermal, dry tropical
Esmeraldas	0–3488 masl	Deltas, plains, coastline, mountain range	Wet mesothermal, semi-wet mesothermal
Guayas	0–1686	Deltas, coastline, savannas	Tropical savannah, tropical monsoon
Los Ríos	0–2159 masl	Plains, valleys, swamps	Tropical monsoon
Manabí	0–1200 masl	Coastal plains with some isolated hills	Dry subtropical to wet tropical
Santa Elena	0-40 masl	Coastal plains with some isolated hills	Dry subtropical to wet tropical
Sucumbios	100-2027 masl	Deltas, plains, mountain range	Tropical monsoon
Orellana	50-3732	Deltas, plains, mountain range	Tropical monsoon

Soils

According to the National Information System on Rural Lands and Technological Infrastructure,[23] there is a 1:25,000 scale cartography available for the entire country, including a map of soil orders using the United States Department of Agriculture (USDA) 2006 categories, which is one of the best-known systems.

For the project areas, the following soil orders are present: alfisols, inceptisols, molisols and vertisols. See brief descriptions below based on the National Soil Map Technical Report:[24]

Alfisols: Mineral soils, with a clear surface horizon over a clay horizon. Soil fertility is moderate to high, with deep layers. Very hard layers can prevent deep penetration of tree roots. Base saturation is at least 35% and generally in terrain free from recent disturbances.

[23] http://www.sigtierras.gob.ec/mapa-de-ordenes-de-suelos/

[24] PDD Supporting documents/Technical Information/MEMORIA_MAPA_ORDENES_DE_SUELOS

Inceptisols: Soils with medium development, with very variable physical and chemical properties, such as drainage and pH. Base saturation is around 60%. These soils occur widely throughout the country, and current grow cocoa, corn, African palm and banana crops.

Molisols: Soils with a surface horizon rich in organic matter, dark in color, with soil base exchange possible and favorable properties for root development. Mostly present in the structural terrains and tertiary hills located on the Pacific coast.

Vertisols: Poorly developed mineral soils with high swelling clay content (>30%). They present an extremely deep A horizon and are prone to flooding in the rainy season. With good management, crops give high productivity rates in these soils.

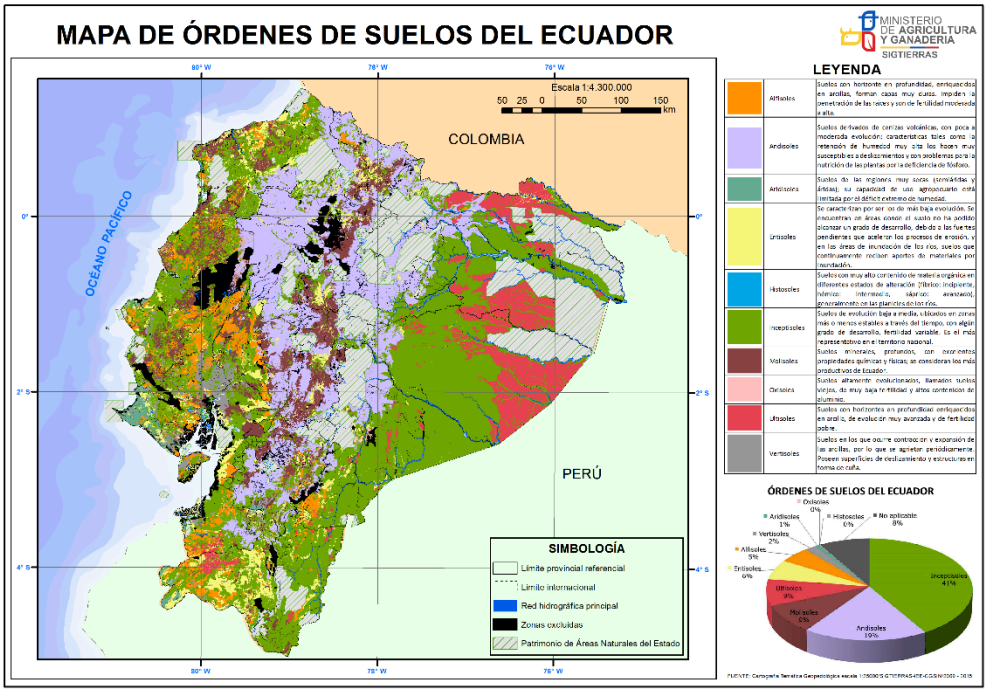


Figure 8. Land orders for Ecuador (Source: Sistema Nacional de Información de Tierras Rurales e Infraestructura tecnológica, 2015)

Biota

Below is a brief description of the natural regions of Ecuador (Coast, Andes and Amazon) and the ecosystems present in the coastal region, as classified by the Ministry of Environment in 2012.[25]

[25] PDD Supporting documents/Technical Information/Leyenda_ecosistemas_Ecuador2

In the vicinity, but outside of the project plots, are some forest relics. In the Esmeraldas River basin, there are lowland evergreen forests dominated by the families Myristicaceae, Moraceae, Fabaceae and Meliaceae. There is dense understory composed of species belonging to the Rubiaceae family and small palms of the Geonoma genus.

On the coast, there are concentrations of mangroves. Relatively close to some of the project's plots, is a mangrove forest in the bay of Cojimíes.

To the south, there are lowland deciduous forests up to 400 meters above sea level. They present strong anthropic pressure, since many populated areas are developing around them. Such forests are located on steep mountain slopes, of between 40% and 50%, with very stony soils. The tree stratum can reach up to 20–25 meters in height, with abundant shrubs and herbaceous plants in the rainy season.

The western part of the province of Guayas tends towards aridity as it approaches the sea. The vegetation is scrub, thorny forest and deciduous or semi-deciduous mountain forest.

A.3. **Technologies and/or measures**

The purpose of the project is to establish the DAF system in small farmer's plots and convert monocultures, conventional cocoa crops and underutilized areas into better managed areas. The DAF system aims to improve and implement sustainable production of cacao in areas of small farmers with the inclusion of native species in the agroforestry systems. In this way, dependence on agrochemicals can be reduced, family income and food security can be improved, and cocoa productivity is expected to increase.

Further, all woody species included in the DAF plots, from cocoa to timber, fruit trees and biomass species, represent the carbon sequestration accounting for the project. It is essential to note that the project does not involve any harvesting processes or extractive wood or timber activities (except a very small number of *Cedrela odorata* (MU-DAF 2020-2023) and *Gmelina arborea* (MU-DAF 2018-2019) because the systems are intended to be implemented with conservation objectives.

Dynamic agroforestry is a system based on understanding the natural structure and succession of ecosystems. The method is based on the knowledge of Indigenous Peoples in Latin America and was compiled by Swiss farmer and researcher Ernst Götsh in the 1980s and '90s. In the 1990s the German development service (then known as *Deutscher Entwicklungsdienst*, or DED) promoted the development of this system.[26] The main characteristics of DAF are:[27]

- High diversity and density of planting, stratification and high energy flows without external inputs. Different applications of management practices such as pruning (training, maintenance, rehabilitation). Selective cleaning and selection of healthy and productive propagation materials.
- Simulating natural succession, different species are planted at the same time, in groups classified as 'pioneers', 'secondary' species and 'primary' species. Such a model can include cassava or pigeon pea as pioneer species. These species are then replaced by secondary species, such as pineapple, papaya and banana, among other slower-growing species, which can develop in the shade of other trees.

For the project plots, a series of species have been defined according to their ecosystem function. See the following list in Table 5:

Table 5. Species that can be used in the system (Source: South Pole, 2022)

Category	Local name	Scientific name
Main species:	Cocoa	(*Theobroma cocoa*)
Timber species:	Pachacho	*(Schizolobium parahybum)*
	Cedar	(*Cederela odorata)*
	Mahogany	(*Swietenia macrophilla*)
	Oak	(*Quercus petraea*)
	Melina	(*Gmelina arbórea)*
	Yellow guaiac	(*Handroanthus chrisanthus*)
	Guachapele	(*Samanea saman*)
	Laurel	(*Laurus nobilis*)
	Pink Guayacan	(*Tabebuia rosea*)
	Fernan Sanchez	(*Triplaris cumingiana*)

[26] http://www.fao.org/3/CA0647EN/ca0647en.pdf

[27] PDD supporting documents/Technical information/Andres, Christian & Comoé, Hermann & Beerli, Anna & Schneider, Monika & Rist, Stephan & Jacobi, Johanna. (2016). Cocoa in Monoculture and Dynamic Agroforestry. 10.1007/978-3-319-26777-7_3.

Fruit trees:	Mango	(*Mangifera indicates*)
	Avocado	(*Persea Americana*)
	Achotillo	(*Nephelium lappaceum*)
	Soursop	(*Annona muricata*)
	Lemon subtle	(*Citrus aurantifolia*)
	Orange	(*Citrus sinensis)*
	Tangerine	(*Citrus reticulata*)
	Zapote	(*Pouteria sapota*)
Biomass species:	Guama	(*Inga spp. – Inga edulis*)
	Achiote	(*Bixa orellana*)
Pioneering species:	Banana	(*Muse of Paradise*)
	Banana	(*Musa sp.*)
	Guineo	(*Musa sp.)*
	Silk Guineo	(*Musa sp.)*
	Cassava	(*Manihot esculenta*)
	Bean	(*Phaseolus spp.*)
	Ahuyama	(*Cucurbita spp.*)
	Gandul	(*Cajanus cajan*)
	Corn	(*Zea mays*)

After several years, secondary species dominate the system and will eventually be replaced by primary species. The species that complete their life cycles are cut and replanted in the plot so that their biomass is incorporated into the soil, in order to maintain and increase the physical and chemical properties of the soil.

The arboreal stratum of the system can be occupied by timber trees, the middle stratum can include palms (oil and coconut) and the lower stratum has fruit trees, including the cocoa tree. This diversity of species and strata also considers diversity in the root system, so that competition for resources (e.g., space, nutrients and water) is minimized and different niches and depths in the soil are occupied. In this way, synergies are strengthened and at each successive stage, the current species can contribute to the successful development of the next.

Cocoa is included within the primary species of the system. The cocoa plant is of great importance worldwide for the production of cocoa beans and its derivatives. The

species is native to the Amazon understory and was traditionally cultivated in the shade of large trees.

The sequence of project activities is briefly described below:[28]

First, previous use of the intervention area is identified (e.g., type of crop, whether or not it has cocoa). This process is recorded in a database.[29] Later, the participant will agree on the new species to be established, based on the standard planting scheme (Figure 9). The re-establishment of cocoa cultivation (i.e., renewal or new plantation) is programmed according to the existing crop. Once this is determined, the land is cleared and demarcated according to where cocoa and some species of Musaceae (which may include banana, plantain, banana or guineo[30]) will be planted.

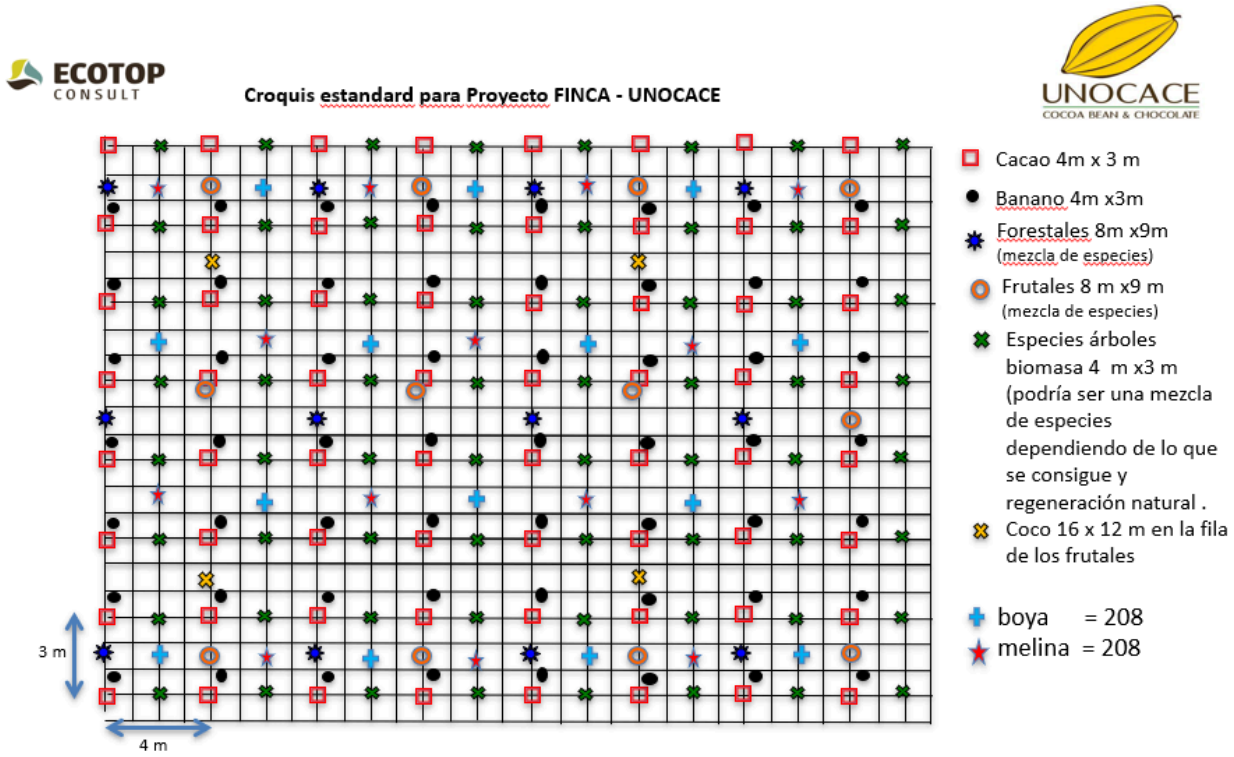


Figure 9. Reference diagram for the project (Source: South Pole, 2022)

[28] for more detail see the document Technical Methodology for Establishing a Plot, available in the supporting files: Supporting documents/Technical information/Metodología técnica para el establecimiento de una parcela

[29] https://ee.kobotoolbox.org/x/N3tgX12T

[30] Local type of banana

After the establishment of the larger species, the smaller species follow, which include cassava, corn, beans, vine guava, annatto, corn, Peruvian tumbe and orange. Species such as pumpkin, cane or vegetables such as cucumber, tomato or radish can also be included. The soil should be covered with vegetable residues from the previous crop to conserve humidity, avoid erosion and supply nutrients through slow decomposition of the waste.

Figure 10. Series of photographs illustrating the installation of a DAF plot (Source: South Pole, 2022)

Table 6. Details of the number of trees per ha (Source: South Pole, 2022)

Pre-2020 – Dynamic agroforestry system 2018–2020	Post-2020 – Dynamic agroforestry system 2020–2023

<table>
<tr>
<td>Timber species (planting density per ha)
S. parahybum (7)
S. macrophylla (7)
Q. petraea (7)
G. arbórea (7)
H. chrysanthus (7)
S. saman/A. guachapele (7)
L. nobilis (7)
T. rosea (7)
T. cumingiana (7)
C. odorata (7)
Total timber trees/ha = 70</td>
<td>Timber species (planting density per ha)
S. macrophylla (17)
Q. petraea (18)
H. chrysanthus (17)
S. saman/A. guachapele (18)
L. nobilis (17)
T. rosea (18)
T. cumingiana (17)
C. odorata (18)
Total timber species/ha = 140</td>
</tr>
<tr>
<td colspan="2">Fruit trees (planting density per ha)
M. indica (23)
P. americana (23)
C. cinensis (24)
Total fruit trees/ha = 70</td>
</tr>
<tr>
<td colspan="2">Biomass trees (planting density per ha)
Inga spp. (833)</td>
</tr>
<tr>
<td colspan="2">Cocoa (planting density per ha)
T. cacao (833)</td>
</tr>
</table>

Note: The model pre-2020 includes thinning for *Gmelina arborea* in year 10, and for model Post-2020 the species *G. arborea* is not included and the *Cedrela odorata* includes a thinning of 50% of trees.

During each cycle the timber trees, fruit trees, biomass trees and cocoa grow, sequester greenhouse gas (GHG) emissions and store carbon in their trunk, branches, and leaves, to be accounted.

SDG contributions

As described above, three SDGs are related to the project activities: 2 Zero hunger, 13 climate action, and 15 Life on land.

Monitoring for SDG 2 involves the measurement of net gain income from the selling of food production per plot. Due to the DAF systems implemented, an increase in the production of cocoa and recognition due to good production practices.

SDG 13 reflects the amount of CO_2 removed by the project activities and the monitoring and verification required for the project.

Finally, the project impact related to SDG 15 will be represented by increasing the reforested areas with DAF systems. More detailed information is presented in B.6.1 Explanation of methodological choices/approaches for estimating the SDG impact.

A.4. **Scale of the project**

According to the "Rule Update: Smallholder, Small scale and Microscale definitions and requirements for Land-Use and Forestry (LUF) Projects", the project is classified as small scale. This is because the project is developed on smallholder-owned[31] land, and the farmers selected will have more than 50% of farm work done by family members, cooperative members, or neighbors. The project is also considered small scale as the emissions reductions expected annually will be less than 16,000 tCO_2 in a project area larger than 500 ha (in this case, 510 ha).

A.5. **Funding sources of the project**

The funds to finance the project come from different sources. For the first years GIZ, UNOCACE, COOP and Chocolats Halba will contribute. All contributions will be made in cash until 2022. Following this, the financing will be done by COOP for CO_2 capture until the end of 2029 and UNOCACE.

The total budget of the project is EUR 4,784,046, allocated in four components: establishment and management of DAF plots (86%), professionalization and organizational development of UNOCACE (9%), establishment of value chains (2%) and inclusion of young talent and women (3%). 49% of the budget is estimated to be used between 2020 and 2022, and the remaining 51% between 2023 to 2029.

For more information, please refer to the explanatory document which is available in the media folder[32], with the corresponding cash flows and amounts per participant. Also the contracts signed between parties are attached in the supporting documents[33].

[31] Smallholders are farmes that have more than 50% of farm work is done by family members, cooperative members or neighbours.

[32] PDD Supporting Documents/Financial/20200402_Budget_FINCAPlus finale version_EN_South Pole

[33] PDD Supporting Documents/Contracts

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. **Reference of approved methodology(ies)**

The methodology selected for the calculation of the GHG sequestration is Gold Standard (GS) Afforestation/Reforestation (A/R), GHG emission reduction and sequestration methodology, Version 1 published in July 2017.

B.2. **Applicability of methodology(ies)**

1. Planting of trees on land that does not meet the definition of forest.

According to the GS, an A/R project is eligible where trees are planted on land that does not meet the definition of a forest at the start of planting, and the planted area should not have been forest for at least 10 years prior to the start of planting. The eligibility analysis was performed as described in section A.2 Eligibility of the project under Gold Standard. The forest definition used was the one in FREL for Ecuador (2001–2014) published in 2020, which is the same as the one used for the UNFCCC DNA for Ecuador.[34] According to Ecuador's forest definition for A/R project activities, forest means a land with tree crown cover of more than 30 percent and an area of more than 1 ha. The trees should be able to reach a minimum height of 5 m

The project eligibility analysis was conducted in two phases. The first phase was a cartographic evaluation, in reference to the official information available (Analysis from the Reference Level of Forest Emissions from Deforestation in Ecuador: period 2001–2014).[35] Bearing in mind that this source only covers up to 2014 and the project start date is 2018, a second phase of analysis was carried out. This second phase evaluates on-site information sources for each project plot, confirming the vegetation coverage for the period 2014–2018, and thereby completing the 10-year time frame of analysis as required by the Gold Standard.

First phase: Cartographic analysis 2009–2014

This analysis confirmed that the area where the plantations are established does not correspond to the definition of forest, both at the time of planting and at 10 years before the start date of planting.[36] The main input used to define the eligible areas

[34] http://cdm.unfccc.int/DNA/index.html

[35] https://redd.unfccc.int/files/06.01.2020_nivel_de_referencia_de_emisiones_forestales_de_ecuador_2001-2014.pdf

[36] According to the definition of forest for the country, for afforestation and reforestation activities, a forest is equivalent to an area of at least one (1) hectare, with canopy cover of more than 30% and a minimum height of five meters. (http://cdm.unfccc.int/DNA/index.html)

was the cartographic information in the Analysis from the Reference Level of Forest Emissions from Deforestation in Ecuador: period 2001–2014.

This reference level comprises a forest cover gain and loss map for the period 2001–2014, produced by the Ministry of Environment of Ecuador and based on Landsat TM, ETM+ and OLI images, with a pixel size of 30 m and a cartographic scale of 1:100,000. To perform the eligibility analysis, the following raster layers were used to form the reference level: forest cover at pixel level (0% to 100% cover) for the year 2000 and the annual forest loss for the period 2009–2014. The following raster layers were used to obtain the Forest/Non-Forest layers for the years 2009 and 2014:

- **Reclassification of forest cover by the year 2000**, according to the FAO definition of forest (30% canopy cover); and
- **Reclassification of the annual forest loss for the years 2009–2014** to obtain the accumulated forest loss layers in 2009 and 2014.

Using the Forest/Non-Forest layers (2009 and 2014), an overlap of both periods was created to determine eligible areas. The information was reclassified to show that eligible areas were those with non-forest coverage between 2009 and 2014, and, for the same period, ineligible areas were those with stable forest or forest gains/losses.

Finally, with the results obtained in raster format, a post-processing was carried out consisting of changing the projection coordinate system from WGS84 to the flat system WGS84-UTM17S, then transforming the resulting layer (raster) to a vector format. This eliminated any isolated polygons of less than 0.1 ha and produced an attribute table with the following categories: eligible (1) and non-eligible (0), along with an excerpt of the layer showing the limits of the project area.

The official documents employed for eligibility were reviewed for accuracy using the Independent Accuracy Report of the Land Cover and Land Use Change Map for the Period 2008-2014 (Methodology of the second reference level of emissions from deforestation). Global accuracy, obtained from the area proportions and calculated with the sum of the coincidences of the area proportions, equals 87.36%. The supporting document "*Evaluación independiente de la exactitud de los datos de*

actividad para los pagos por resultados del programa REDD Early Movers (REM) Ecuador" is attached as evidence.[37]

The project developer is aware that the accuracy value obtained is below the 90% minimum required by the LUF Activity Requirements. However, sources used are the best available, and images from other sources consulted such LandSat, Sentinel and Google Earth were found to be covered by clouds. For those reasons, the project developer requested a Deviation, which was sent to the GS during the preliminary review stage.

Second phase: On-site analysis years 2015–2018

Because of the absence of official cartographic information after 2014, it was necessary to complete the analysis of each plot through on-site inspections. This enables verification that the coverage during the period analyzed is not defined as forest.

Since UNOCACE has had a presence in the region for a number of years and almost all the current producers participating in the project had links with UNOCACE before joining the carbon project, there are records of UNOCACE technical visits to the farms. These show the existence of agricultural crops before the start of the carbon project.

These records also prove that areas that were not forest in 2014, according to the first phase of the eligibility analysis, remained non-forest in the following years, until the project start date in 2018. The records assessed include:

- Baseline forms: evaluation prior to establishing the DAF plot. They describe the existing vegetation cover, and record the presence/absence of timber and fruit trees.
- Follow-up forms: periodic reports verifying the development of activities in the plots.
- Other forms: diagnosis, new-plot, and renewal forms.

For all eligible plots, visitation logs are listed[38] and individual documents are available for review in UNOCACE archives. Also, during the auditor's validation visit, from

[37] See page 19 document named *Informe Evaluación Cambio de Uso.*

[38] Supporting_documents/Eligibility/Onsite_inspections_eligibility

January 17 to 20, 2022, some visits to the plots were recorded and the photos are available in the attached documents.[39]

Finally, information collected during site visits to plots in the preliminary eligible area identified in the first phase analysis for the years without available information (2015–2018) confirmed that the cover continued to be non-forest.

Using this evidence, the entire area identified as eligible in Phase 1 was confirmed to still be eligible in Phase 2. These areas, along with the remaining areas to be planted in the years 2020, 2021 and 2022, make up the total project area, amounting to 510 ha.

It is important to clarify that of the total eligible hectares from phase 1 and phase 2, some partially eligible plots were excluded. In order to facilitate management and monitoring, it was decided that areas to be included in the project were those with a surface area greater than 0.2 ha and that were at least 50% eligible.

As the project activities have been implemented it have been addressed sixth deliveries of eligibility analysis. The results presented in Table 7. Summary of eligibility results are based on the attribute table from the Geographic Information System (GIS) analysis:

Table 7. Summary of eligibility results including phase 1 and 2 (Source: South Pole, 2023)[40]

Deliverables	Planting year	Total area analyzed (ha)	Eligible area (ha)
First	2018-2020	143.0	119.8
Second	2018-2020	37.2	35.6
Third	2021	186.3	178.9
Fourth	2021 - 2022	15.2	15.2
Fifth	2023	27.0	26.8
Sixth	2023	57.4	55.2
TOTAL		**466.2**	**431.5**

[39] Supporting_documents/photos auditor validation
[40] Supporting documents/Eligibility "second phase"/20230911_Eligibilidad_09_14_Consolidado

A total of 466.2 ha was assessed, of which 431.5 ha was classified as eligible.

2. Project areas shall not be on wetlands

Ramsar defines wetlands as: "areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six meters."[41]

The eligible project areas planted in 2018–2023 were not established in wetlands as they have been used for agriculture production for more than 10 years. Table 8below shows the hectares of baseline of eligible areas already planted. Baseline data collection was done through questionnaires for the cocoa producers and includes a photo registry[42].

Table 8. Project area planted under different baselines (Source: South Pole, 2023)

Year	Cocoa with trees (ha)	Cocoa monoculture (ha)	Fruit tree orchard and other perennial crops (ha)	Grassland (ha)	Maize (ha)	Shrub (ha)	Total (ha)
2018	21.08	1.25	6.79	3.57	12.00	1.79	46.47
2019	14.67	3.44	4.33	10.57	9.29	10.03	52.33
2020	27.94	5.42	2.42	6.09	5.34	9.42	56.62
2021	25.08	44.11	48.59	14.37	9.62	37.19	178.96
2022	1.82	6.41	1.76	1.43	5.52	6.35	23.29
2023*	62.3	9.6	13.8	19.6	26.9	20.1	152.34
Total	152.9	70.2	77.7	55.6	68.7	84.9	**510.00**

*At the time of the preliminary review, not all these areas were measured. This will be completed by design certification.

In addition, Figure 11 indicates that wetlands are not present in the project areas.

[41] RAMSAR (2016). *An introduction to the Ramsar convention on wetlands* (5th Ed.). Ramsar handbooks.
[42] Please see "Baseline_covers_eligible_areas.xlsx" in folder Carbon estimations, as a result of processing the forms "eligibility second phase".

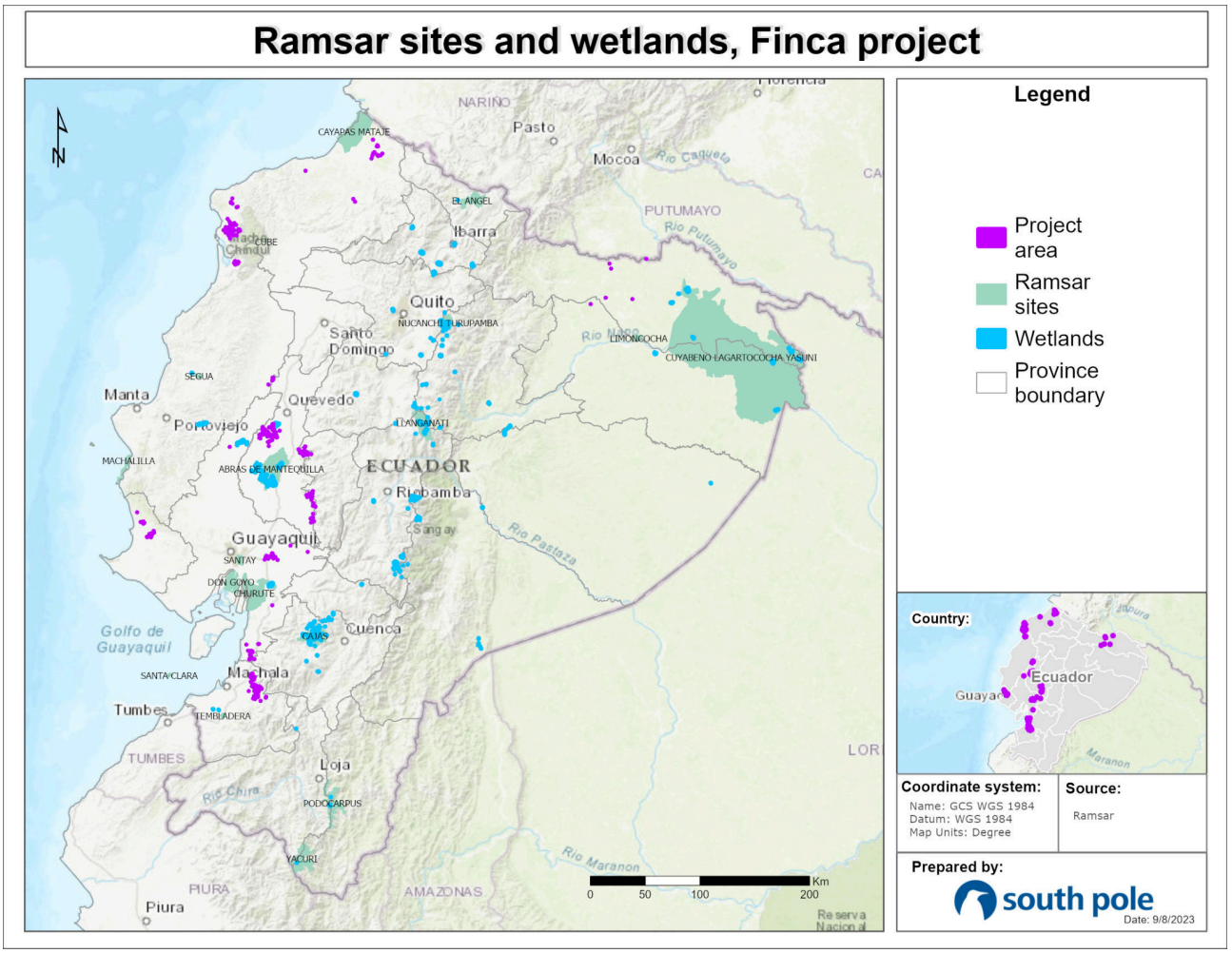


Figure 11. Wetlands (Source: South Pole, 2023)

For the areas to be planted in 2023, the implementation team will ensure that they are not established on wetlands. A similar situation is expected for the plots established during 2018–2023, as the new beneficiaries will be cocoa farmers interested in renovating their plantations or establishing new ones in other cultivated areas.

3. Project areas with organic soils shall not be drained or irrigated (except for irrigation for planting)

The GS methodology defines organic soils as follows:

1. If the soil is never saturated with water for more than a few days and contains more than 20% organic carbon in terms of weight (35% organic matter).
2. If the soil is subject to water saturation episodes and has either:
    - 12% (by weight) organic carbon (20% organic matter) if it has no clay;
    - 18% (by weight) organic carbon (30% organic matter) if it has more than 60% clay; or

- A proportional lower limit of organic carbon content between 12 and 18% if the clay content of the mineral fraction is between 0 and 60%.

According to the above definitions, the soil type present in the project area is not organic. In addition, the areas already planted were not drained, as a preparation for planting. Similarly, areas to be planted in the future will not be drained.

4. Soil disturbance (through ploughing, digging of pits, stump removals, infrastructure, etc.) on organic soils shall be in less than 10% of the area that is submitted to certification (not 10% of the entire project area).

For the areas planted during 2018–2023 the soil disturbance was minimal. No ploughing was applied for the establishment of the DAFs. In preparation for planting, biomass was chopped and left in the area as mulch. Big trees were not cut. The digging of pits for tree planting was done manually and no machinery was used for soil preparation. A minimal disturbance of soil is also expected for areas to be planted in 2023, since the same methodology as for already planted areas will be followed.

5. The most likely scenario without the project (baseline scenario) shall be defined for the project area. This scenario shall not show any significant increase of the baseline biomass (tree and non-tree).

Based on the historical use of the land and conversations with local extension workers, the potential baseline scenario is described below:

- Cocoa monoculture is expected to continue as cocoa monoculture: this shadeless intensive agricultural system requires large amounts of chemical inputs as fertilizers. It has an environmental impact on the soil and is an additional expenditure for cocoa farmers. Without the project, cocoa farmers would not have a source for income diversification to help them improve their living conditions and increase their resilience to climate change.
- Cocoa with trees is expected to continue as cocoa with trees. Cocoa under this production system requires little chemical input and maintenance. Cocoa producers will probably keep it, as they can still obtain income from cocoa and other fruit trees with minimum input. Establishment of new areas is very unlikely since that would represent additional costs for small producers with existing economic constraints.
- Fruit tree orchard and other perennial crops is expected to continue as fruit tree orchard and other perennial crops. Fruit tree orchards and other perennial crops

such as banana, palms and coffee provide food to the household and may represent a source of income. Fruit tree orchards and other perennial crops may be replaced by crops depending on the expected income and economic incentive to produce another crop.

- Grassland is expected to remain as grassland. The area of grassland may increase under a baseline scenario depending on the demand for livestock.
- Maize cultivation is expected to remain as maize. Maize is a staple crop that can be easily planted in combination with other crops.
- Shrubland are expected to continue as shrub. They can potentially be converted into pasture for livestock or crops depending on the demand of crop and livestock production.

B.3. **Project boundary**

The project includes approximately 500 small cocoa farmers that will establish DAF systems in areas smaller than 5 ha. The project beneficiaries (cocoa producers) are distributed in nine provinces in Ecuador: Bolívar, El Oro, Esmeraldas, Guayas, Los Ríos, Manabí, Santa Elena, Sucumbios and Orellana, from Southwest to Northwest, alongside the Andes Mountain range. The specific location of the project beneficiaries is shown in maps of the plots already selected and is provided in a separate folder.

A diagram based on the description of technologies is presented in the below Figure 12:

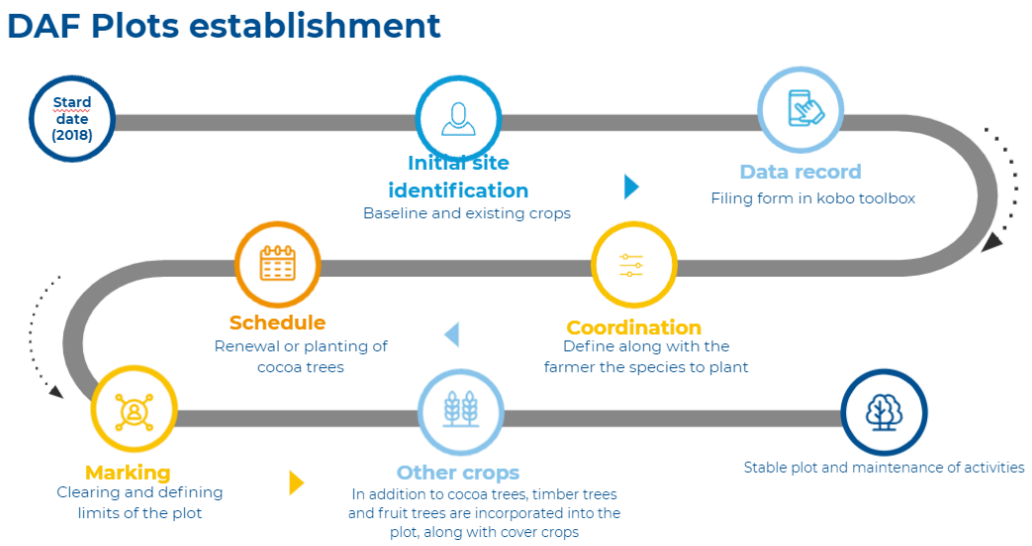


Figure 12. Project boundary diagram (Source: South Pole, 2022)

Table 9. Source and GHGs. (Source: South Pole, 2022).

Source		GHGs	Included?	Justification/explanation
Baseline scenario	Source 1: Above-ground biomass	CO_2	Yes	All tree, grass and shrubs existing in the plantable area will be accounted for following the requirements for baseline carbon stocks estimation put forward by the methodology.
		CH_4	No	N/A
		N_2O	No	N/A
	Source 2: Below-ground biomass	CO_2	Yes	Below-ground biomass will be included in the baseline estimations in accordance with the applied methodology.
		CH_4	No	N/A
		N_2O	No	N/A
	Source 3: Soil	CO_2	No	The project does not include disturbance of organic soils. Emissions are not expected from this activity.
	Source 4: Other emissions	N_2O	No	Using a conservative approach, it was assumed that the baseline scenario does not include the use of fertilizers. Emissions are not expected from this activity.
Project scenario	Source 1: Above-ground biomass	CO_2	Yes	During each cycle, trees will grow, sequester and stock carbon in their trunk, branches and leaves.
		CH_4	N/A	N/A
		N_2O	N/A	N/A
	Source 2: Below-ground biomass	CO_2	Yes	While trees are growing they will capture and store carbon in their roots. Those roots will not be removed; as a result, the carbon will remain stored.
		CH_4	N/A	N/A
		N_2O	N/A	N/A
	Source 3: Soil	CH_4	N/A	N/A
		N_2O	N/A	N/A
	Source 4: Other emissions	N_2O	No	No emissions from N_2O are expected

B.4. **Establishment and description of baseline scenario**

The project used the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities". The identification of the baseline scenario followed the steps suggested by the methodology.

The baseline scenario was developed in accordance with the GS4GG Principles & Requirements. The baseline scenario was defined according to pre-project land uses, which involved traditional cocoa production. This is the most plausible scenario that would be sustained in the absence of project activities as this activity engages with traditional use and subsistence for local communities. This aspect is presented in more detail in the folder: Supporting documents, Additionality document, section Step 1 for the full development of the baseline scenario.

In terms of additionality, the main conclusion is that the scenario of the implementation of the DAF system faces several constraints, principally due to the lack of tradition and the financial requirements that implied this type of activity, especially at the beginning. (More detailed information is presented in the Additionality document).

B.5. Demonstration of additionality

As presented in the previous section, the project is additional because without it the business-as-usual scenario would continue. Due to the economic restrictions of local communities and lack of government subsidies and/or funding options appropriate for local communities, the project would not happen without the economic support provided by COOP (through programme managed by WWF). COOP is funding the implementation costs for 10 years (by contract) which includes the establishment of the DAF system that would otherwise not be possible for local communities to establish.

The DAF system will generate benefits early from its establishment. In the first years, crops growing in combination with cocoa will be a source of food for the household with the potential to generate an additional source of income from the sale of surplus production. Income from cacao is expected to start after year six. Producers may obtain some additional income from the sale of a small number of *Gmelina arborea* in MU-DAF 2018–2020 or *Cedrela odorata* in MU-DAF 2021–2023. Fruit trees from the DAF will also contribute to diversified food security and be a potential source of income from surplus sales.

Adequate management of the DAF system is expected to increase cocoa yields (in comparison to a poorly managed agroforestry system) with no need for agrochemical inputs. This is expected to represent an increase in the income for the producer. The

project activities will increase income diversification of local communities, increasing their resilience and improving landscape connectivity by using native species.

For further evidence on additionality, please refer to folder: Supporting documents, Additionality document.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities, version 01.
Describe how the proposed project meets the criteria for deemed additionality.	In Ecuador, cocoa production represents an essential source of income for small farms and is a traditional product activity for the local economy. This practice, as was mentioned before, implies a monoculture of the variety CCN51 and a full-sun exposition system. However, this traditional production sometimes means insufficient enough revenue for the farms. Such is the case as recorded in northwestern Ecuador, where cocoa production was not profitable and did not compensate for the expenses of the family laborers, the maintenance of the plantation and the use of fertilizer (Worku Ketema et al., 2021 and Castañeda-Ccori et al., 2020). The DAF system is not traditional or common for the specific project area, and no activity of this type is currently operating in the project area (Villacis et al., 2022). The local and Indigenous communities present a similar system in the Amazon region. These systems imply the "*chagras*" areas, where the combination of the traditional cocoa and other plants could be detected (Castellon, 2015). However, "*chagras*" generally include only agriculture products, while the DAF system includes trees. For this reason the project is considered "the first of its kind" in the project area. This traditional activity does not encounter any substantial barriers impeding its continuous establishment. On the other hand, the change from traditional cocoa production to the implementation of the DAF system implies new management of cocoa and some technical requirements, besides of the initial investment to be implemented. As a result there are some constraints, particularly regarding the lack of investment and other financial incentives. These particularities represent real barriers to the development and thriving of this type of land use, putting forest cover conservation and development at risk in this area. Please refer to the Gold Standard for the Global Goals – Additionality Requirements for Afforestation/Reforestation Projects – Version 1 (July 2017) to see the full additionality analysis development.

B.5.1. Prior consideration

As this is a retroactively registered project, the following supporting documents are attached to demonstrate prior consideration:

a. Carbon project feasibility study;[43]
b. Annual report for 2018 for the UNOCACE – Chocolats Halba – Swiss Contact agreement, describing the project activities and its financing;[44] and
c. Emails between UNOCACE and Chocolats Halba discussing carbon.[45]

B.5.2. Ongoing financial need

This section and the investment analysis demonstrating the project's additionality is not required as the additionality assessment was evaluated using the barrier analysis (see the supporting documentation).

B.6. SDG outcomes

Table 10. Relevant target/indicator for each of the three SDGs Relevant target/indicator for each of the three SDGs

SDG targeted	Most relevant SDG target	SDG impact
		Indicator (Proposed or SDG indicator)
2 Zero hunger	2.4 By 2030, ensure the sustainability of food production systems and implement resilient agricultural practices that increase productivity and production, contribute to the maintenance of ecosystems, strengthen the capacity to adapt to climate change, extreme weather events, droughts, floods and other disasters, and progressively improve soil and land quality.	Income due to the selling of food production per plot (excluding cocoa). Cocoa yield (kg/ha)
13 Climate action (mandatory)	13.b Promote mechanisms for raising capacity for effective climate change-related planning and management in the least developed countries and small island developing states, including focusing on women, youth and local and marginalized communities.	Tonnes of CO_2 removed by the project activities/

[43] PDD Supporting Documents/Start date and prior consideration/Proyecto Finca Feasibility Report

[44] PDD Supporting Documents/Start date and prior consideration/Finca_annual_report_2018_FIN

[45] PDD Supporting Documents/Start date and prior consideration/proof of carbon consideration pre-2018

15 Life on land	15.2 By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.	Area reforested annually with DAF systems (ha/year)/

According to the 2022 Sustainable Development Report, which describes each country's progress towards achieving the SDGs, Ecuador ranks 63 out of a total of 163[46] countries. This progress is monitored using multisectoral official data from international organizations like the World Bank, Organization for Economic Cooperation and Development (OECD), World Health Organization (WHO), Food and Agriculture Organization (FAO), International Labor Organization (ILO), United Nations Children's Fund (UNICEF), and others. As a result, this index offers an aggregated rating and is presented as a guideline for each country in the development sector.

Ecuador has done comparatively well in terms of achieving the SDGs. Compared to the average value record for Latin America (68.5), Ecuador has an above average performance score of 71.5, reflecting the work of Ecuador related to international agreements.

Ecuador occupies a relatively good position in the progress of SDGs 6, 7, 11, and 13 as compared to other countries. This highlights the importance and relevance of continued efforts in climate action, the development of projects to reduce emissions in the country, and the government's policy on the right of natural services, including carbon issues.

Overall, Ecuador has more initiatives on SDG 13 than other goals, and a positive result has been recorded (as indicated in Figure 13). SDG 13 is reported through indexes like the reduction of CO_2 emissions from fossil fuel combustion and cement production (tCO_2/capita) and the reductions of emissions embodied in imports. Unfortunately, this is not the case for the other SDGs. For example, progress on SDGs 2 and 15 remains unchanged, with issues related to the prevalence of undernourishment and a low increase in the area protected for terrestrial sites important to biodiversity.[47]

[46] https://dashboards.sdgindex.org/profiles/ecuador
[47] https://dashboards.sdgindex.org/profiles/ecuador/indicators

Figure 13. SDG dashboards and trends (2022) (Source: Sustainable Development Report, 2022)

B.6.1. Explanation of methodological choices/approaches for estimating the SDG impact

All the SDGs will be monitored. However, only the effects on SDG 13 will be estimated, using a methodology presented in the Gold Standard Afforestation/Reforestation (A/R) GHG Emissions (Reduction & Sequestration Methodology).

SDG 2 Zero Hunger

There is no GS-defined methodology for determining the particular contributions for this SDG. However, after considering the activities of the project, two indicators were defined to evaluate the impact of activities on cocoa and food production for each producer:

- Income per food production (excluding cocoa)

$$PAl\ (USD/plot) = \frac{\sum_{i=0}^{n} P1+P2+....+Pn}{A\ (ha)}$$

PAl = income per food production per plot without cocoa. The global unit is USD. Each product has its own measurement unit that should be translated into USD. See monitoring plan in section B.7.2 for more detail.
P1, P2,...Pn = income from food products produced on the plot. The unit is USD.
A = plot area

The baseline for food crops (except cocoa) is zero, according to the monitoring plan the incomes projected are described in SDG 2 - PAI data and parameters fixed ex-ante.

- Cocoa Production

$$PC\ (kg/ha) = \frac{P\ cacao}{A\ (ha)}$$

PC = cocoa production per plot (Kg/ha)
$P\ cacao$= Cacao production (kg)
A = plot area

SDG 13: Climate action

The proposed project activity is a mechanism for raising capacity for effective climate management in the country. The suggested way of measuring the contribution of the project to this SDG and specific target is by estimating the amount of CO_2 removed

through the project activities. The methodology used for its estimation is presented in the Gold Standard Afforestation/Reforestation (A/R) GHG Emissions (Reduction and Sequestration Methodology) and following the Uncertainty of LUF parameters presented in Annex A[48]. This is summarized below:

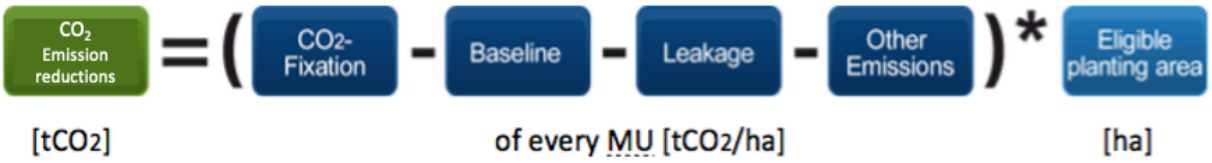
Calculation of CO_2 emission reductions

Table 11. The MU identified for the project (Source: South Pole, 2022)

Planting design	Planting year	Cocoa with trees	Cocoa monoculture	Fruit tree and other perennial crops	Grass	Maize	Shrub
Pre-2020	2018	X	X	X	X	X	X
	2019	X	X	X	X	X	X
Post-2020	2020	X	X	X	X	X	X
	2021	X	X	X	X	X	X
	2022	X	X	X	X	X	X
	2023	X	X	X	X	X	X

These MUs, shown in Table 11, correspond to the combination for implementing the DAF in areas where the common activity was a traditional cocoa plantation, cocoa with trees, fruit trees and other perennial crops, grasslands, maize and shrublands. This was presented in section B.5 in the baseline scenario defined for the project area.

The number of CO_2 emission reductions is determined for every year (t) of the crediting period using the following formula:



The CO_2-fixation of every MU is determined and its portion of the total baseline deducted. The project will not generate emissions from leakage or other emissions. All the CO_2-emission reductions of the MUs are added to make up the total CO_2-emission reductions of the entire project.

[48] Please see "Halba Ecuador carbon assessment version_SEP23" in the folder Carbon estimations.

The total CO_2 certificates for the project area are estimated as follows:

$$CO_{2-certificates\ Project\ area} = \sum_{MU=1}^{MUs} \sum_{t=1}^{CP} CO_{2\ certificates\ MU,t}$$

Where:

CO_2–certificates project area, t = [tCO_2] CO_2-certificates of a project area in year t

CO_2 certificates MU,t = [tCO_2] CO_2-certificates of a MU in year t

MUs = 1, 2, 3, … MUs of a project area

t = 1, 2, 3, … Years of the crediting period

CP = [ ] Year the crediting period ends

The estimations were done separately for each MU of the project. The pre-2020 and post-2020 planting design differ in the number of timber trees planted and the timber species as presented in the below Table 12.

Table 12. Project planting design

Pre-2020 – Dynamic agroforestry system 2018–2020	**Post-2020 – Dynamic agroforestry system 2020–2023**
Timber species (planting density per ha) • *S. parahybum (7)* • *S. macrophylla (7)* • *Q. petraea (7)* • *G. arbórea (7)* • *H. chrysanthus (7)* • *S. saman/A. guachapele (7)* • *L. nobilis (7)* • *T. rosea (7)* • *T. cumingiana (7)* • *C. odorata (7)* Total timber trees/ha = 70	**Timber species (planting density per ha)** • *S. macrophylla (17)* • *Q. petraea (18)* • *H. chrysanthus (17)* • *S. saman/A. guachapele (18)* • *L. nobilis (17)* • *T. rosea (18)* • *T. cumingiana (17)* • *C. odorata (18)* Total timber species/ha = 140
Fruit trees (planting density per ha) • *M. indica (23)* • *P. americana (23)* • *C. cinensis (24)* Total fruit trees/ha = 70	
Biomass trees (planting density per ha) • *Inga spp. (833)*	
Cocoa (planting density per ha) • *T. cacao (833)*	

1. Carbon fixation

Carbon fixation was calculated for each of the MUs based on the species selected, growth models and adequate conversion factors to obtain annual and cumulative

values in tCO_2. Both above- and below-ground biomass were considered. The equations used to calculate net benefits are presented in Table 13.

Table 13. CO_2 fixation per species (Source: South Pole, 2022)

Type of tree	Species	CO_2 fixation (annual)
Timber	*S. parahybum* *S. macrophylla* *Q. petraea* *G. arborea* *H. chrysanthus* *S. saman/A. guachapele* *L. nobilis* *T. rosea* *T. cumingiana*	• Annual growth = total biomass/age of tree in a sample • Total biomass = AGB + (root to shoot ratio*AGB) • AGB[49] = 21.297 – 6.953 * (DBH) + 0.740 * $(DBH)^2$ • DBH = the average DBH of a sample of trees in the project area. The year of the trees was known. • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * timber trees/ha
	C. odorata	• Annual growth = total biomass/age of tree in a sample • Total biomass = Volume * density[50] • Volume[51] = log10y = –3.86 + 2.52 * log10 (DBH) • DBH = the DBH to estimate annual growth was measured in the field from C. odorata individuals planted in Ghana in 2018 as part of other agroforestry project. The environmental conditions of Ghana are considered similar to those in Ecuador in the cocoa growing area.[52] • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * Cedrela trees/ha

[49] PDD supporting documents/Technical information/IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods)

[50] PDD supporting documents/Technical information/IPCC Chapter 3 "LUCF sector good practice guidance" Table 3A.1.9-2

[51] PDD supporting documents/Technical information/Henry, M., Picard, N., Trotta, C., Manlay, R.J., Valentini, R., Bernoux, M. & Saint-André, L. 2011. Estimating tree biomass of sub-Saharan African forests: a review of available allometric equations. p528

[52] This information can be reviewed in Supporting Documents/Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xlsx

Fruit trees	*M. indica* *P. americana*	Data on DBH and height was collected by South Pole from a sample in Ghana. This data was used to create models to estimate the DBH and height, based on age. Biomass was calculated using the allometric equations as follows: • DBH < 8, AGB, DBH * 1 • DBH>8 to < 32, AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3 • AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 • DBH > 32, AGB = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) • AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 • (DBH)^2 • tCO_2/tree/yr = annual biomass * 0.5 * 3.67 • Annual biomass = AGB + (root to shoot ratio * AGB) • tCO_2/ha/year=tCO_2/tree/yr * fruit trees/ha
	C. Sinensis	Data on DBH and height was collected by South Pole from a sample in Tanzania at year 20.[53] The AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3 • AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 • Annual growth = total biomass/age of tree in a sample • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * C.sinensis trees/ha
Biomass	*Inga spp.*	• AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 from IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods). • The DBH to estimate annual growth for each of the tree species were measured in the field at the research station in Ecuador. • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * *Inga spp.* trees/ha

[53] This information can be found in supporting Documents/Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xlsx

Cocoa	*T. cacao*	Annual growth tC/ha: • 1<yr<8: Isaac, M.E., Timmer, V.R., Quashie-Sam, S.J. (2007). Shade tree effects in an 8-year-old cocoa agroforestry system: biomass and nutrient diagnosis of Theobroma cacao by vector analysis. *Nutrient Cycling in Agroecosystems 78*(2), 155-65. • 8<yr<15: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23-31. • 15<yr<25: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23–31. • tCO2/ha/yr = annual growth tC * 3.66666666666667

2. Baseline annual

With the applicability conditions, and in accordance with the methodology, the baseline was only deducted in year 1 (t=1). The baseline was calculated using values obtained from default values from the IPCC, research papers and official governmental reports for each of the baseline categories identified in the project. For trees remaining in the baseline for the entire period of the project, their tCO_2 stored was also included in the model in year 2. Table 14 below presents the specific equations used as well as a summary of the parameters and the sources used for the calculations.

Table 14. Baseline Calculation (Source: South Pole, 2022)

Baseline	Calculation
Cocoa with trees	• **Cocoa:** the allometric equation used to calculate the AGB was obtained from Rojas-Garcia *et al.* (2009). *Ecuaciones alometricas para estimar biomasa aérea en Mexico.* ECOSUR-CONAFOR. Inedito. The DBH for the allometric equations were obtained from a sample in the project area. The value reported here is the average of the sample. • **Non-cocoa fruit trees**: all the fruit trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated as follows, depending on the DBH value: - DBH < 8, AGB = DBH * 1 - DBH>8 to < 32 = AGB of citrus from IPCC. Chapter 4. Table 4.A.3 - DBH > 32, ABG = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods) - AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 - AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 • **Non-cocoa timber trees**: all the timber trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated using IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods) • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2 = total biomass * 0.5 * 3.66666666666667
Cocoa monoculture	The allometric equation used to calculate the AGB was obtained from Rojas-Garcia *et al.* (2009). *Ecuaciones alometricas para estimar biomasa aérea en Mexico*. ECOSUR-CONAFOR. Inedito. The DBH for the allometric equations were obtained from a sample in the project area. The value reported here is the average of the sample[54]. • total biomass = AGB + (root to shoot ratio * AGB) • tCO_2 = total biomass * 0.5 *3.66666666666667

[54] The results on sample plot in cocoa monoculture are in the tab "Cocoa monoculture baseline data" in the folder Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xlsx

Baseline	Calculation
Fruit tree orchard and other perennial crops	• **fruit trees**: all the fruit trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated as follows, depending on the DBH value: - DBH < 8, AGB = DBH * 1 - DBH>8 to < 32 AGB = of citrus from IPCC. Chapter 4. Table 4.A.3 - DBH > 32, ABG = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) - AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 - AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 • **timber trees**: all the timber trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated using IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) • total biomass= AGB + (root to shoot ratio * AGB) • tCO_2= total biomass * 0.5 * 3.66666666666667
Grassland	**Non-tree biomass** • tCO_2 = total biomass[55] * 0.5 * 3.66666666666667.
Maize	**Non-tree biomass** • tCO_2 = total biomass[56] * 0.5 * 3.66666666666667
Shrub	**Non-tree biomass** • tCO_2 = AGB[57] * root to shoot ratio[58] * 0.5 * 3.67

[55] PDD supporting documents/Technical information/IPCC Chapter 6 Table 6.4. Total AGB and BGB for temperate, warm, wet.

[56] PDD supporting documents/Technical information/IPCC Chapter 5 Table 5.1. Total AGB and BGB for crops

[57] PDD supporting documents/Technical information/Narcisa et al. (2018). Diversidad de plantas, estructura de la comunidad y biomasa aérea en un páramo del sur del Ecuador. Bosques Latitud Cero.

[58] PDD supporting documents/Technical information/IPCC. Ch6 Table 6.1

3. Leakage

No leakage is expected as a result of the implementation of the project.

- No timber will be harvested (except a very small number of *Cedrela odorata* (MU-DAF 2020-2023) and *Gmelina arborea* (MU-DAF 2018-2019) as part of project activities during the crediting period of the project, at year 10. None of the plots included in the project were previously used for timber production.
- 18% of the area already planted in 2018 to 2022 was cultivated with maize and was replaced by the DAF. Farmers will agree on the new species to be established, based on the standard planting design. In addition, the farmers are focused on cocoa production and the establishment of this type of small-scale subsistence farming will be done in the combination with the same DAF implemented on their lands (see Figure 3).
- The areas where DAF was established were not used for livestock.

4. Other emissions

- Site preparation: burning will not be conducted for land preparation. The land will be manually prepared and the plant residues will remain in the area to decompose.
- Fertilizer: no fertilizer or agrochemicals (inorganic or organic) will be used. Only biomass trees will be cut and chopped to be used as mulch to return nutrients to the soil.

SDG 15 Life on Land

There is no GS-determined methodology for calculating contributions to this SDG. Therefore, an indicator was defined according to the activities of the project:

$$AR = \frac{\sum_{i=0}^{n} A(ha)}{year}$$

AR = area in hectares, annually reforested with DAF systems

A = area per plot in ha

The baseline scenario of SDG 15 is accounted as zero, since without the implementation of the project activities there would be no increase of the ha of DAF system in the project area.

B.6.2. Data and parameters **fixed ex ante**

SDG 2

Data/parameter	PAI
Unit	USD/ha
Description	Income for selling the complementary food production (excluding cocoa), measured in USD per ha.
Source of data	UNOCACE's internal control system information.
Value(s) applied	A preliminary estimate[59] has been completed for the following products: cassava, lemon, orange and tangerine. Selling such products would generate the following: Year 1: 1,117 USD/ha Year 2: 1,204 USD/ha Year 3: 1,261 USD/ha Year 4: 1,525 USD/ha Year 5: 2,258 USD/ha
Choice of data or measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Purpose of data	Determining the amount of food produced per plot is required to calculate the incomes and determine the contribution to SDG 2.
Additional comment	Different crops have different measurement units. The corresponding conversions will be made at the time of monitoring.

Data/parameter	Pn
Unit	USD
Description	Food products produced on the plot, represented in income for selling the complementary food production (excluding cocoa), measured in USD.
Source of data	UNOCACE's internal control system information.

[59] PDD Supporting documents/Technical Information/Ingresos_variedades_17112020

Value(s) applied	A preliminary estimate[60] has been completed for the following products: cassava, lemon, orange and tangerine. Selling such products would generate the following: Year 1: 1,117 USD/ha Year 2: 1,204 USD/ha Year 3: 1,261 USD/ha Year 4: 1,525 USD/ha Year 5: 2,258 USD/ha
Choice of data or measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Purpose of data	Determining the amount of food produced per plot is required to calculate the incomes and determine the contribution to SDG 2.
Additional comment	None

Data/parameter	PC
Unit	kg/ha
Description	Cocoa production
Source of data	Measurements made by project participants and registered by UNOCACE technicians.

[60] PDD Supporting documents/Technical Information/Ingresos_variedades_17112020

Value(s) applied	<table><tr><th>Age (yrs)</th><th>Baseline data[61] (kg/ha)</th><th>. Projection of project activities[62] (kg/ha)</th></tr><tr><td>5</td><td>138</td><td>368</td></tr><tr><td>10</td><td>368</td><td>552</td></tr><tr><td>15</td><td>460</td><td>736</td></tr><tr><td>20</td><td>598</td><td>920</td></tr><tr><td>25</td><td>690</td><td>1104</td></tr><tr><td>30</td><td>828</td><td>1242</td></tr></table>
Choice of data or Measurement methods and procedures	Scales are used in which the product is weighed, by each participant in charge of each plot. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	P_{cacao}
Unit	Kg
Description	Quantity of cocoa in kilograms produced per plot.
Source of data	Measurements made by project participants and registered by UNOCACE technicians.
Value(s) applied	0.0
Choice of data or Measurement methods and procedures	Scales are used in which the product is weighed, by each participant in charge of each plot. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application.

[61] Production of cocoa of national varieties (fine aroma) BUT under a management system typical of the zone that is **outside** the plots that are within the Finca Project. Please see Comparacion produccion de cacao.xlxs in the folder Technical information.

[62] Agroforestry system (the Finca Project), with national varieties of cocoa (fine aroma) with the benefits of participating in the project (such as training and selection of more productive varieties). Please see Comparacion produccion de cacao.xlxs in the folder Technical information.

Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	A
Unit	ha
Description	annually reforested with the DAF
Source of data	GIS
Value(s) applied	0
Choice of data or Measurement methods and procedures	See section A.1.1 Eligibility of the project under Gold Standard.
Purpose of data	Estimation of the annually reforested with DAF systems.
Additional comment	None

SDG 13

Data/parameter	Wood density
Unit	g/cm^3
Description	Mass per unit volume of wood.
Source of data	IPCC Chapter 3 “LUCF sector good practice guidance” Table 3A.1.9-2
Value(s) applied	0.43
Choice of data or measurement methods and procedures	The density was reported for C. odorata in Central America.
Purpose of data	Estimation of GHG emission reductions and removals.
Additional comment	None

Data/parameter	Biomass Expansion Factor (BEF)
Unit	Dimensionless
Description	Ratio of above-ground, oven-dry biomass to the oven-dry biomass of the steam.
Source of data	Gold Standard Afforestation/Reforestation (A/R) GHG Emissions Reduction & Sequestration Methodology
Value(s) applied	1.1
Choice of data or Measurement methods and procedures	The above-ground tree biomass is calculated using the BEF1 in connection to increment in volume.
Purpose of data	Estimation of GHG emission reductions and removals.
Additional comment	None

Data/parameter	Root-to-shoot ratio (R)
Unit	Dimensionless
Description	Ratio of the weight of the roots to the weight of the top of the tree. Used for below-ground tree biomass estimation.
Source of data	• Gold Standard Afforestation/Reforestation (A/R) GHG Emissions Reduction & Sequestration Methodology • IPCC, 2006. Guidelines for National Greenhouse Gas Inventories Chapter 6 Grassland. Table 6.1
Value(s) applied	• Tropical Rain forest = 0.20 • Grasslands = 2.8
Choice of data or Measurement methods and procedures	Below-ground biomass is usually estimated as sampling is destructive and expensive.
Purpose of data	Estimation of GHG emission reductions and removals.
Additional comment	None

Data/parameter	Carbon fraction tC/t d.m.

Unit	Percentage
Description	Percentage of the biomass of the tree that is carbon.
Source of data	Gold Standard Afforestation/Reforestation (A/R) GHG Emissions Reduction & Sequestration Methodology
Value(s) applied	0.5
Choice of data or Measurement methods and procedures	Estimation of GHG emission reductions and removals and baseline emissions.
Purpose of data	Carbon fraction (CF)
Additional comment	None

Data/parameter	C to CO2e
Unit	tCO_2/tC
Description	Factor applied to convert the tree carbon sequestered to tree CO_2e sequestered.
Source of data	Gold Standard Afforestation/Reforestation (A/R) GHG emissions reduction and sequestration methodology
Value(s) applied	44/12
Choice of data or Measurement methods and procedures	IPCC default value.
Purpose of data	Estimation of GHG emission reductions and removals and baseline emissions.
Additional comment	None

Data/parameter	**Biomass baseline data**
Unit	tdm/ha

Description	There are six main baseline scenarios: • Cocoa with trees • Cocoa monoculture • Fruit trees orchard and other perennial crops • Grassland • Maize • Shrub

Source of data

Baseline cover scenario	AG biomass (tonnes dry matter/ha)	Root to shoot (tonnes dry matter/ha)	Total biomass per ha (tdm/ha).
Shrub	Urgiles et al. (2018)	IPCC 2006. Chapter 6. Grassland. Table 6.1. Value for shrubs	-
Grassland			IPCC 2006. Chapter 6. Grassland. Table 6.4. Total AGB and BGB for tropical moist and wet.
Maize			IPCC 2006. Chapter 5. Croplands 5.1.

Baseline cover scenario	Total CO_2/ha
Cocoa monoculture	Rojas-Garcia et al. (2009). Ecuaciones alométricas para estimar biomasa aérea en Mexico. ECOSUR-CONAFOR. Inedito.
Tree component of the cocoa with trees and fruit tree orchard and other perennial crops baseline categories	Fruit trees: allometric equation for citrus from IPCC. Chapter 4. Table 4.A.3. Timber trees: IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods).

Value(s) applied

	AG biomass (tonnes dry matter/ha)	Root to shoot (tonnes dry matter/ha)	Total biomass per ha tdm/ha
Shrub	14.76	2.8	41.328

Grassland	16.1
Maize	10

Baseline cover scenario	t CO_2/ha
Fruit tree orchard and other perennial crops (except cocoa)	25.67
Cocoa monoculture	48.98
Cocoa with trees (cocoa trees + shade trees)	69.36

Baseline cover scenario	AG biomass (tonnes dry matter/ha)	Root to shoot (tonnes dry matter/ha)	Total biomass per ha tdm/ha
Shrub	Study conducted in Ecuador in an area with similar climatic characteristics as San Pablo del Lago	Default value	-
Grassland			Default value
Maize			Default value
Pastures			Default value

Choice of data or Measurement methods and procedures

Baseline cover scenario	t CO_2/ha
Cocoa	Allometric equation from Rojas-Garcia et al. (2009). Ecuaciones alometricas para estimar biomasa aérea en Mexico. ECOSUR-CONAFOR. Inedito. A sample of DBH from cocoa trees in the project area were measured to estimate the biomass.
Tree component of cocoa with trees and fruit tree orchard and other perennial crops baseline categories.	Allometric equations from the IPCC were used due to a lack of more specific equations for the species selected. The DBH used in the allometric equations correspond to the trees measured in the field for the totality of the plots included in the project in 2018, 2019 and 2020[63].

Purpose of data	Calculation of baseline scenario
Additional comment	The approach identified for quantification baseline biomass according to the Annex A - Uncertainty of LUF parameters is the approach 2 and the "Special guidance for smallholder projects": - Shrub, grassland, maize and pastures data are obtained from peer reviewed published literature data from studies conducted under similar climate and site conditions. - Fruit tree orchards and other perennial crops and cocoa with trees data were found by collecting data from trees in the field (project area) and using allometric equations for estimation of biomass, complying with the parameters of uncertainty[64]. - Cocoa tree data were found by collecting DBH from trees in the field (plots within the project area) and using allometric equations for estimation of biomass, complying with the parameters of uncertainty[65]. - According to above, only for fruit orchards and other perennial crops deductions were included in the calculation of biomass baseline.

[63] Please see "Formulario_linea_base version 18 September 2023" in the folder Carbon estimations.
[64] Please see tab "Uncertainty" in "Formulario_linea_base version 18 September 2023.xlsx" in the folder Carbon estimations for Uncertainty assessment.
[65] Please see tab "Cocoa monoculture baseline data" in "Halba Ecuador carbon assessment clean version_NOV23.xlsx" in the folder Carbon estimations for Uncertainty assessment.

<table>
<tr><td>Data/parameter</td><td>Biomass (AGB) of project trees</td></tr>
<tr><td>Unit</td><td>tdm/tree</td></tr>
<tr><td>Description</td><td>These are for type of trees:
Timber species: S. parahybum, S. macrophylla, Q. petraea, G. arborea, H. chrysanthus, S. saman/A. guachapele, L. nobilis, T. rosea, T. cumingiana.
Biomass: Inga spp.</td></tr>
<tr><td>Source of data</td><td>
<table>
<tr><td>Type of tree</td><td>Biomass (AGB) tonnes per tree</td></tr>
<tr><td>Timber species: S. parahybum, S. macrophylla, Q. petraea, G. arborea, H. chrysanthus, S. saman/A. guachapele, L. nobilis, T. rosea, T. cumingiana.</td><td>IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2. (for tropical wet hardwoods).</td></tr>
<tr><td>Biomass: Inga spp.</td><td>IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods).</td></tr>
</table>
</td></tr>
<tr><td>Value(s) applied</td><td>
<table>
<tr><td>Type of tree</td><td>Biomass (AGB) tonnes d.m. per tree</td><td>Age of trees (years)</td></tr>
<tr><td>Timber species:</td><td></td><td></td></tr>
<tr><td>S. parahybum</td><td>1.03</td><td>14</td></tr>
<tr><td>S. macrophylla</td><td>0.59</td><td>13</td></tr>
<tr><td>Q. petraea</td><td>0.59</td><td>13</td></tr>
<tr><td>G. arborea</td><td>1.85</td><td>11</td></tr>
<tr><td>H. chrysanthus</td><td>0.29</td><td>12</td></tr>
<tr><td>S. saman/A. guachapele</td><td>0.59</td><td>13</td></tr>
<tr><td>L. nobilis</td><td>0.53</td><td>11</td></tr>
<tr><td>T. rosea</td><td>0.29</td><td>12</td></tr>
<tr><td>T. cumingiana.</td><td>0.21</td><td>12</td></tr>
<tr><td>Biomass: Inga spp.</td><td>0.02</td><td>3</td></tr>
</table>
</td></tr>
</table>

Choice of data or Measurement methods and procedures	<table><tr><td>Type of trees</td><td>Biomass (AGB) tonnes per tree</td></tr><tr><td>**Timber species:** *S. parahybum, S. macrophylla, Q. petraea, G. arborea, H. chrysanthus, S. saman/A. guachapele, L. nobilis, T. rosea, T. cumingiana.*</td><td>Allometric equation for ABG Tropical wet hardwoods from IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2, was used to estimate AGB. The DBH used in the allometric equations correspond to the trees measured in the project area or regions with similar conditions to project area. The age of trees is known. Y = 21.297 – 6.953 • (DBH) + 0.740 • (DBH)^2</td></tr><tr><td>**Biomass:** *Inga spp.*</td><td>Allometric equation for AGB from IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods). The DBH to estimate annual growth for each of the tree species was measured in the field at the research station in Ecuador. The age of trees is known. Y = 21.297 – 6.953 • (DBH) + 0.740 • (DBH)^2</td></tr></table>
Purpose of data	Estimation of GHG emission reductions and removals
Additional comment	The approach identified for quantification project activity biomass according to the Annex A - Uncertainty of LUF parameters is the approach 2 and the “Special guidance for smallholder projects”: - For timber and biomass species data were found by collecting DBH from trees in the project area and using allometric equations for estimation of biomass. - According to above, in some cases deductions were included in the calculation of project ex-ante trees biomass accordingly[66]. - For details of the calculation process please see table 15.

Data/parameter	Volume (m^3) of **project trees**
Unit	m^3/tree

[66] Please see the tab for each species in “Halba Ecuador carbon assessment version_SEP23.xlsx” in the folder Carbon estimations for Uncertainty assessment.

<table>
<tr><td>Description</td><td>There are for type of trees:
• Timber species: C. odorata.</td></tr>
<tr><td>Source of data</td><td>
<table>
<tr><td>Type of tree</td><td>Volume (m^3) per tree</td></tr>
<tr><td>Timber species: C. odorata.</td><td>Henry, M., Picard, N., Trotta, C., Manlay, R.J., Valentini, R., Bernoux, M. & Saint-André, L. 2011. Estimating tree biomass of sub-Saharan African forests: a review of available allometric equations. p528.</td></tr>
</table>
</td></tr>
<tr><td>Value(s) applied</td><td>
<table>
<tr><td>Type of tree</td><td>Volume (m^3) per tree</td><td>Age of trees (years)</td></tr>
<tr><td>Timber species: C. odorata.</td><td>1.9</td><td>15</td></tr>
</table>
</td></tr>
<tr><td>Choice of data or Measurement methods and procedures</td><td>
<table>
<tr><td>Type of trees</td><td>Volume (m^3) per tree</td></tr>
<tr><td>Timber species: C. odorata.</td><td>Allometric equation from Henry, M., Picard, N., Trotta, C., Manlay, R.J., Valentini, R., Bernoux, M. & Saint-André, L. 2011. Estimating tree biomass of sub-Saharan African forests: a review of available allometric equations. p528. The DBH to estimate annual growth was measured in the field from C. odorata individuals planted in Ghana in 2018 as part of other agroforestry project. The environmental conditions of Ghana are considered similar to those in Ecuador in the cocoa growing area. The age of trees is known.

Volume = 10^(-3.86 + 2.52 x log10 (DBH)</td></tr>
</table>
</td></tr>
<tr><td>Purpose of data</td><td>Estimation of GHG emission reductions and removals</td></tr>
</table>

Additional comment	The approach identified for project activity biomass according to the Annex A - Uncertainty of LUF parameters is the approach 2 and the "Special guidance for smallholder projects": For this species the model was found by peer reviewed published literature and data collected from studies conducted under similar climate and site conditions. Then, no deductions were included in the calculation of ex-ante project trees biomass. For details of the calculation process please see table 15

Data/parameter	MAI biomass (AGB) of **project trees**
Unit	tdm/tree/yr
Description	These are for type of trees: • Fruit trees: *M. indica and P. americana* • Fruit trees: *C. Sinensis.*
Source of data	

Type of tree	MAI Biomass (AGB) tonnes d.m. per tree per year
Fruit trees: *M. indica, P. americana.*	IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) and Annex 4A.3 (for citrus).
Fruit trees: *C. sinensis.*	IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.3 (for citrus).

Value(s) applied

Type of tree	Biomass (AGB) tonnes d.m. per tree per year (td/tree/yr)
Fruit trees: *M. indica* *P. americana.*	 0.0352 0.0340
Fruit trees: *C. Sinensis.*	0.0061

<table>
<tr><td>Choice of data or Measurement methods and procedures</td><td>
<table>
<tr><td>Type of trees</td><td>Biomass (AGB) tonnes d.m. per tree per year</td></tr>
<tr><td>Fruit trees: M. indica, P. americana.</td><td>The allometric equations for AGB estimation depends on DBH (cm) size:
DBH < 8, AGB, DBH * 1
DBH>8 to < 32, AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3
DBH > 32, AGB equation from IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods)
Data on DBH and height was collected by South Pole from a sample in Ghana. This data was used to create models to estimate the DBH and height, based on age. The environmental conditions of Ghana are considered similar to those in Ecuador in the project area. The age of trees is known.</td></tr>
<tr><td>Fruit trees: C. sinensis.</td><td>The allometric equation for AGB of citrus from IPCC. Chapter 4. Table 4.A.3. Data on DBH and height was collected by South Pole from a sample in Tanzania at year 20. The environmental conditions of Tanzania are considered similar to those in Ecuador in the project area.</td></tr>
</table>
</td></tr>
<tr><td>Purpose of data</td><td>Estimation of GHG emission reductions and removals</td></tr>
<tr><td>Additional comment</td><td>The approach identified for quantification project activity biomass according to the Annex A -Uncertainty of LUF parameters is the approach 2 and the “Special guidance for smallholder projects”: For the species M. indica, P. americana and C. Sinencis, the models to estimate biomass were obtained from peer reviewed published literature and the field data were collected in sites with similar conditions (climate, rainfall, vegetation). Then, no deductions were included in the calculation ex-ante of project trees biomass. For details of the calculation process please see table 15</td></tr>
</table>

Data/parameter	Total biomass Cocoa
Unit	t C/ha
Description	There are for type of trees: • Cocoa: *T. cacao.*

Source of data	Total biomass t C/ha Isaac, M.E., Timmer, V.R., Quashie-Sam, S.J. (2007). Shade tree effects in an 8-year-old cocoa agroforestry system: biomass and nutrient diagnosis of Theobroma cacao by vector analysis. *Nutrient Cycling in Agroecosystems 78*(2), 155-65. Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23-31.
Value(s) applied	Total biomass t C/ha 1<yr<8: 8.57 t C/ha 8<yr<15: 13.44 t C/ha 15<yr<25: 15.9 t C/ha
Choice of data or Measurement methods and procedures	Total biomass Annual growth tC/ha: • 1<yr<8: Isaac, M.E., Timmer, V.R., Quashie-Sam, S.J. (2007). Shade tree effects in an 8-year-old cocoa agroforestry system: biomass and nutrient diagnosis of Theobroma cacao by vector analysis. *Nutrient Cycling in Agroecosystems 78*(2), 155-65. • 8<yr<15: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23-31. • 15<yr<25: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23–31.
Purpose of data	Estimation of GHG emission reductions and removals

Additional comment	The approach identified for quantification project activity biomass according to the Annex A -Uncertainty of LUF parameters is the approach 2 and the “Special guidance for smallholder projects”: For the species *T. cacao* the models to estimate biomass were obtained from peer reviewed published literature in studies conducted in sites with similar conditions (climate, rainfall, vegetation). Then, no deductions were included in the calculation ex-ante of project trees biomass. For details of the calculation process please see table 15

Data/parameter	Eligible planting area
Unit	ha
Description	The total eligible planting area is 431.54 ha which is where project activities will take place.
Source of data	The areas were selected based on the eligibility assessment following the methodology described in sections A.1.1 and B.2.
Value(s) applied	431.54 ha
Choice of data or Measurement methods and procedures	The no-forest analysis follows the requirements of the Gold Standard. See sections A.1.1 and B.2. for a detailed description of the methodology and procedures
Purpose of data	Project areas
Additional comment	At the time of the preliminary review, not all planted areas were measured. The target of the eligible area is 510 ha as the project area. This will be completed by design certification.

Data/parameter	Area per strata
Unit	ha
Description	Projet area per strata
Source of data	GIS and forms of “eligibility second phase”
Value(s) applied	

Strata	Year	Baseline scenarios	Management system	Area (ha)
1	2018	Cocoa with trees (ha)	DAF - Model per ha pre2020	21.1
2	2018	Cocoa monoculture (ha)	DAF - Model per ha pre2020	1.2

3	2018	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha pre2020	6.8
4	2018	Grassland (ha)	DAF - Model per ha pre2020	3.6
5	2018	Maize (ha)	DAF - Model per ha pre2020	12.0
6	2018	Shrub (ha)	DAF - Model per ha pre2020	1.8
7	2019	Cocoa with trees (ha)	DAF - Model per ha pre2020	14.7
8	2019	Cocoa monoculture (ha)	DAF - Model per ha pre2020	3.4
9	2019	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha pre2020	4.3
10	2019	Grassland (ha)	DAF - Model per ha pre2020	10.6
11	2019	Maize (ha)	DAF - Model per ha pre2020	9.3
12	2019	Shrub (ha)	DAF - Model per ha post2020	10.0
13	2020	Cocoa with trees (ha)	DAF - Model per ha post2020	27.9
14	2020	Cocoa monoculture (ha)	DAF - Model per ha post2020	5.4
15	2020	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha post2020	2.4
16	2020	Grassland (ha)	DAF - Model per ha post2020	6.1
17	2020	Maize (ha)	DAF - Model per ha post2020	5.3
18	2020	Shrub (ha)	DAF - Model per ha post2020	9.4
19	2021	Cocoa with trees (ha)	DAF - Model per ha post2020	25.1
20	2021	Cocoa monoculture (ha)	DAF - Model per ha post2020	44.1
21	2021	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha post2020	48.6

	22	2021	Grassland (ha)	DAF - Model per ha post2020	14.4
	23	2021	Maize (ha)	DAF - Model per ha post2020	9.6
	24	2021	Shrub (ha)	DAF - Model per ha post2020	37.2
	25	2022	Cocoa with trees (ha)	DAF - Model per ha post2020	1.8
	26	2022	Cocoa monoculture (ha)	DAF - Model per ha post2020	6.4
	27	2022	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha post2020	1.8
	28	2022	Grassland (ha)	DAF - Model per ha post2020	1.4
	29	2022	Maize (ha)	DAF - Model per ha post2020	5.5
	30	2022	Shrub (ha)	DAF - Model per ha post2020	6.4
	31	2023*	Cocoa with trees (ha)	DAF - Model per ha post2020	62.3
	32	2023*	Cocoa monoculture (ha)	DAF - Model per ha post2020	9.6
	33	2023*	Fruit tree and other perennial crops (except cocoa) (ha)	DAF - Model per ha post2020	13.8
	34	2023*	Grassland (ha)	DAF - Model per ha post2020	19.6
	35	2023*	Maize (ha)	DAF - Model per ha post2020	26.9
	36	2023*	Shrub (ha)	DAF - Model per ha post2020	20.1
	The values for 2023 were extrapolated using the proportion of historical data for the baseline scenarios.				
Choice of data or Measurement methods and procedures	See section B.2. for a detailed description of the methodology and procedures				
Purpose of data	Estimation of GHG Emission Reductions and Removals and Baseline emissions				
Additional comment	NA				

Table 15. CO_2 fixation per species (Source: South Pole, 2022)

Type of tree	Species	CO_2 fixation (annual)
Timber	*S. parahybum* *S. macrophylla* *Q. petraea* *G. arborea* *H. chrysanthus* *S. saman/A. guachapele* *L. nobilis* *T. rosea* *T. cumingiana*	• Annual growth = total biomass/age of tree in a sample • Total biomass = AGB + (root to shoot ratio*AGB) • AGB[67] = 21.297 – 6.953 * (DBH) + 0.740 * $(DBH)^2$ • DBH = the average DBH of a sample of trees in the project area. The year of the trees was known. • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * timber trees/ha
	C. odorata	• Annual growth = total biomass/age of tree in a sample • Total biomass = Volume * density[68] • Volume[69] = log10y = –3.86 + 2.52 * log10 (DBH) • DBH = the DBH to estimate annual growth was measured in the field from C. odorata individuals planted in Ghana in 2018 as part of other agroforestry project. The environmental conditions of Ghana are considered similar to those in Ecuador in the cocoa growing area.[70] • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * Cedrela trees/ha

[67] PDD supporting documents/Technical information/IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods)

[68] PDD supporting documents/Technical information/IPCC Chapter 3 "LUCF sector good practice guidance" Table 3A.1.9-2

[69] PDD supporting documents/Technical information/Rueda, A. Et al (2014). Estimación de biomasa aérea en plantaciones de Cedrela odorata L. y Swietenia macrophylla King. Revista Mexicana de Ciencias Forestales. 5 (25).Henry, M., Picard, N., Trotta, C., Manlay, R.J., Valentini, R., Bernoux, M. & Saint-André, L. 2011. Estimating tree biomass of sub-Saharan African forests: a review of available allometric equations. p528

[70] This information can be reviewed in Supporting Documents/Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xls

Fruit trees	*M. indica* *P. americana*	Data on DBH and height was collected by South Pole from a sample in Ghana. This data was used to create models to estimate the DBH and height, based on age. Biomass was calculated using the allometric equations as follows: • DBH < 8, AGB, DBH * 1 • DBH>8 to < 32, AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3 • AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 • DBH > 32, AGB = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) • AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 • (DBH)^2 • tCO_2/tree/yr = annual biomass * 0.5 * 3.67 • Annual biomass = AGB + (root to shoot ratio * AGB) • tCO_2/ha/year=tCO_2/tree/yr * fruit trees/ha
	C. Sinensis	Data on DBH and height was collected by South Pole from a sample in Tanzania at year 20.[71] The AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3 • AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 • Annual growth = total biomass/age of tree in a sample • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * C.sinensis trees/ha
Biomass	*Inga spp.*	• AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 from IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods). • The DBH to estimate annual growth for each of the tree species were measured in the field at the research station in Ecuador. • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2/tree/yr = annual growth * 0.5 * 3.66666666666667 • tCO_2/ha/year=tCO_2/tree/yr * *Inga spp.* trees/ha

[71] This information can be found in supporting Documents/Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xls

Cocoa	*T. cacao*	Annual growth tC/ha: ● 1<yr<8: Isaac, M.E., Timmer, V.R., Quashie-Sam, S.J. (2007). Shade tree effects in an 8-year-old cocoa agroforestry system: biomass and nutrient diagnosis of Theobroma cacao by vector analysis. *Nutrient Cycling in Agroecosystems 78*(2), 155-65. ● 8<yr<15: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23-31. ● 15<yr<25: Isaac, M., Gordon, A., Thevathasan, N., Oppong, S., Quashie-Sam, J. (2005). Temporal changes in soil carbon and nitrogen in west African multistrata agroforestry systems: a chronosequence of pools and fluxes. *Agroforestry Systems 65*(1), 23–31. ● tCO2/ha/yr = annual growth tC * 3.66666666666667

Table 16. Baseline Calculation (Source: South Pole, 2022)

Baseline	Calculation
Cocoa with trees	• **Cocoa:** the allometric equation used to calculate the AGB was obtained from Rojas-Garcia *et al.* (2009). *Ecuaciones alometricas para estimar biomasa aérea en Mexico.* ECOSUR-CONAFOR. Inedito. The DBH for the allometric equations were obtained from a sample in the project area. The value reported here is the average of the sample. • **Non-cocoa fruit trees**: all the fruit trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated as follows, depending on the DBH value: - DBH < 8, AGB = DBH * 1 - DBH>8 to < 32 AGB equation citrus from IPCC. Chapter 4. Table 4.A.3 - DBH > 32, ABG = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods) - AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 - AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 • **Non-cocoa timber trees**: all the timber trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated using IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for Tropical wet hardwoods) • Total biomass = AGB + (root to shoot ratio * AGB) • tCO_2 = total biomass * 0.5 * 3.66666666666667
Cocoa monoculture	The allometric equation used to calculate the AGB was obtained from Rojas-Garcia *et al.* (2009). *Ecuaciones alometricas para estimar biomasa aérea en Mexico*. ECOSUR-CONAFOR. Inedito. The DBH for the allometric equations were obtained from a sample in the project area. The value reported here is the average of the sample[72]. • total biomass = AGB + (root to shoot ratio * AGB) • tCO_2 = total biomass * 0.5 * 3.66666666666667

[72] The results on sample plot in cocoa monoculture are in the tab "Cocoa monoculture baseline data" in the folder Carbon calculations/ Halba Ecuador carbon assessment clean version _NOV23.xlsx

Baseline	Calculation
Fruit tree orchard and other perennial crops	• **fruit trees**: all the fruit trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated as follows, depending on the DBH value: - DBH < 8, AGB = DBH * 1 - DBH>8 to < 32 AGB equation of citrus from IPCC. Chapter 4. Table 4.A.3 - DBH > 32, ABG = IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) - AGB citrus = –6.64 + 0.279 * Basal Area + 0.000514 * Basal Area^2 - AGB for tropical wet woods = 21.297 – 6.953 * (DBH) + 0.740 * (DBH)^2 • **timber trees**: all the timber trees in the eligible plots that participate in the project were measured (DBH). The AGB was estimated using IPCC Good Practice Guidance for LULUCF. Chapter 4: Supplementary Methods and Good Practice Guidance Arising from the Kyoto Protocol. Annex 4A.2 (for tropical wet hardwoods) • total biomass= AGB + (root to shoot ratio * AGB) • tCO_2= total biomass * 0.5 * 3.66666666666667
Grassland	**Non-tree biomass** • tCO_2 = total biomass[73] * 0.5 * 3.66666666666667.
Maize	**Non-tree biomass** • tCO_2 = total biomass[74] * 0.5 * 3.66666666666667
Shrub	**Non-tree biomass** • tCO_2 = AGB[75] * root to shoot ratio[76] * 0.5 * 3.67

[73] PDD supporting documents/Technical information/IPCC Chapter 6 Table 6.4. Total AGB and BGB for temperate, warm, wet.

[74] PDD supporting documents/Technical information/IPCC Chapter 5 Table 5.1.Total AGB and BGB for crops

[75] PDD supporting documents/Technical information/Narcisa et al. (2018). Diversidad de plantas, estructura de la comunidad y biomasa aérea en un páramo del sur del Ecuador. Bosques Latitud Cero.

[76] PDD supporting documents/Technical information/IPCC. Ch6 Table 6.1

SDG 15

Data/parameter	AR
Unit	ha
Description	Annually reforested area.
Source of data	GIS
Value(s) applied	0
Choice of data or Measurement methods and procedures	See section A.1.1 Eligibility of the project under Gold Standard
Purpose of data	Estimation of the annually reforested with DAF systems
Additional comment	None

B.6.3. Ex ante estimation of SDG impact

SDG 2

- Income per food production (excluding cocoa)

As was mentioned in section B.6.2, for the case of income per food production, excluding the cocoa crops, the ex-ante estimations used some reported values[77] for crops like cassava, lemon, orange and tangerine. Selling such foods would generate the following incomes:

Year 1: 1,117 USD/ha
Year 2: 1,204 USD/ha
Year 3: 1,261 USD/ha
Year 4: 1,525 USD/ha
Year 5: 2,258 USD/ha

In future stages of the project, income per food production will be recorded by UNOCACE technicians during periodic visits, and they will save the values in the KoboToolBox application.

[77] PDD Supporting documents/Technical Information/Ingresos_variedades_17112020

- Cocoa production

The cocoa production for the ex-ante estimations assumed a value of zero because the project activity (DAF systems) have not been implemented before in the project area. The project areas will be established either on land where no cocoa was previously grown (shrub, grassland, maize, fruit orchard and other perennial crops) or on old cocoa plantations (cocoa with trees and cocoa monoculture) where cocoa production is negligible. For the purpose of reference, average cocoa production (for national varieties at year 10 after planting under typical management system) amongst Unocace farmers is ~400 kg/ha/yr. In future stages of the project, cocoa production will be recorded by UNOCACE technicians during periodic visits, and they will save the values in the KoboToolBox application.

It is important to mention that the DAF systems in some study cases recorded in Bolivia and Côte d'Ivoire, some of the biggest cocoa producers in the world, have recorded an increase between 12–46% for cocoa yields in agroforestry systems in comparison with monoculture systems.

In addition, significantly higher total system yields for the DAF systems have been recorded compared to monocultures due to the additional yields of the other species and crops included in this system.

The project is expected to have a potential increase in the production of the DAF compared to the traditional one, in the range of three times the original amount, after five years of establishing the systems[78].

SDG 13

CO_2 emission reductions were estimated for the DAF for the period 2018–2020 and 2021–2023. Carbon estimations were developed following the A/R methodology. The process is described briefly in section B.6. The detailed estimations are presented in the supporting folder ER estimations. The table below shows the project's estimate tCO_2 sequestered and net benefits (tCO_2) per year. The estimates do not include a deduction of the buffer.

[78] This potential comparison is presented in PDD supporting documents/Technical information/Comparacion produccion de cacao

The ex-ante of SDG 13 is calculated with the Calculation of CO_2 certificates equation below.

$$CO_2 certificates = \left(CO_{2-fixation} - Baseline - Leakage - Other\ Emissions\right) * Eligible\ planting\ area$$

Table 17. SDG 13 ex-ante (Source: South Pole, 2023)

Year	CO2-Fixation (tCO2e/year)	Baseline (tCO2e/year)	Other emissions (tCO2e/year)	CO2 certificates (tCO2e/year)
1	1,092	2,158	0	(1,066)
2	4,490	2,537	0	861
3	9,577	3,255	0	1,832
4	20,894	8,559	0	2,758
5	35,671	1,109	0	13,688
6	52,009	7,739	0	8,599
7	71,484	0	0	19,478
8	88,536	0	0	17,050
9	105,984	0	0	17,448
10	123,744	0	0	17,759
11	141,444	0	0	17,701
12	158,492	0	0	17,048
13	175,589	0	0	17,097
14	192,194	0	0	16,605
15	208,902	0	0	16,708

16	225,449	0	0	16,547
17	241,984	0	0	16,536
18	258,596	0	0	16,611
19	274,674	0	0	16,079
20	290,757	0	0	16,083
21	306,454	0	0	15,697
22	322,073	0	0	15,619
23	337,511	0	0	15,438
24	353,669	0	0	15,158
25	367,280	0	0	14.611
26	381,564	0	0	14,284
29	422,484	0	0	13,312
27	395,424	0	0	13,860
28	409,171	0	0	13,748
30	435,673	0	0	13,189

SDG 15 does not include any monetary transactions. The relevant areas of the project aim to contribute to the improvement of biodiversity recorded through the implementation of the DAF systems. Table 1 presents a summary of the contribution and goals of the project for each SDG included in the assessment.

B.6.4. Summary of ex ante estimates of each SDG impact

SDG 2

The ex-ante estimation for SDG 2, related to income per food production (excluding cocoa) and cocoa production, are presented in section B.6.3. The baseline for both cases is determined by primary data from UNOCACE.

SDG 13

The outcomes corresponding to SDG 13 are presented in Table 18. After estimating the long-term average according to the GS methodology, a total of 410,316 tCO_2 was estimated as the total carbon reduction generated by the project.

The ex-ante of SDG 13 is calculated with the calculation of CO_2 certificates equation below.

$$CO_2 certificates = \left(CO_{2-fixation} - Baseline - Leakage - Other\ Emissions\right) * Eligible\ planting\ area$$

Table 18. SDG 13 Outcomes (Source: South Pole, 2022)

Year	Baseline estimate (tCO2)	Project estimate (tCO2/year)	Net benefit (tCO2)
Year 1	2,158	1,092	(1,066)
Year 2	2,537	4,490	861
Year 3	3,255	9,577	1,832
Year 4	8,559	20,894	2,758
Year 5	1,109	35,671	13,688
Year 6	7,739	52,009	8,599
Year 7	0	71,484	19,478
Year 8	0	88,536	17,050
Year 9	0	105,984	17,448

Year 10	0	123,744	17,759
Year 11	0	141,444	17,701
Year 12	0	158,492	17,048
Year 13	0	175,589	17,097
Year 14	0	192,194	16,605
Year 15	0	208,902	16,708
Year 16	0	225,449	16,547
Year 17	0	241,984	16,536
Year 18	0	258,596	16,611
Year 19	0	274,674	16,079
Year 20	0	290,757	16,083
Year 21	0	306,454	15,697
Year 22	0	322,073	15,619
Year 23	0	337,511	15,438
Year 24	0	353,669	15,158
Year 25	0	367,280	14.611
Year 26	0	381,564	14,284
Year 27	0	422,484	13,312
Year 28	0	395,424	13,860
Year 29	0	409,171	13,748
Year 30	0	435,673	13,189

Total	25,357		410,316
Total number of crediting years	30		
Annual average over the crediting period	13,677		

SDG 15 – Project scenario ex ante

It is estimated that a series of plots will be established annually to cover the entire project area, calculated to be 510 ha. The following estimates have been determined, based on planning:

Table 19. Estimated preliminary project contributions to the Life on Land SDG indicator (hectares in DAF)

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	46.47	46.47
Year 2	0	52.33	52.33
Year 3	0	56.62	56.62
Year 4	0	178.96	178.96
Year 5	0	23.29	23.29
Year 6	0	152.34	152.34
…	0	0	0
…	0	0	0
Year 29	0	0	0
Year 30	0	0	0
Total	0	0	510.00
Total number of crediting years	30	510.00	510.00
Annual average over the crediting period	0	17.00	17.00

B.7. **Monitoring plan**

B.7.1. Data and parameters to be monitored

SDG 2

Data/parameter	PAI
Unit	USD/ha
Description	Income for selling complementary food production (excluding cocoa), measured in USD per ha.
Source of data	UNOCACE's internal control system information.
Value(s) applied	To be determined in each performance certification with an annual registry.
Measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, the project area database will be reviewed at the time of monitoring.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	Pn
Unit	USD
Description	Food products produced on the plot, represented in income for selling complementary food production (excluding cocoa), measured in USD.
Source of data	UNOCACE's internal control system information.
Value(s) applied	To be determined during each performance certification with an annual registry.
Measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.

Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, for every monitoring the project area database will be reviewed.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	PC
Unit	kg/ha
Description	Cocoa production
Source of data	UNOCACE's internal control system information.
Value(s) applied	To be determined in each performance certification with an annual registry.
Measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, for every monitoring the project area database will be reviewed.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	P_{cacao}
Unit	Kg
Description	Quantity of cocoa in kilograms, produced per plot.
Source of data	UNOCACE's internal control system information.
Value(s) applied	To be determined in each performance certification with an annual registry.
Measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Monitoring frequency	For every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, for every monitoring the project area database will be reviewed.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2.
Additional comment	None

Data/parameter	A
Unit	ha
Description	Annually reforested with the DAF
Source of data	Satellite imagery and registration of implemented forest areas with the project activities.
Value(s) applied	To be determined annually.
Measurement methods and procedures	Annual records of the planted area will be consulted. GIS information can also be consulted.
Monitoring frequency	Annually
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, for every monitoring the planting area database will be reviewed.
Purpose of data	Estimation of the annually reforested with DAF systems

Additional comment	None

SDG 13

Data/parameter	Diameter at breast height (DBH)
Unit	cm or m
Description	The DBH of a tree is the diameter of its trunk measured at 1.3 m height, over the bark.
Source of data	Forest inventory
Value(s) applied	To be estimated at each performance certification.
Measurement methods and procedures	The DBH will be measured using a measuring tape in each forest inventory plot. Temporal plots will be used for sampling carbon stock in all the areas. The sample plots will be used to take measurements such as DBH as well as list species type. For all trees, the DBH measurement will be taken at a height of 1.3 m. Circular temporal plots of e.g., 250 m2 (radius of 8.92 meters) or rectangular temporal plots of 20 m by 25 m may be used before each performance certification, and recorded and kept in a plot file. Temporal plots will be geo-referenced. Note: it might be possible to reduce the size of temporal plots to 100 m2 or to increase the plot size. More information is presented in the SOP in the supporting information folder.
Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	Carbon estimations
Additional comment	None

Data/parameter	Total height (Ht)
Unit	m
Description	Measured in temporal sample plots, represented by the total height of the trees presented in the sampling plots.

Source of data	Measured by the project proponent – forest inventory.
Value(s) applied	To be estimated at each performance certification.
Measurement methods and procedures	The Ht will be measured using a measuring tape in each forest inventory plot. Temporal plots will be used for sampling carbon stock in all the areas. The sample plots will be used to take measurements such as DBH as well as list species type. For all trees, the DBH measurement will be taken at a height of 1.3 m. Circular temporal plots of e.g. 250 m2 (radius of 8.92 meters) or rectangular temporal plots of 20 m by 25 m may be used before each performance certification, and recorded and kept in a plot file. Temporal plots will be geo-referenced. Note: it might be possible to reduce the size of temporal plots to 100 m2 or to increase the plot size. More information is presented in the SOP in the supporting information folder.
Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	To assess the emission reduction potential of the project by capturing carbon through the establishment of forest plantations.
Additional comment	None

SDG 15

Data/parameter	AR
Unit	ha
Description	Annually reforested area.
Source of data	GIS
Value(s) applied	To be determined annually.
Measurement methods and procedures	Annual records of the planted area will be consulted. GIS information can also be consulted.
Monitoring frequency	Annually
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, the planting area database will be reviewed at time of monitoring.
Purpose of data	Estimation of the annually reforested with DAF systems.
Additional comment	None

Parameters associated with the Safeguarding Principles Assessment

(Principle 3. Parameter #1). Number of trainings with farmers

Data/parameter	Number of trainings with farmers
Unit	Number of trainings with farmers
Description	Number of training carried out by UNOCACE in the framework of the project, with the purpose of ongoing training.
Source of data	UNOCACE records
Value(s) applied	To be defined in each performance certification.
Measurement methods and procedures	The records of the training carried out by UNOCACE will be consulted and the theme and participation will be verified.
Monitoring frequency	Annual
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	To record the number of people attending these sessions. To safeguard Principle 3 Projects involving the production, harvesting and/or management of living natural resources by smallholders shall adopt sustainable, appropriate and culturally sensitive resource management practices.

Additional comment	None

(Principle 4.4. Parameter #1). Number of community integration events

Data/parameter	Number of community integration events
Unit	Number of community integration events
Description	Number of community integration events (family days, sports, cultural days and others) carried out by UNOCACE in the framework of the project, with the purpose of community integration.
Source of data	UNOCACE records
Value(s) applied	To be defined in each performance certification.
Measurement methods and procedures	The records of the events carried out by UNOCACE will be consulted and the theme and participation will be verified.
Monitoring frequency	Annual
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	To record the number of people attending these sessions. To safeguard Principle 3 Projects involving the production, harvesting and/or management of living natural resources by smallholders and/or local communities shall adopt sustainable, appropriate and culturally sensitive resource management practices.
Additional comment	None

(Principle 4.4. Parameter #2). Number of events for producers belonging to the Chachi community

Data/parameter	Number of events for producers belonging to the Chachi community
Unit	Number of events
Description	Number of community and technical events (family days, sports, cultural days and others) carried out by UNOCACE within the framework of the project, for producers belonging to the Chachi community.
Source of data	UNOCACE records
Value(s) applied	To be defined in each performance certification.
Measurement methods and procedures	The records of events carried out by UNOCACE will be consulted and their subject matter and participation will be verified.
Monitoring frequency	Annual
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	To safeguard Principle 4.4 – Indigenous Peoples
Additional comment	None

(Principle 4.4. Parameter #1). Number of participants in community and technical events carried out by UNOCACE for producers belonging to the Chachi community

Data/parameter	Number of Chachi participants in the community and technical events
Unit	Number of participants
Description	Number of participants in community and technical events (family days, sports, cultural days and others) carried out by UNOCACE within the framework of the project, for producers belonging to the Chachi community.
Source of data	UNOCACE records
Value(s) applied	To be defined in each performance certification.
Measurement methods and procedures	The records of events carried out by UNOCACE will be consulted and their subject matter and participation will be verified.

Monitoring frequency	Annual
QA/QC procedures	See section B.7.3 Other elements of monitoring plan.
Purpose of data	To safeguard Principle 4.4 – Indigenous People
Additional comment	None

(Principle 6.2 Parameter #1) Negative economic consequences

Data/parameter	PC
Unit	kg/ha
Description	Cocoa production
Source of data	UNOCACE's internal control system information.
Value(s) applied	To be determined in each performance certification with an annual registry.
Measurement methods and procedures	Scales used for weight or units are counted according to the product to be quantified. The information is collected by UNOCACE technicians during periodic visits and recorded in the KoboToolBox application. The conversion and reports are made by UNOCACE.
Monitoring frequency	Every performance certification.
QA/QC procedures	See section B.7.3 Other elements of monitoring plan. In addition to this, for every monitoring the project area database will be reviewed.
Purpose of data	Determination of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2 and evidence the Principle 6.2. Negative economic consequences are not impacting the producers.
Additional comment	None

(Principle 8.1. Parameter #1). Number of existing artesian wells in the project area that are used for the project

Data/parameter	Number of wells
Unit	Number of wells

Description	Number of existing artesian wells in the project area that are used for the project. As there is no national regulation on their management, a registry of wells will be kept.
Source of data	UNOCACE records
Value(s) applied	To be defined in each performance certification.
Measurement methods and procedures	The records of UNOCACE's internal control system will be consulted.
Monitoring frequency	Annual
QA/QC procedures	The revision of documents will be done by personnel other than the one in charge of on-site visits.
Purpose of data	To safeguard Principle 8.1 – Impact on natural water patterns/flows.
Additional comment	None

B.7.2. Sampling plan

This monitoring plan provides guidance on monitoring and standard operational procedures for the reforestation project activity. This monitoring plan fulfills the requirement that the project activity should have credible and accurate monitoring procedures in place to enable the evaluation of project performance and verification of the net anthropogenic GHG emission removals.

Only the above-ground biomass of trees and shrubs will be measured. Therefore, only the individual growth of trees will be monitored in the inventory plots.

The below-ground carbon content and standing dead wood will also be assessed. These will be estimated by using default values and suggested methods as per the Gold Standard Afforestation/Reforestation (A/R) GHG Emissions Reduction & Sequestration Methodology. Root-to-shoot ratio (R) will be used for estimation of below-ground biomass given the above-ground measures.

SDG 2
Agricultural production indicators to estimate the contribution to SDGs.

Two methods will be used to measure SDG 2. Both involve sampling. The first one consists of monitoring the incomes from cocoa and banana, using the accountability

systems by UNOCACE, for the plots from 2016, 2017, 2018 and 2019. With this information it is possible to estimate the incomes from a representative number of plots sized 1.5 ha each, then extrapolate to the rest of the plots.

The second method, aimed at the other products from complementary foods, consists of monitoring 10 areas within the project in the DAF system which total 2.5 ha. These areas were established in 2021 and will be measured with a frequency of four times per year. UNOCACE technicians will collect these measurements of harvest quantity during farmer visits. Data will be kept on KoboToolBox and the quantification will be by units or weight, depending on the product harvested. After determining quantities, the selling price will be evaluated per each product and multiplied by the amounts to obtain the total incomes.

The units that are handled locally are:

Table 20. Local measurement units (Source: South Pole, 2022)

Local name in Spanish	Metric system equivalent	Product
Quintal	46 kg	Gandul
Libra (Pound)	456 g	Frejol, sesame, tumbe
Racimo (Cluster)	N/A	Plantain, bananas
Moño	N/A	Vegetables
Sacos	75 kg	Cassava
Cientos	N/A	Oranges
Gabetas	18 kg	Lemons

Registration will be done using tablets and the KoboToolBox application, which generates an online database and records historic data from each provider. It also includes photographs and records geographic coordinates.

SDG 13

Sampling frame and stratification

The data sampling will be designed based on the DAF database and according to strata defined for the project according to the planting date and the different DAF designs. Thereafter, the sample plots will be established in each stratum.

Table 21. The MUs and species considered (Source: South Pole, 2022)

Planting design	Planting year	Cocoa with trees	Cocoa monoculture	Fruit tree and other perennial crops	Grass	Maize	Shrub
Pre-2020	2018	X	X	X	X	X	X
	2019	X	X	X	X	X	X
Post-2020	2020	X	X	X	X	X	X
	2021	X	X	X	X	X	X
	2022	X	X	X	X	X	X
	2023	X	X	X	X	X	X

The DAF areas will be monitored to identify different growth patterns inside each stratum that can lead to unification between strata or a more detailed stratification. The areas will be periodically monitored using the monitoring criteria established for the project boundaries. The process seeks to identify whether the stratum defined ex ante needs to be changed. According to changes in the accumulation of carbon during each carbon monitoring period, a new stratification based on biomass growth can be created.

Plot type and size

Temporal plots will be used for sampling carbon stock in all areas. The sample plots will be used to take measurements such as DBH as well as list species type. For all trees, the DBH measurement will be taken at a height of 1.3 m. Circular temporal plots of e.g., 250 m^2 (radius of 8.92 m) or rectangular temporal plots of 20 m by 25 m may be used before each performance certification, and recorded and kept in a plot file. Temporal plots will be geo-referenced. It may be possible to reduce the size of temporal plots to 100 m^2 or to increase the plot size.

Number of sample plots

The number of sample plots for the forest inventory will meet an MU precision with minimum errors of +/-20% with a 90% confidence level. Sampling error will be estimated using the following equation:[79]

[79] Pearson, T., Walker, S., & Brown, S. (2005). Sourcebook for land use, land-use change and forestry projects. Winrock International and the BioCarbon Fund of the World Bank, 57.

$$E^2 = \left[\frac{\left(\sum_{h=1}^{L} N_h * S_h \right)^2}{n} - \left(\sum_{h=1}^{L} N_h * S_h^2 \right) \right] * \frac{t^2}{N^2}$$

Where:

E = allowable error. Calculated by multiplying the mean carbon stock by the desired precision (0.2 as per Gold Standard rules)

t = the sample statistic from the t-distribution for the 90% confidence level

Nh = number of sampling units for stratum h (= area of stratum in hectares or area of the plot in hectares)

N = number of sampling units in the population

Sh = standard deviation of carbon stock within stratum h.

L = number of strata

n = sample size

SDG 15 Life on land

Indicators of reforested areas

According to UNOCACE planning, measurement records of new areas planted yearly will be kept, using GPS or remote sensors. It is considered a census of all areas.

B.7.3. Other elements of monitoring plan

Procedures for internal auditing and QA/QC

QA/QC procedures will be implemented to ensure that net GHG removals are measured and monitored precisely, credibly, verifiably, and transparently. The project will follow the IPCC GPG method, using two procedures to ensure that the inventory estimates and their contributing data are high quality. As a QA/QC plan is fundamental to creating credibility, an appropriate plan will be developed that outlines QA/QC activities. It will include timeframes and outline steps from preparation to final reporting. The plan will describe specific QC procedures in addition to special QA review procedures.

The QA/QC plan is an internal document used to organize, plan, and implement QA/QC activities and will only be summarized here. Below are abstracts from the QA/QC plan:

- Standard Operating Procedures (SOP) will be established for all procedures such as GIS analysis, field measurements, data entry, data documentation and data storage.
- Training courses will be held for all relevant personnel on all data collection and analysis procedures.
- Steps will be taken to control errors in sampling and data analysis. This is done to develop a credible plan for measuring and monitoring carbon stock change in the project context. The same procedures will be used throughout the project life to ensure continuity.

The personnel involved in the measurement of carbon pools will be fully trained in field data collection and analysis. SOPs will be developed for each step of the field measurements and followed so that measurements are comparable over time. If different interpretations of the SOPs exist among the field teams, they will be jointly revised to ensure clear, cohesive guidance. This procedure will be repeated during the field data collection stage.

To verify that plots have been installed and the measurements are taken correctly, a 10% minimum of randomly selected plots will be remeasured by a supervisor and a team that was not involved in the initial measurement sampling.

The remeasurement data will be compared with the original measurement data with a simple relation between the two measurements to detect any outliers in the data. Any errors found will be corrected and recorded. The variation in the data will be calculated and reported using the following equation:

$$Data\ difference\ = \frac{Estimate1 - Estimate2}{Estimate2}$$

The proper entry of data into the data analysis spreadsheets is required to produce reliable carbon estimates. All data sheets will include a 'Data recorded by' field. Communication between all personnel involved in measuring and analyzing data will be used to resolve any apparent anomalies before final analysis of the monitoring data can be completed.

If there are any problems with the monitoring plot data that cannot be resolved, the plot will not be used in the analysis. Expert judgement and comparison with independent data will be used to ensure data results are in line with expectations. Additionally, field data will be reviewed by the leader of the monitoring team, further ensuring that the data and analysis are accurate.

Due to the long length of the project and the speed at which technology changes, raw data archiving is essential. Data will be archived in several forms and copies of all data will be provided to each project participant. Original copies of the field measurements (data sheets and electronic files) will be stored in a secure location. Copies of all data analysis and models, the final estimate for the carbon sequestered, any GIS products, and measuring and monitoring reports will be stored in a dedicated safe place (preferably offsite).

Electronic copies of all data and reports will be updated periodically and converted to any new format required by future software or hardware. An employee involved in the field measurements will be tasked with implementing this update. The data collected shall be archived for a period of at least two years after the end of the last crediting period of the project activity. The main activities to be developed for the QA/QC process are described in Table 22.

Table 22. Verification and checklist used to guarantee the quality of information gathered and its management (Source: South Pole, 2022)

QC activity	Procedures
Check that assumptions and criteria for the selection of activity data, emission factors and other estimation parameters are documented	Crosscheck descriptions of activity data, emission factors and other estimation parameters with information on source and sink categories, and ensure that these are properly recorded and archived.
Check for transcription errors in data input and reference	Confirm that bibliographical data references are properly cited in the internal documentation. Crosscheck a sample of input data from each source category (either measurements or parameters used in calculations) for transcription errors.
Check that emissions and removals are calculated correctly	Reproduce a representative sample of emission or removal calculations. Selectively mimic complex model calculations with abbreviated calculations to judge relative accuracy.

QC activity	Procedures
Check that parameter and units are correctly recorded and that appropriate conversion factors are used	At each design certification check that: • Units are properly labelled in calculation sheets and are correctly carried through from beginning to end of calculations; • Conversion factors are correct; and • Temporal and spatial adjustment factors are used correctly.
Check the integrity of database files	At each field data collection: • Confirm that the appropriate data processing steps are correctly used in the database; • Confirm that data relationships are correctly represented in the database; • Ensure that data fields are properly labelled and have the correct design specifications; and • Ensure that adequate documentation of database, model structure and operation are archived.
Check for consistency in data between categories	Identify parameters, such as activity data and constants, that occur in multiple categories of sources and sinks, and confirm that the values used for these parameters are consistent in the emissions calculations.
Check that the movement of inventory data among processing steps is correct	At each design certification check that: • Emission and removal data are correctly aggregated through lower reporting levels to higher ones, when preparing summaries; and • Emission and removal data are correctly transcribed between different intermediate products.
Check that uncertainties in emissions and removals are estimated or calculated correctly	At each design certification check: • That the individuals providing expert judgement for uncertainty estimates are suitably qualified; and • Qualifications, assumptions and expert judgements are recorded. Check that calculated uncertainties are complete and calculated correctly.
Undertake review of internal documentation	At each field data collection check that: • There is detailed internal documentation to support the estimates and to enable reproduction of the emission, removal and uncertainty estimates; and • Inventory data, supporting data and inventory records are archived and stored to facilitate detailed review.
Check time series consistency	At each field data collection: • Check for temporal consistency in time series input data for each category of sources and sinks; and • Check for consistency in the algorithm and method used for calculations throughout the time series.

QC activity	Procedures
Undertake completeness checks	At each design certification: • Confirm that estimates are reported for all categories of sources and sinks and for all years; and • Check that known data gaps that may result in incomplete emissions estimates are documented and treated conservatively.
Compare estimates to previous estimates	For each category, where available, current inventory estimates should be compared to previous estimates. If there are significant changes or departures from expected trends, recheck estimates and explain the difference.

>> Uncertainty assessment

The project follows methods from IPCC GPG for LULUCF (GPG 2003) and procedures for A/R project activities to estimate baseline and project net GHG removal. This covers removals by sinks, leakage, actual net GHG removal by sinks and net anthropogenic removal by sinks. In the context of this methodology, the major sources of uncertainties are related to changes in carbon stock in the living biomass pool. They include natural factors such as fire and pest outbreaks, as well as stand variables such as variation in the yield tables, allometric equation, biomass expansion factor (BEF), wood density, carbon fraction and errors in measurement. Uncertainty estimates will be developed for all land-use categories in the inventory part of the monitoring.

Regarding the Annex A – Uncertainty of LUF Parameters, the project activity follows the Approach 1 where data is measured within each stratum and shall follow accepted sampling and analysis protocols, in case the uncertainty of data is equal to or greater than 20% the deduction values will be applied according to the guideline. On the other hand, the baseline estimations follow Approach 2 and the "Special Guidance for Smallholder Projects" please see the section B.6.2. for each parameter.

SECTION C. DURATION AND CREDITING PERIOD

C.1. **Duration of project**

C.1.1. Start date of project

03/01/2018. Corresponding to the establishment of the first plot[80].

C.1.2. Expected operational lifetime of project

[80] PDD Supporting documents/start date and prior consideration/First plot

The operational lifetime of the project that was chosen following the study[81] is 35 years, which suggests that an optimum turn for tropical timber trees as C. odorata and S. macrophylla is achieved between 25 and 35 years, to reach a diameter at breast height (DBH) in the range of 50 to 60 cm, as an optimal condition for succession in tropical forest.

C.2. **Crediting period of project**

C.2.1. Start date of crediting period

03/01/2018

According to the document "Land use and forests activity requirements" version 1.2.1[82] published in April 2020, retroactive projects (3.1.14) shall submit the required documents to Gold Standard within five years of its start date. In this instance, the timeframe is still possible). If carbon certificates are to be issued, documents shall be submitted within three years (3.1.17). However, this project will not issue carbon credits, so this latter timeframe is not relevant.

[81] PDD Supporting documents/Technical information/Da Cunha T., Guimarães C., and H. Hasenauer, (2016.) Tree basal area increment models for Cedrela, Amburana, Copaifera and Swietenia growing in the Amazon rain forests. Forest Ecology and Management. Volume 365. Pages 174-183. ISSN 0378-1127. https://doi.org/10.1016/j.foreco.2015.12.031.

[82] https://globalgoals.goldstandard.org/203-ar-luf-activity-requirements/.

C.2.2. Total length of crediting period
30 years

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1. Ongoing monitoring is summarized below.

Table 23. Principles and mitigation measures (Source: South Pole, 2022)

Principles	Mitigation measures added to the monitoring plan
Principle 3	Registering all training carried out by UNOCACE for the farmers. The registry includes technical visits and recommendations made to project activities and prevent accidents. This is aimed to care for the trees and the plot of the agroforestry system.
Principle 4.4	Registering all events (family days, sports days, cultural days and others) carried out by UNOCACE for the Indigenous community, the Chachis. The registry also includes technical events and the number of attendants. This is intended to guarantee the inclusion of the community in the project and provide an opportunity to listen to their comments and suggestions.
Principle 6.2	Registering of the amount of cocoa produced per plot is required to calculate productivity per hectare and to determine the contribution to SDG 2 and evidence the Principle 6.2. Negative economic consequences are not impacting the producers.
Principle 8.1	Registering all the artesian wells existing in the project area and employed for irrigating the project plots. This is to keep records and prepare documentation, in case any new relevant national regulation comes into force.

D.2. Assessment that project complies with GS4GG gender sensitive requirements

Question 1 – Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	UNOCACE's corporate policies promote non-discrimination, regardless of ethnicity, gender or religion. Similarly, UNOCACE seeks that its employees do not participate in any practices that discriminate against women and girls, nor violate their rights. Likewise, the family of each producer participating in the project is encouraged to participate, including women, in the training processes and in the administrative processes of the project. Therefore, the Gold Standard Gender Equality guidelines are met during the design and validation process of the carbon project. Regarding the preparation of consultation meetings, ensuring the participation of women is a priority, especially those who family members of the participants. This allows them to articulate their comments and suggestions in the design of the project, as direct beneficiaries of participation. In these meetings, 212 men and 79 women attended, for a total of 291 attendees. These statistics showed a participation of 27% of women, despite their participation was lower than the men, this value is higher than the national average of land ownership by women in Ecuador (12.71% for women and 87.29% for men[83]), placing their participation above 15% of national average, those figures are aligned with UNOCACE gender policy, in which it encourages contractors and suppliers to be aware of and share UNOCACE's commitments to equal opportunities, gender equity, and work-life balance. The producers, which include both women and men, and their families, were given open invitations to the LSC meetings through physical letters, as the project area is predominantly family-owned land. Also, cocoa producers associated with the project are not only men, but also women who are heads of households and have title to the land; all were encouraged to participate.

[83] Mujeres rurales y tierra en Ecuador. Es hora de cerrar las brechas de género en campo. In folder: Gender policy.

Question 2 – Explain how the project aligns with existing country policies, strategies and best practices.	In accordance with the Constitution (2008), art. 70: "The State will formulate and execute policies to achieve equality between women and men, through the specialized mechanism in accordance with the law, and will incorporate the gender perspective in plans and programs and will provide technical assistance for its mandatory application in the public sector." Further, in accordance with the framework of the Gender Equality Policy of Ecuador (2018), UNOCACE has aligned all its cocoa production procedures with the laws and standards established by Ecuador's government, by means of its Corporate Equality Gender Policy. Likewise, UNOCACE's corporate policy complies with the first two specific objectives of the National Policy for Gender Equality: Specific Objective 1: To prevent and eliminate all types of acts that generate discrimination or gender violence in the workplace or in the activities carried out as a result of the employment relationship, as well as to pay special attention to those situations that require specialized protection, such as in the cases of victims of gender violence and labor or sexual harassment. Specific Objective 2: To promote and strengthen equality of labor and professional rights by reducing the inequalities that may persist in the field of the professional career of civil servants, and to support the reconciliation of personal, family and working life and the co-responsibility between men and women. Specific Objective 3: Promote and facilitate the participation of company members and union organizations in decision-making, implementation, and evaluation instances of plans, programs, goals, and outcomes that materialize the fulfillment of the strategic axes. Specific Objective 4: Encourage contractors and suppliers to be aware of and share UNOCACE's commitments to equal opportunities, gender equity, and work-life balance. This objective is aimed to the participants of the Also, all entities participating in the project are aligned with the laws and standards established by Ecuador's government. This includes Chocolats Halba, Ecotop, WWF Switzerland, South Pole and GIZ. On the other hand, Coop has Fundamental requirements for all business partners[84], "the business partner will take appropriate measures to ensure that the country-specific regulations and the relevant conventions and principles of the United Nations, OECD and the International Labour Organization are observed. The business partner also guarantees that, prior to any activity that could adversely affect the rights, land, resources, livelihoods and food security of indigenous peoples and local communities, their Free, Prior and Informed Consent (FPIC) is obtained. Business partners also undertake to protect the rights of environmental and human rights activists, whistleblowers, complainants and community spokespersons, and guarantee their anonymity".

[84] Please see "COOP's Sustainable_business_policy" in the folder: Gender policy.

Question 3 – Is an Expert required for the Gender Safeguarding Principles & Requirements?	Not required
Question 4 – Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	Not required

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

Below is a summary of Step 2 in the GS4GG consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. **Summary of stakeholder mitigation measures**

1. Concerns were raised about the risks that producers participating in the project would be exposed to from activities such as thinning (use of machinery for tree felling).
2. The presence of indigenous communities (Chachi) in proximity to the project areas in Esmeraldas was identified as a safeguarding issue that should be continuously taken into consideration and monitored by the project.
3. Concerns raised by producers that DAf could potentially decrease the cocoa yield per hectare.
4. Producers raised concerns that DAF could potentially impact on water availability.
5. Producers raised concern that DAF could lead to soil degradation.

E.2. **Final continuous input/grievance mechanism**

The following channels to receive input/grievance from the cocoa producers were announced in the physical meetings for stakeholder consultation held from January to March 2021:

	Method chosen (include all known details e.g., location of book, phone number, identity of mediator)	**Justification of choice (best practice)**
Message text	WhatsApp group[85]	UNOCACE has set up a WhatsApp group to provide greater access to the grievance mechanism for producers.

[85] Please find the evidence of whatsapp group on: UNOCACE documents/02 Evidencia quejas.

	Method chosen (include all known details e.g., location of book, phone number, identity of mediator)	Justification of choice (best practice)
GS conta (mandatory)	help@goldstandard.org	
Telephone acce (optional)	(+593) 99 416 02 29	
Internet/email access (optional)	Website: https://unocace.com/ Email: info@unocace.com cristhian.roman@unocace.com	Will serve as records for formal communications.
Field contact	Field technicians	The field technicians will act as an intermediary betwee the community and the company in case of comment or complaints about the project. Any communit requests will be made through them.
Other	Recinto El Deseo Km 30 vía Duran – Boliche – Milagro, Ecuador.	Some people are in regular proximity to the UNOCACE office. Therefore it will be easier for them to communicate with this entity.

Field technicians are assigned by UNOCACE and are involved in the project FINCA. Field technicians are one channel to address participant grievances and inputs. In cases where producers do not have access to other channels, technicians will be available in the field, and they can help producers submit their comments easily. Field technicians will play a transmission role, and are intended to upload complaints at time of complaint, via WhatsApp, mail, phone and other communication channels described above, to ensure neutral transmission.

The measure that project has to ensure that technicians are neutral transmitters is by the "Compromiso para evitar Conflicto de interés - confidencialidad de la información"[86] signed by technicians. Also there is the email, where producers can send their comments and grievances anonymously. The process to address and track the grievances by UNOCACE is provided in the supporting documents[87].

[86] Please find the evidence of "Compromiso para evitar Conflicto de interés - confidencialidad de la información" on: UNOCACE documents/03 Conflicto de intereses
[87] Please see the "Flujo procesos reclamos" on UNOCACE documents/07 Diagrama mecanismo quejas y reclamos.

No grievances have been received through any of the mentioned channels.

Appendix 1 – Safeguarding Principles Assessment

Complete the Assessment below and copy all mitigation measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form below.

Assessment questions/requirements	Justification of relevance (Yes/potentially/no)	How the project will achieve requirements through design, management or risk mitigation.	Mitigation measures added to the monitoring plan (if required)
Principle 1. Human rights			
1. The project developer and the project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The project shall not discriminate with regards to participation and inclusion	No	The project complies with provisions set forth in the Universal Declaration of Human Rights. UNOCACE's principles and criteria embody the respect for human rights as provided for in common, national and international human rights laws. The project is developed by linking small farmers through the entire process of marketing and producing cocoa, and associated certified organic and fair-trade crops, which respects and empowers all partners and managers. Structural and associative growth is used as a strategy. The processes advanced by UNOCACE include all members of the producer's family to empower partners and their families. Likewise, partners participate in managerial positions of the organizations without discrimination on gender, race or religious grounds.	Not required
Principle 2. Gender equality			

1. The project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women. 2. Projects shall apply the principles of non-discrimination, equal treatment and equal pay for equal work. 3. The project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks. 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	The project does not expose girls and/or women to situations of slavery, labor exploitation or punishments. All family members are included in the project activities. Furthermore, families agree on the tasks to be performed among men, women and children according to their abilities and their time availability. UNOCACE policies endorse and guarantee the sound execution of processes, where employees and associated producers must ensure non-discrimination in the performance of their tasks. These policies are implemented following the National Guidelines of the Ministry of Foreign Affairs and Human Mobility of Ecuador.[88] The project respects and embraces the national gender strategy. UNOCACE's operating policies promote non-discrimination by race, gender or religion. According to this criterion, all members of the producer's family are included in the training activities[89] so they can be linked later to the technical team. In the same way, the benefits from the certification will be equally used within the families. The project abides by the gender policies established in the country, as detailed in section A8 of the project design document. In addition, the project considers and respects all cultural differences, roles, and responsibilities, as well as statutory rights. Therefore, throughout the development of the project, it will promote a gender approach among partner families so that results achieved are equitably reflected both on men and women.	Not required
Principle 3. Community health, safety and working conditions			

[88] PDD Supporting documents/Safeguards/Política UNOCACE

[89] PDD Supporting documents/Safeguards/Capacitaciones individuales

1. The project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes	UNOCACE relies on an occupational safety protocol for its direct employees. The risks to which producers and their families may be exposed are associated with project activities like thinning. To avoid risk, the farmers will be trained to adequately carry out project activities and prevent accidents[90], such as is presented in the monitoring parameters in section B.7. In addition, all UNOCACE employees included in the project have health protection in case of any event.[91]	Yes
Principle 4.1 Sites of cultural and historical heritage			

[90] Please find attached a summary of training and monitoring visits to farmers in PDD Supporting documents/05 Capacitaciones/Capacitaciones (Normativas orgánicas, análisis de riesgos)

[91] PDD Supporting documents/Safeguards/seguro_entrada_técnico

Does the project area include sites, structures or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The areas where the project is being developed do not include cultural and historical heritage significant sites that may be affected by the project activities. This is endorsed by the organic production certifications[92] granted to current crops.	Not required
>>The project shall not involve or be complicit in the alteration, damage or removal of any sites, objects or structures of significant cultural heritage.		The benefits from the project are expressed in the technical support (training inputs)[93] for the strengthening of plantations, organizations and partners linked to the project. These benefits further the strengthening of productive and commercial chains to export cocoa. A map with the location of Indigenous Peoples in the country is included. It is based on data generated by the Geoportal from the Military Geographic Institute.[94] Although the map does not record the presence of the self-identified Indigenous group, the Chachis, the project recognizes their presence and their rights. The project will keep special records for all the activities performed in the Chachi community and will include them in the project area and project activities.	

[92] PDD Supporting documents/UNOCACE certifications. The organic certification is suspended as of June 2022, currently UNOCACE is working on recertification.

[93] PDD Supporting documents/technical Information

[94] PDD_Supporting_documents/Safeguards/Indigenous people

>>Where a project proposes to utilize cultural heritage, including the knowledge, innovations, or practices of local communities, affected communities shall be informed of their rights, the scope and the consequences of the activity.		The project will not utilize cultural heritage sites for project activities.	Not required
>>The project shall provide for equitable sharing of benefits from commercialization of such knowledge, innovation, or practice, consistent with their customs and traditions.		The project will not commercialize cultural knowledge.	Not required
>>The opinions and recommendations of an expert stakeholder shall be sought and demonstrated as being included in the project design		The project areas do not include areas of cultural knowledge, therefore no issues regarding it occur in the project area.	Not required
Principle 4.2 Forced Eviction and Displacement			
Does the project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project is developed in territories with land ownership. Most of the participants are the owners. Communities are not displaced for activities, because activities are integrated within communities.	Not required
Principle 4.3 Land Tenure and other rights			

Does the project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	Landowners will join the project on a voluntary basis depending on their interests in the project	Not required
>>For projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?		Only landowners with title deed allotment letters will participate in the exercise especially because it is a long-term project. However, in some cases, the owner holds evidentiary documentation for their property right. Such documents may include land titles or tax bills. Such tax bills indicate that the landowner inhabits the land and pays taxes to the government.	
The project developer shall identify all such sites/matters potentially affected by the project. For all such sites/matters identified the project shall respect and safeguard: (a) Legal rights, or (b) Customary rights, or Special cultural, ecological, economic, religious or spiritual significance of people shall be demonstrably promoted/protected.		The project is developed in territories with private land ownership. Most of the participants are the owners themselves. Communities have not been displaced to leave room for activities, because these are integrated within them.	

Changes in legal arrangements must be in line with relevant law and regulation and must be carried out in strict adherence with such laws. All legal disputes must be resolved prior to the project being carried out in such areas. All such changes must be demonstrated as having been agreed with free, prior and informed consent	No	The project does not involve any change in land tenure.	Not required
The project developer must hold uncontested land title for the entire project boundary to complete project design certification.	--	The project developer does not own the land, as the land belongs to the local communities. Each landowner holds documentation for evidencing their property right. Such documents may include land titles or tax bills. Such tax bills indicate that the landowner inhabits the land and pay the taxes to the government.	Not required
The opinions and recommendations of an expert stakeholder(s) shall be sought and demonstrated as being included in the project design		To be included at the design certification stage.	Not required
Principle 4.4 Indigenous Peoples			
Are Indigenous Peoples present in or within the area of influence of the project and/or is the project located on land/territory claimed by Indigenous Peoples?	No	To date, there are no officially recognized Indigenous communities involved in the project. However, UNOCACE recognizes and respects the self-identified Indigenous community, the Chachis.	
The project developer shall identify all communities of Indigenous Peoples within the project area of influence who may be affected directly or indirectly by the project	Yes	The project has identified the Indigenous community, the Chachis, in the province of Esmeraldas.	Constant update on the Indigenous communities' plots.

The project developer shall recognize and respect the Indigenous People's collective rights to own, use, and develop and control the lands, resources and territories that they have traditionally owned, occupied or otherwise used or acquired, including lands and territories for which they do not yet possess title	Yes	The project recognizes and respects Indigenous People's collective rights to own, use, and develop and control the lands, resources and territories that they have traditionally owned, occupied or otherwise used or acquired, including lands and territories for which they do not yet possess title.	Constant update on the Indigenous communities plots.
The project developer shall respect, protect, conserve and shall not take the cultural, intellectual, religious and spiritual property of Indigenous Peoples without their free, prior and informed consent (FPIC).	No	The project does not take any cultural, intellectual, religious or spiritual property. All the intellectual knowledge from the project is based on the DAF system.	Does not required

The project developer shall ensure that the Indigenous People are provided with the equitable sharing of benefits to be derived from utilization and/or commercial development of natural resources on lands and territories or use of their traditional knowledge and practices by the project. This shall be done in a manner that is culturally appropriate and inclusive and that does not impede land rights or equal access to basic services including health services, clean water, energy, education, safe and decent working conditions and housing	No	No utilization or commercial development of natural resources on lands and territories will be conducted by the project. There is no usage of traditional knowledge by the project.	Not required
The opinions and recommendations of an expert stakeholder(s) shall be sought and demonstrated as being included in the project design	Yes	Official sources, such as the map derived from the last census in the country (2010),[95] were consulted to verify the presence of Indigenous communities. Although according these sources, there are no Indigenous communities in the project area, UNOCACE recognizes and respects the self-identified Indigenous community, the Chachis.	
Principle 5. Corruption			

[95] Supporting documents/safeguards/indigenous self-identification map

1. The project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt projects	No	UNOCACE has developed an internal control system for the production and marketing activities of organic cocoa, fair trade and Biosuisse,[96] to ensure that cocoa production and marketing activities are carried out as provided for in the Internal Rules of the General Regulations to Promote and Regulate Organic Production in Ecuador.[97] All the members of UNOCACE are fully trained to follow the policies, including the ones against the corruption, presented above. Additionally, the entire project was informed by FPIC and all the participants agree that the process was transparent.	Not required
Principle 6.1 Labor rights			

[96] PDD Supporting documents/Safeguards/Control interno para el comercio justo

[97] General Regulations to Promote and Regulate Organic Production in Ecuador https://services.controlunion.com/organic/product/ecuador-organic/ and organic certifications are in folder "UNOCACE certifications".

1. The project developer shall ensure that all employment is in compliance with national labor occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions. 2. Workers shall be able to establish and join labor organizations.[98] 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND	No	Producers are not considered employees of the project, since they will implement the activities on their private properties, for their own benefit. In this case, producers voluntarily sign-in into an association and in addition sign a commitment contract[99]– to be linked to the project – that states both the responsibilities of the owner and UNOCACE, as well as the duration of the contract. • Minimum age for a producer to sign the commitment contract to be part of the project is 18 years. • The project participants are trained in usage of tools. • The project participants holds health insurance provided by the government, called “Seguro Campesino”. • None of the landowners have employed workers. All the tasks will be performed by the family members. • The landowners and family members will carry out the project activities, therefore no additional labor force is required. • By traditional and cultural practice, and because farms may have the presence of family members, children are expected to help the family in their productive activities and household chores. Children over 12 years of age will help planting trees on weekends, because it is illegal to deprive them of schooling. Tasks to be carried out by children include planting activities and weeding. • Training is delivered to producers before starting planting activities[100][101].	Not required

[98] PDD Supporting documents/Safeguards/Contrato de Trabajo Asistente técnico
[99] PDD Supporting documents/Safeguards/Formato Contrato Productores
[100] Please find attached a summary of training and monitoring visits to farmers in PDD Supporting documents/06 Monitoreo/seguimiento parcelario
[101] Please find attached a summary of training and monitoring visits to farmers in PDD Supporting documents/05 Monitoreo/capacitaciones (Normativa orgánica, producción de parcela, control de malezas).

d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employees, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labor is allowed (exceptions for children working on their families' property requires an expert stakeholder opinion). 5. The project developer shall ensure the use of appropriate equipment; training of workers; documentation and reporting of accidents and incidents; and emergency preparedness and response measures.			
Principle 6.2 Negative economic consequences			

1. Does the project cause negative economic consequences during and after project implementation?	Potentially	The project does not cause any economic damage to producers. However, the project will record the amount of cocoa produced per plot to monitor negative economic consequences that are not impacting the producers. On the contrary, the resources obtained by the certification will be earmarked for the strengthening of crops and the technical and organizational training of partners to ensure the sustainability of the processes.	Yes
>>			
Principle 7.1 Emissions			
Will the project increase greenhouse gas emissions over the baseline scenario?	No	The project is based on planting timber and fruit trees. It does not involve the removal of trees, and it does not use nitrogen fertilizers or combustion machinery producing greenhouse gas emissions.	Not required
>>			
Principle 7.2 Energy supply			
Will the project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project activities do not involve the use of machinery or tools powered by energy from a local grid.	Not required
>>			
Principle 8.1 Impact on natural water patterns/flows			

Will the project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	Potentially	According to the FAO,[102] agroforestry systems can contribute to improving the availability and quality of water at basin level, through water being harvested by trees, reducing run-off and improving infiltration.[103] The risks of the project negatively impacting the catchment are minimal because the project is dispersed throughout many basins. As the activity is not concentrated in one single place, the overall effects are negligible. Also, thanks to the size of the plots and the fact that most activities are manually carried out, the project does not produce erosion and does not compromise the stability of water bodies.	Some project participants have wells on their properties. Such wells are small scale, but records will be kept in order to survey any potential effect on the catchments.
>>			

[102] http://www.fao.org/forestry/agroforestry/89999/en/

[103] http://www.fao.org/3/xii/ms20-e.htm

		The project consists of small plots, widely spread throughout different provinces in Ecuador, with permanent vegetation cover in regions where rainfall exceeds 1000 mm/year. This means that the water consumed by the crop does not pose any threat to the basin. There will be no massive or extensive planting, and the cover composition of the agroforestry system is similar to that of a natural forest, which does not negatively impact water flows of the basin[104]. Additionally, there will be no clear-cutting. Project activities do not include the discharge of wastewater, or abstractions, except for some users who have wells in their properties. The activity of wells is not nationally regulated and, additionally, the impact will be very insignificant at basin level, considering the dispersion of the project's plots. There is a protocol to look after water resources, which is attached in the supporting documents.[105] The project area is located in an area with low water stress, according to the Atlas of Natural Areas Exposed to Risks and Emergencies of Ecuador, prepared by the National Service of Risk and Emergency Management.[106]	
Principle 8.2 Erosion and/or water body instability			

[104] http://www.fao.org/3/xii/ms20-e.htm

[105] PDD Supporting documents/Safeguards/Water/Protocolo Ambiental UNOCACE

[106] PDD Supporting Documents/Risks/Atlas amenazas naturales

Could the project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project guarantees soil protection through the following actions: implementation of green fences, companion crops, permanent vegetation cover on the soil, non-mechanized tilling and soil dressing as provided for in organic certification regulations.[107,108] Due to the nature of the project activities, as well as their scale and dispersion, there is no impact on water bodies or erosion.	Not required
>> Is the project's area of influence susceptible to excessive erosion and/or water body instability?			

[107] PDD Supporting documents/UNOCACE certifications. The organic certification is suspended as of June 2022, currently UNOCACE is working on recertification.

[108] PDD Supporting documents/Technical information/Metodología técnica de establecimiento de una parcela.

>> The risk of the project negatively impacting the catchment and any risks impacting the project's success shall be assessed and addressed. The project shall ensure its ongoing, long-term viability and impact on surrounding social-economic and environmental assets. The project shall assess the sensitivity of the physical area of influence due to low percentage of impervious cover in a project (e.g., basin, catchment), susceptibility to erosion and water body instability, and lack of terrestrial habitat connectivity. The project can use mapping tools, or other appropriate nationally recognized tools, academic or published studies on the relevant area. The recommended methods include online tools, visual inspection, engineering or physical assessment, historical land use records, aerial photographs, and verbal or written surveys with local agencies and residents on the characterisation of geomorphology of water bodies		With the implementation of project activities associated with reforestation, it is expected to positively impact the watershed, increasing coverage, reducing the risk of erosion and improving soil structure which in turn, improves infiltration.	

>> The project shall demonstrate that measures to ensure soil protection and minimized erosion are in place prior to the commencement of the project		To reduce the risk of erosion of the planted areas, slopes in areas included do not exceed the load capacity of the soil. Also, tree planting itself is an activity that ensures soil protection and minimizes erosion, especially as compared with the baseline scenario.	
>> The project shall demonstrate that measures will be undertaken to ensure that surface and ground waters are protected from erosion and that these measures are in place prior to the commencement of the project		Surface and groundwater will not be susceptible to any erosion caused by the project. On the contrary, tree planting is expected to reduce potential erosion.	
>> Measures shall be incorporated to reduce soil erosion on slopes (e.g., hedge and tree rows, natural terracing, infiltration strips, permanent ground cover). For these measures, the concept of the effective slope length shall be taken into account		The forest arrangements design is oriented in the direction of the contour lines and with densities of trees corresponding to the load capacity of the soil.	
>> Impact shall be reassessed at a frequency appropriate to the context of the ecosystem affected. The monitoring approach and frequency shall be justified by reference to natural patterns and variations		The increase in coverage will be monitored according to the monitoring plan. In addition, during follow up visits, the soil will be checked for changes that need to be managed.	

>> Where the project takes place in a water scarce or water stressed area, the opinions and recommendations of an Expert Stakeholder shall be sought and demonstrated as being considered and incorporated into the project design		According to the Aqueduct Water Risk Atlas tool,[109] the basin is in the Low category (<10%) in relation to water stress, so project activities are expected to contribute to improving soil infiltration by improving coverage.	
Principle 9.1 Landscape modification and soil			
Does the project involve the use of land and soil for production of crops or other products?	No	The project guarantees soil protection through the following actions: implementation of green fences, companion crops, permanent vegetation cover on the soil, non-mechanized tilling, and soil dressing as provided for in organic certification regulations.[110,111] Due to the nature of the project activities, as well as to their scale and dispersion, there is no impact on water bodies or erosion.	The project itself acts as a mitigation measure, since it involves permanent cover, no tillage and no heavy machinery usage. In addition, the species that complete their life cycle are cut and replanted in the plot, so that their biomass is incorporated into the soil to maintain and increase the physical and chemical properties of the soil.
>>			
Principle 9.2 Vulnerability to natural disaster			

[109] https://www.wri.org/applications/aqueduct/water-risk-atlas/#/?advanced=false&basemap=hydro&geoStore=0103742d088cc8b722ac444ed9163f79&indicator=bws_cat&lat=6.053161295714067&lng=-67.14843750000001&mapMode=view&month=1&opacity=0.5&ponderation=DEF&predefined=false&projection=absolute&scenario=optimistic&scope=basins&threshold=40&timeScale=annual&year=baseline&zoom=4

[110] PDD Supporting documents/UNOCACE certifications. The organic certification is suspended as of June 2022, currently UNOCACE is working on recertification.

[111] PDD Supporting documents/Technical information/Metodología técnica de establecimiento de una parcela.

Will the project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project does not increase the vulnerability to man-made hazards, such as landslides or floods. This is because the intervention areas are small and not contiguous, no machinery is being used, the soil is permanently protected with vegetation cover (which mitigates the impact of rainfall on the soil), and the project increases the infiltration capacity and mitigates the impact of landslides and floods.	Not required
>>			
Principle 9.3 Genetic resources			
Could the project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The plant material used in the project activities is native to Ecuador and has no genetic modification. Currently, a GMO cacao variety has not been fully developed.[112,113]	Not required
>> Projects involving the use of GMOs are not eligible for Gold Standard project Design Certification		The project activity does not involve the use of GMOs.	
>> An assessment for the risk of GMO contamination from outside the project area and reasonable and appropriate counter measures should be taken		There is no risk of GMO contamination from outside the project area.	

[112] PDD Supporting documents/Safeguards/GMO/Plant_biotechnology_2018

[113] PDD Supporting documents/Safeguards/GMO/Plant_sciences_2017

Principle 9.4 Release of pollutants			
Could the project potentially result in the release of pollutants to the environment?	No	The project does not use machinery, fuels or chemical pesticides. Also, all fertilizers employed are consistent with organic certifications.[114]. The project follows recommendations and good practices for waste management (separation) and periodic monitoring, including the inspection of waste management.[115]	Not required
>> The project shall avoid the release of pollutants. This applies to the release of pollutants to air, water, and land due to routine, non-routine and accidental circumstances		There will be no use of machinery, chemical fertilizers, or other agrochemicals.	
>> The project developer shall ensure that pollution prevention and control technologies and practices consistent with national regulation or international good practice are applied during the project life cycle		The owners have been informed from the initial phase that the use of agrochemicals is not allowed.	

[114] PDD Supporting documents/UNOCACE certifications (The organic certification is suspended as of June 2022, currently UNOCACE is working on recertification).

[115] PDD Supporting documents/Technical Information/Ficha de Inspección interna

>> All potential pollution sources that may result from the project that cause the degradation of the quality of soil, air, surface and groundwater within the project's area of influence shall be identified. Appropriate mitigation measures and monitoring shall be implemented to ensure the protection of resources. The project can use historical records, ongoing monitoring and reporting through data logging of physical measurements, online sources, government data. The recommended methods include quantitative documentation of all sources and volumes of water abstractions, use of weirs and gauges, flow meters, pump energy consumption, transpiration rates, government data		It has been identified that one potential pollutant is the bags that are used for the production of the seedlings. However, from the project promotes a campaign for owners to engage in collection and proper disposal.[116]	
Principle 9.5 Hazardous and non-hazardous waste			

[116] PDD Supporting information: Safeguards\Water/PROTOCOLO AMBIENTAL UNOCACE

Will the project involve the manufacture, trade, release and/or use of hazardous and non-hazardous chemicals and/or materials?	No	The project does not use machinery, fuels or chemical pesticides. Also, all fertilizers employed are consistent with organic certifications.[117] The project follows recommendations and good practices for waste management (separation) and periodic monitoring, including the inspection of waste management.[118][119][120]	Not required
>> projects shall avoid or, when avoidance is not feasible, minimize and control release of hazardous materials resulting from their production, transportation, handling, storage and use in the project. Where avoidance is not possible, the health risks, including potential differentiated effects on men, women and children, of the potential use of hazardous materials shall be addressed appropriately		Not applicable	

[117] PDD Supporting documents/UNOCACE certifications. The organic certification is suspended as of June 2022, currently UNOCACE is working on recertification.

[118] PDD Supporting documents/UNOCACE documents/06 Monitoreo/seguimiento parcelario

[119] Please find attached a summary of training and monitoring visits to farmers in PDD Supporting documents/05 Monitoreo/capacitaciones (Reglamento orgánico, manejo de residuos inorgánicos).

[120] PDD Supporting documents/Technical Information/Ficha de Inspección interna

>> Projects shall consider the use of less hazardous substitutes for such chemicals and materials and will avoid the manufacture, trade, and use of chemicals and hazardous materials subject to international bans or phase-outs due to their high toxicity to living organisms, environmental persistence, potential for bioaccumulation, or potential for depletion of the ozone layer		Not applicable	
>> All sources of waste and waste products shall be identified and classified. Waste products include, among others: (a) Chemical wastes, and (b) Containers, and (c) Fuels and oils, and (d) Human waste, and (e) Rubbish (including metals, plastics, organic and paper products), and (f) Abandoned buildings, machinery or equipment		It has been identified that one potential pollutant is the bags that are used for the production of the seedlings. However, the project promotes a campaign for owners to collect and conduct proper disposal.[121]	

[121] Supporting information: Safeguards/Water/PROTOCOLO AMBIENTAL UNOCACE

>> Where waste generation may not be avoided, the project shall reduce the generation of waste, and recover and reuse waste in a manner that is safe for human health and the environment		Not applicable	
>> Where waste may not be recovered or reused, it shall be treated, destroyed, or disposed of in an environmentally sound manner that includes the appropriate control of emissions and residues resulting from the handling and processing of the waste material		Not applicable	
>> If the generated waste is considered hazardous, reasonable alternatives for its environmentally sound disposal will be adopted while adhering to the limitations applicable to its transboundary movement		Not applicable	
>> The project shall not make use of chemicals or materials subject to international bans or phase-outs. For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol		Not applicable	

Principle 9.6 Pesticides & fertilizers			
Will the project involve the application of pesticides and/or fertilizers?	No	See response to the question above.	Not required
>>			
Principle 9.7 Harvesting of forests			
Will the project involve the harvesting of forests?	No	The project is not located in forests. In addition, it is essential to note that the project does not involve any harvesting process (except a very small number of Cedrela odorata (MU-DAF 2020-2023) and Gmelina arborea (MU-DAF 2018-2019) because the systems are intended to be implemented with conservation objectives, so no extractive wood or timber activities are included.	Not required
>>			
Principle 9.8 Food			
Does the project modify the quantity or nutritional quality of food available, for example through crop regime alteration or export or economic incentives?	No	On the contrary, the project will enhance the quantity and quality of food, by integrating food crops to the DAF.	Not required
>>			
Principle 9.9 Animal husbandry			
Will the project involve animal husbandry?	No	The project does not involve animal husbandry, but involves the occasional use of animals for transportation of materials.	Not required
>>			
Principle 9.10 High Conservation Value areas and critical habitats			

Does the project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is not located in HCV ecosystems, landscapes or key biodiversity areas. All areas have been historically employed for agriculture.	Not required
>>			
Principle 9.11 Endangered species			
Are there any endangered species identified as potentially being present within the project boundary (including those that may route through the area)? AND/OR Does the project potentially impact other areas where endangered species may be present through transboundary effects?	No	The project is not located in HCV ecosystems, landscapes or key biodiversity areas. All areas have been historically employed for agriculture.	Not required
>>			

Appendix 2- Contact information of project participants

Organization name	UNOCACE – Unión de Organizaciones Campesinas Cacaoteras
Registration number with relevant authority	
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Salutation	Señor
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Middle name	
First name	Fredy
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Personal e-mail	freddy.cabello@unocace.com

Organization name	Chocolats Halba

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Organization name	Ecotop
Registration number with relevant authority	

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Department	
Mobile	
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Personal e-mail	w.garrett@southpole.com

Appendix 3- LUF Additional Information

Risk of change to the project area during project certification period:	The UNOCACE cooperative carries out a selection process prior to granting access to the participants of the project. This process includes a review of land tenure. This information is studied and categorized (owners, tenants, inheritance) and the findings are analyzed in order to guarantee the permanence of the project in the future.
Risk of change to the project activities during project certification period:	The main component of the project activities is the planting of cocoa for export. This export is supported by a high and growing demand for the product, especially organic production. Because the producers manage their crops under organic certification standards (Certified Organic, USDA Organic, Biosuisse organic), the purchase of production is very stable. Similarly, the producers sign a letter of commitment as well as statements to demonstrate their intention to remain within the scheme proposed by the project. This not only ensures production-based income, but also contributes to food production, both for self-consumption and for sale of any surplus.
Land use history and current status of project area:	The plots where the project is established have historically been areas dedicated to agriculture, which contain different crops such as corn, cocoa, coffee, palm, banana or they are disused grass areas. The identification of land use for the baseline was carried out through on-site inspections, since the size of each individual area would demand a very high level of detail from satellite images. The DAF project began in 2016. However, the GS carbon project did not begin until 2018. The plots are established as producers are incorporated into the project.

Socio-economic history:	Peasant family agriculture (AFC) represents 84.5% of the country's agricultural production units with a concentration of 20% of the land and uses 37% of irrigation water. The main objective is to satisfy basic needs.[122] More than 64% of the country's agricultural production is in the hands of small producers, and most of the food consumed comes from AFC, as does the supply of cocoa and coffee as export crops. Despite the importance of the AFC sector for the country, the population of rural areas has been consistently decreasing for several decades, to the point that by 2018 only 36% of the country's population lives in rural areas. This migration is mainly among the young population, who are looking for job opportunities, education and access to technology and communications, which are scarce in the countryside. On the other hand, in terms of cocoa production and export, Ecuador has a long history dating back more than 2,000 years. After the conquest, trade was controlled by Spain, and at the end of the 19th century Switzerland began to play an important role in the production of milk chocolate, creating the industry as we know it today. The great demand for chocolate, especially using fine cocoa, has dynamized the production in the country, which is the third-largest producer worldwide, represented mainly by small producers (70%). Exports of semi-finished products (liquor or paste, powder, butter, cake, nibs) represented a value of USD 47 million.[123]
Forest management applied (past and future):	The Dynamic Agroforestry Systems (DAF) technique is applied, in which the characteristics of the forest successions are reproduced. Different groups of species are planted so that all the sectors of each plot are occupied. The species used are: cocoa, banana, coconut and a number of timber, fruit and biomass tree species, depending on local availability. No agrochemicals are applied, and the plant residues from the same activity are used for fertilization.

[122] (FAO, 2010)
[123] (Anecacao, 2018)

Forest characteristics (including main tree species planted):	Main species: Cocoa (Theobroma cacao) Wood species: Pachacho (Schizolobium parahybum) Cedar (Cederela odorata) Mahogany (Swietenia macrophilla) Oak (Quercus petraea) Melina (Gmelina arborea) Yellow guaiac (Handroanthus chrisanthus) Guachapele (Samanea saman) Laurel (Laurus nobilis) Pink Guayacan (Tabebuia rosea) Fernan Sanchez (Triplaris cumingiana) Pechiche (Vitex giganteum) Fruit trees Mango (Mangifera indica) Avocado (Persea americana) Chocolate (Nephelium lappaceum) Soursop (Annona muricata) Subtle lemon (Citrus aurantifoloa) Orange (Citrus sinensis) Tangerine (Citris reticulata) Zapote (Pouteria sapota) Biomass species Guama (Inga spp. – Inga edulis) Achiote (Bixa rellana) Pioneering species Banana tree (Musa paradisiaca) Banana (Musa sp.) Guineo (Musa sp.) Silk Guineo (Musa sp.) Casava (Manihot esculenta) Beans (Phaseolus spp.) Ahuyama (Cucurbita spp.) Gandul (Cajanus cajan) Corn (Zea mays)

	The plots that make up the project have areas that vary between 0.5 and 2 ha at most. They are planted with a 4 x 3 m spacing between cocoa bushes. They are interspersed with banana or plantain at 4 x 3 m. Cassava is planted in rows between cocoa bushes and other complementary species can be included, according to availability and the characteristics of the land.
Main social impacts (risks and benefits)	Benefits: • Increased income for agricultural producers, through the commercialization of cocoa and other food crops. • Integration of young people into rural productive activities. • Training of young people as extension agents in DAF. • Contribution to the food security of rural producers, through the production of food in the same space as cocoa production. • Increased and improvement of cocoa productivity. Risks: • Possible work accidents during the performance of the work. • Possible market variations affecting the purchase of the cocoa crop.
Main environmental impacts (risks and benefits):	Benefits • Increase in biodiversity of productive systems. • Erosion protection. • Improvement in soil physicochemical condition. Risks: Because the activities do not involve the use of agrochemicals, no environmental risks are identified.
Financial structure:	The project is financed through donations, with technical and administrative support from the following entities: UNOCACE, Chocolats Halba, COOP, GIZ, South Pole and WWF Switzerland. Funding through donations is fixed until 2029. Thereafter, the project is expected to be sustainable, based on income from product sales generated.
Infrastructure (roads/houses etc.):	The project does not involve the construction of hard infrastructure (roads or buildings), since the crops are established on small plots and no specialized machinery or equipment is required for planting or maintenance. All work is done manually.

Water bodies:	The project area is distributed in a dispersed way in different provinces around the country. The hydrography of the country is classified according to the direction of the slope: Pacific and Amazon slopes. The main hydrographic systems where the project areas are located are: • Bolívar: Located within the Guayas basin. • El Oro: The main basins of this province are Zarumilla, Santa Rosa, Jubones and Arenillas. • Esmeraldas: Located within the Verde, Cayapas, Muisne and Esmeraldas basins. • Guayas: Within this province are the Guayas, Zapotal, Taura and Naranjal-Pagua basins. • Los Ríos: Contained within the Guayas Basin. • Manabí: One of the largest provinces in the country, which includes the basins of the Muisne, Jama, Chone, Portoviejo, Jipijapa and Guayas Rivers.
Sites with special significance for Indigenous Peoples and local communities – resulting from the stakeholder consultation:	No special significance sites were identified during the stakeholder consultation.
Where Indigenous Peoples and local communities are situated:	Local communities are located throughout the project area. In the town of Tres Vías, Esmeraldas province, there is an Indigenous group called the Chachis.
Where Indigenous Peoples and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	Within the project areas, no sites of special significance to Indigenous Peoples have been identified.

Appendix 4-Summary of Approved Design Changes

Please refer to Annex A of Principles and Requirements for more information on procedures governing Design Changes

Revision history

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on key project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 year rule) and ongoing financial need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved clarity on SDG contribution/SDG impact terms used throughout Clarity on stakeholder consultation information required Provision of an accompanying guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption